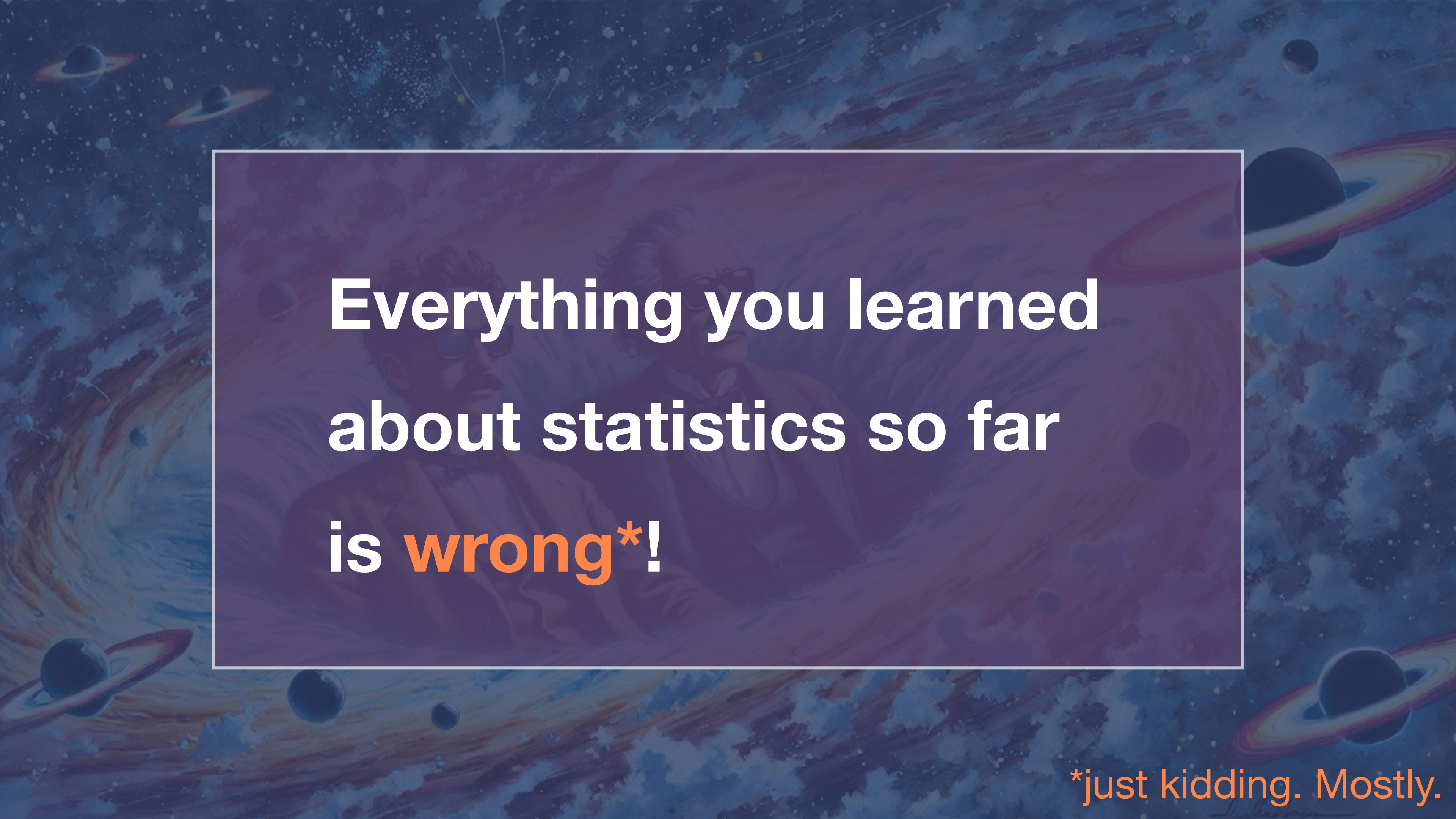




Bayesian Inference



Everything you learned
about statistics so far
is **wrong***!

**just kidding. Mostly.*

Estimating Probabilities for Rare Events



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It's the Cold War during the 1950. The US has lots of nuclear bombs on aeroplanes. How do we estimate the probability of a nuclear bomb exploding by accident?



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Reality: several (16) near-miss accidents had already happened.

What would you do?

What is the probability of a nuclear bomb accident?

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What is the probability of a nuclear bomb accident?



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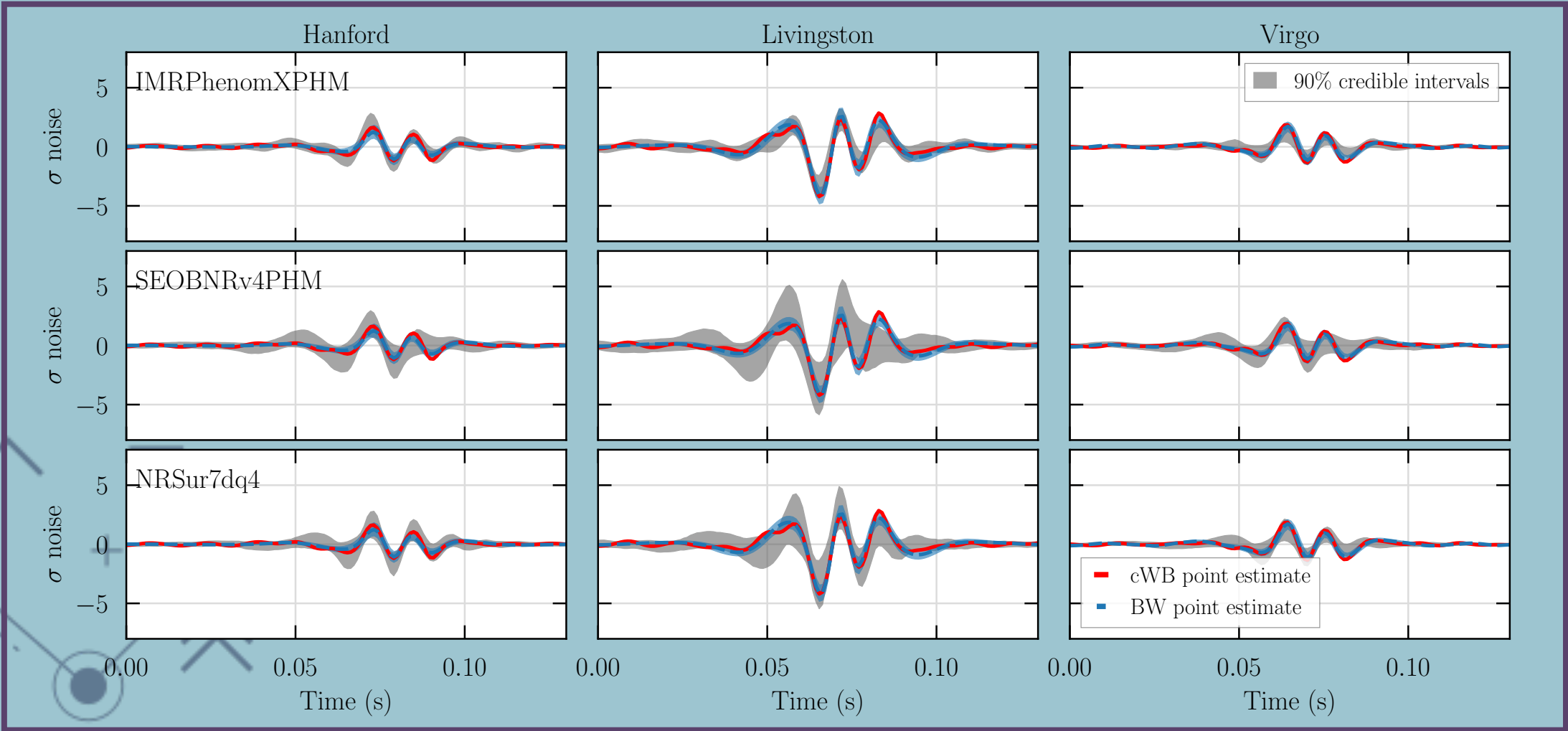
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- Include plans for future number of planes carrying nuclear weapons to calculate the number of “accident opportunities”
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Result: expect 19 “conspicuous weapon accidents” per year. Led to implementation of major safety features.

A related Astronomy
problem:

estimating event rates
in astronomical
surveys



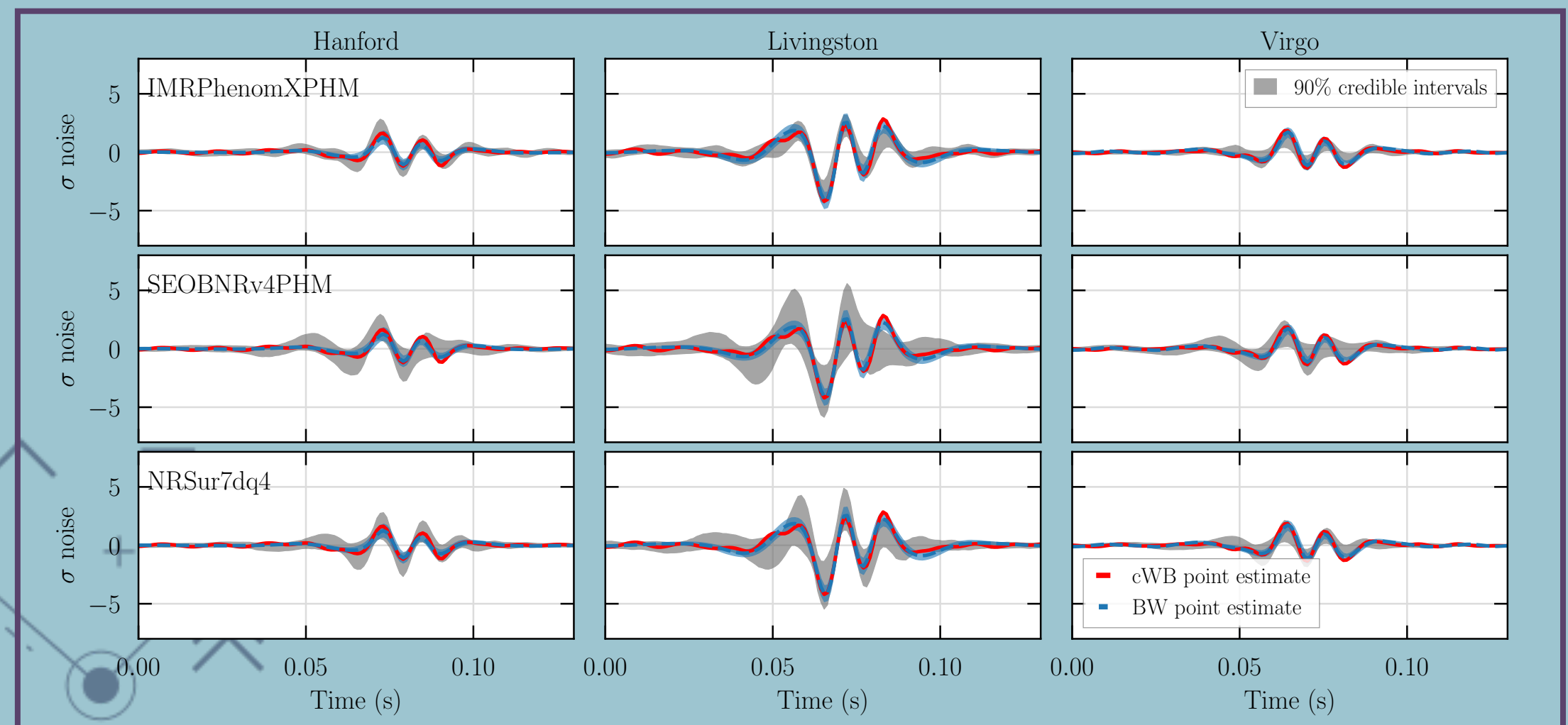
A related Astronomy problem:

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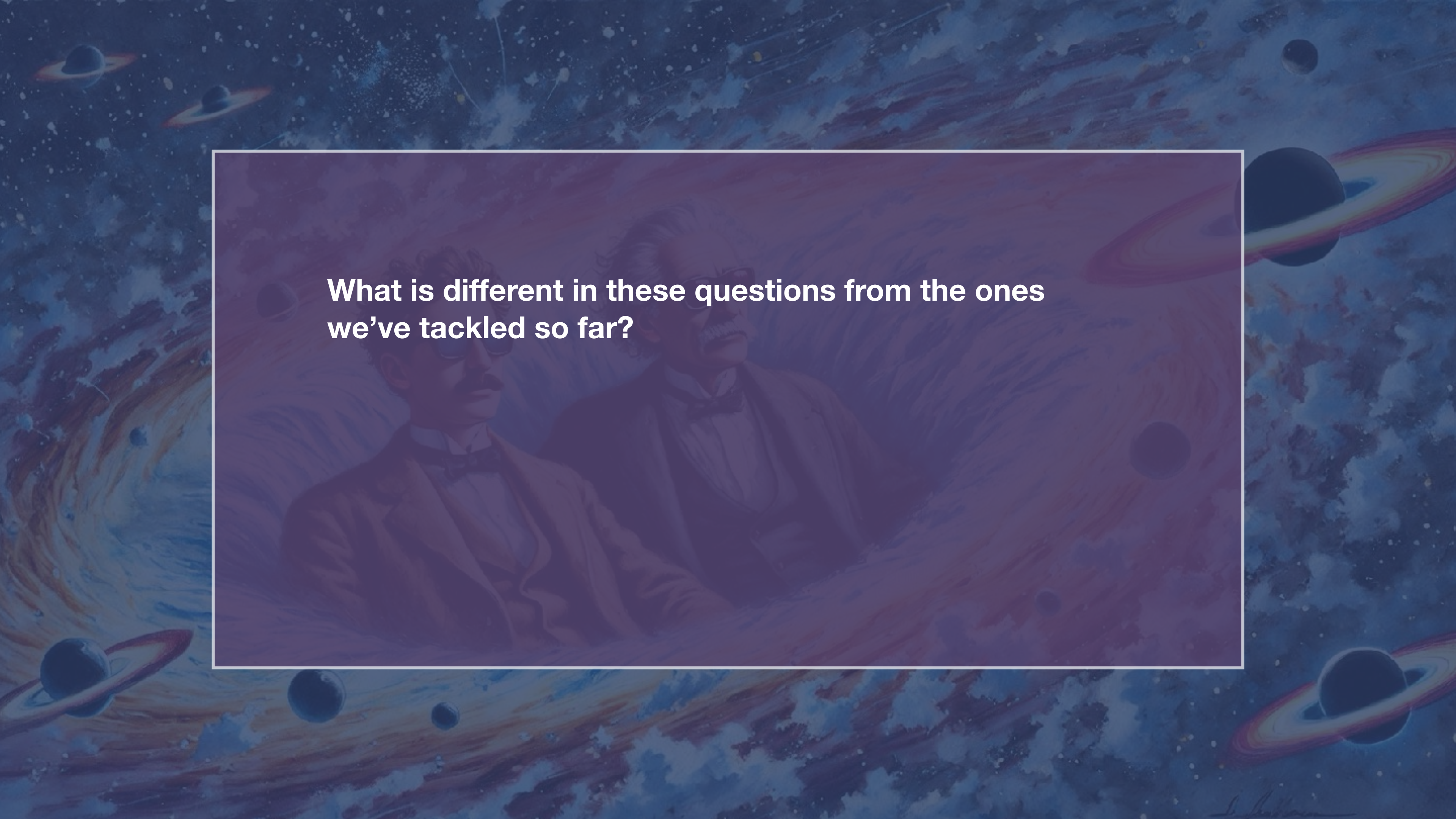
LIGO has found a single GW event from an intermediate-mass black hole.

What is the underlying rate of IMBH mergers?

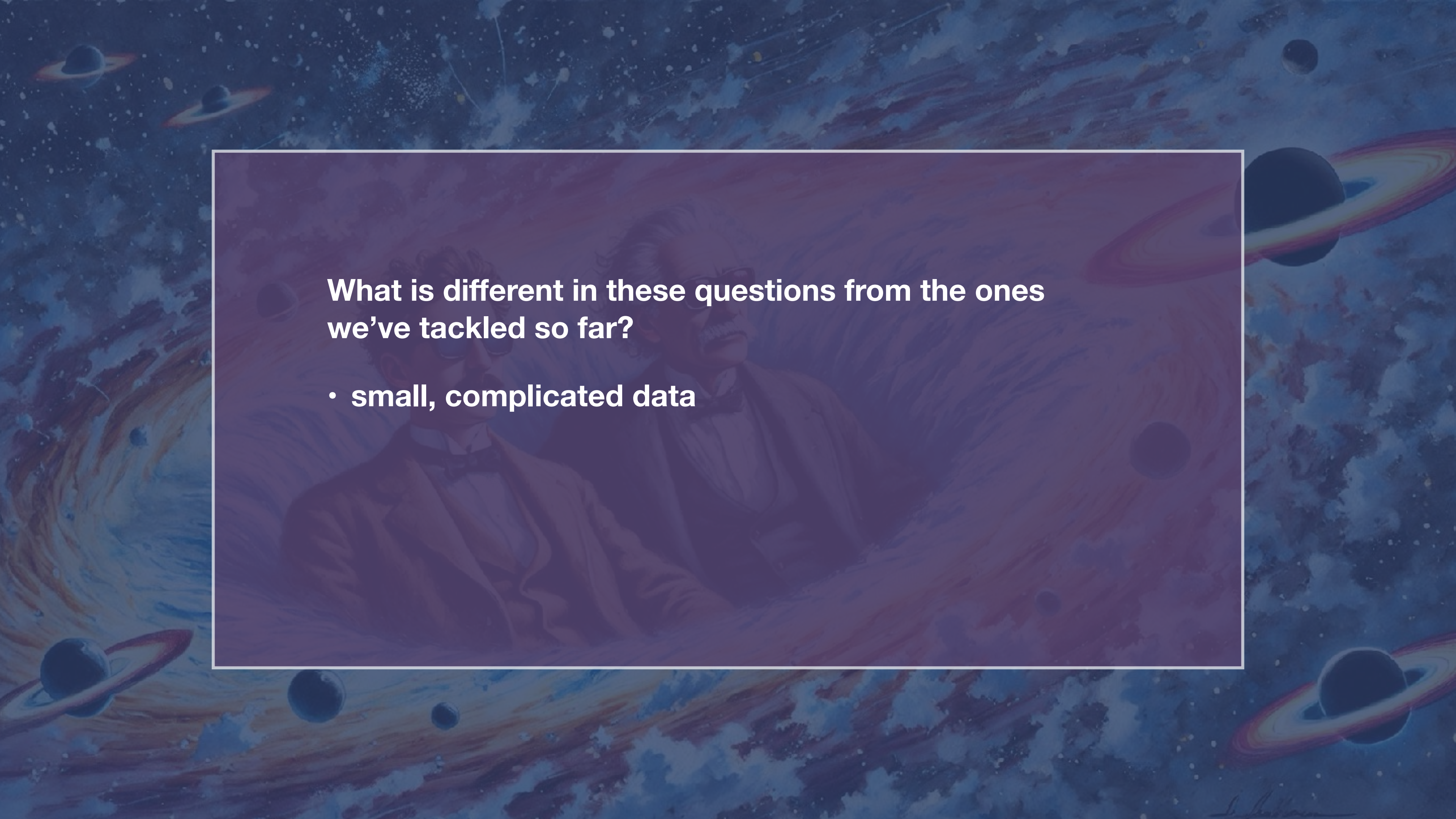
What is the predicted number of IMBH GW events in Advanced LIGO?



The LVK Collaboration (2021)

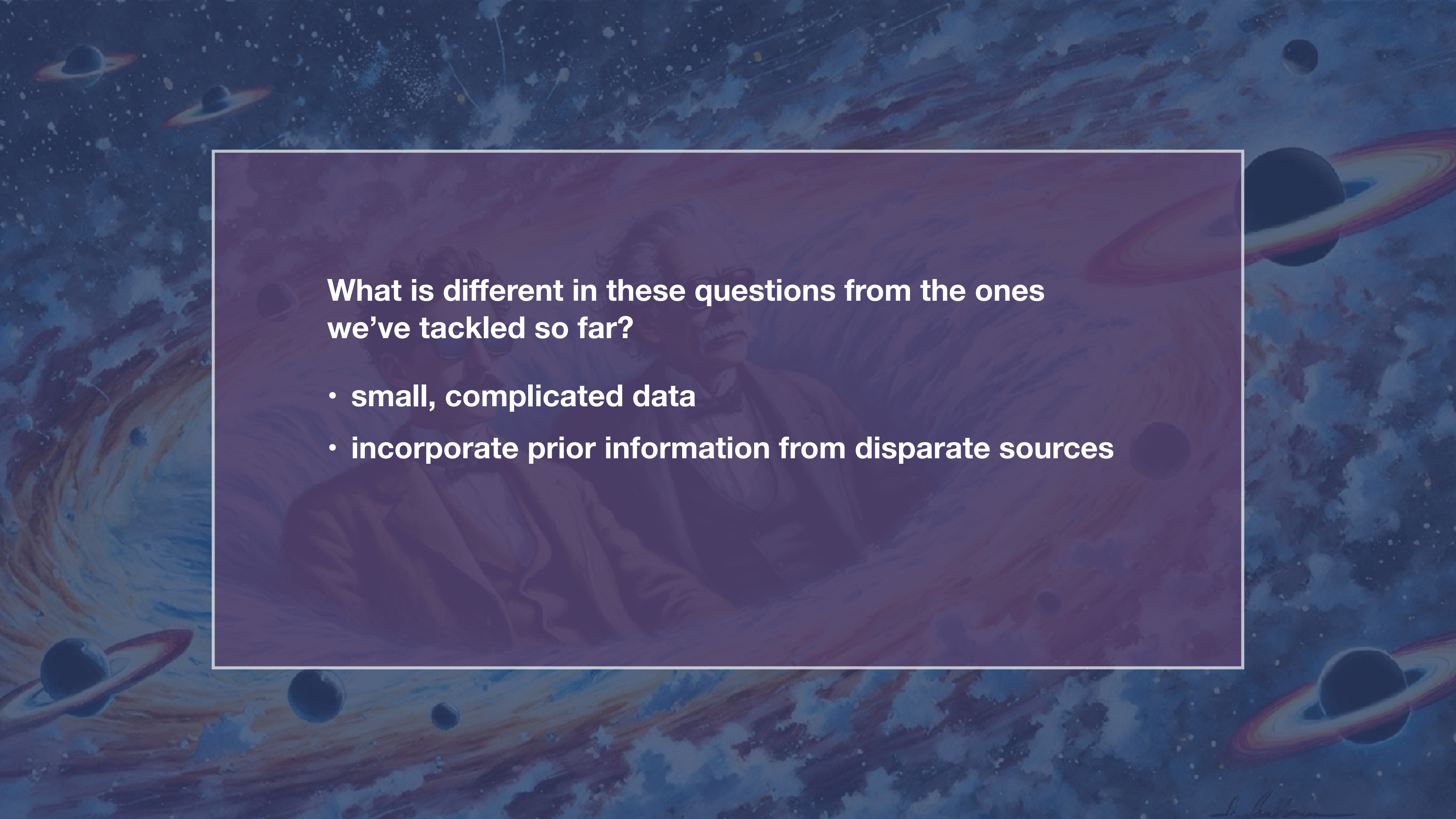
The background of the slide is a vibrant, painterly depiction of outer space. It features swirling galaxies in shades of blue, purple, and orange, interspersed with various celestial bodies, including planets with rings and smaller moons. In the center, a semi-transparent rectangular box with a thin white border contains a painting of two men. The man on the left is Albert Einstein, recognizable by his wild hair and mustache, wearing a dark tuxedo with a white shirt and a dark bow tie. He is looking slightly to his right. The man on the right is also wearing a tuxedo and a bow tie, and is looking towards the left. The overall tone of the image is intellectual and cosmic.

**What is different in these questions from the ones
we've tackled so far?**



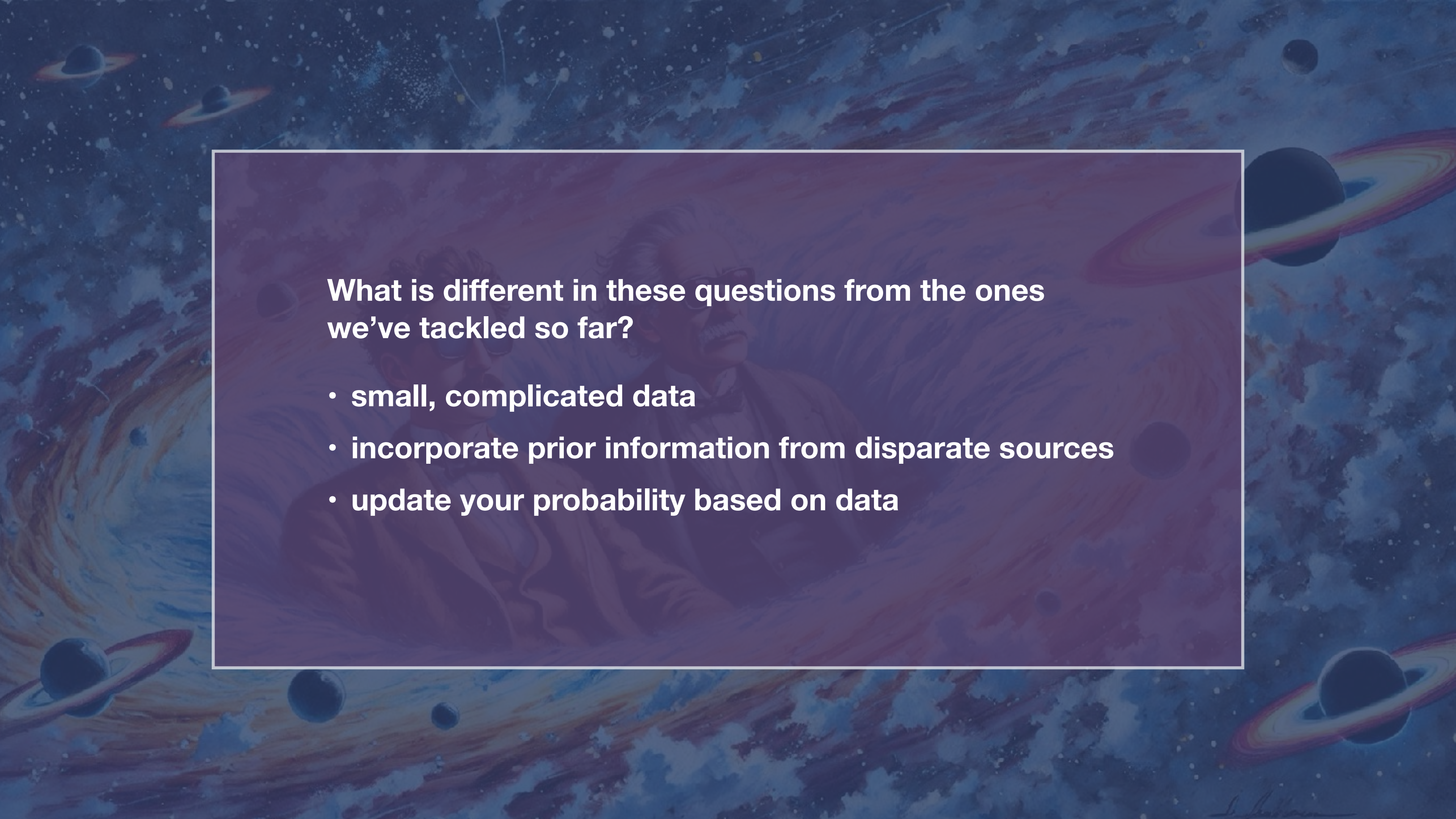
What is different in these questions from the ones we've tackled so far?

- **small, complicated data**



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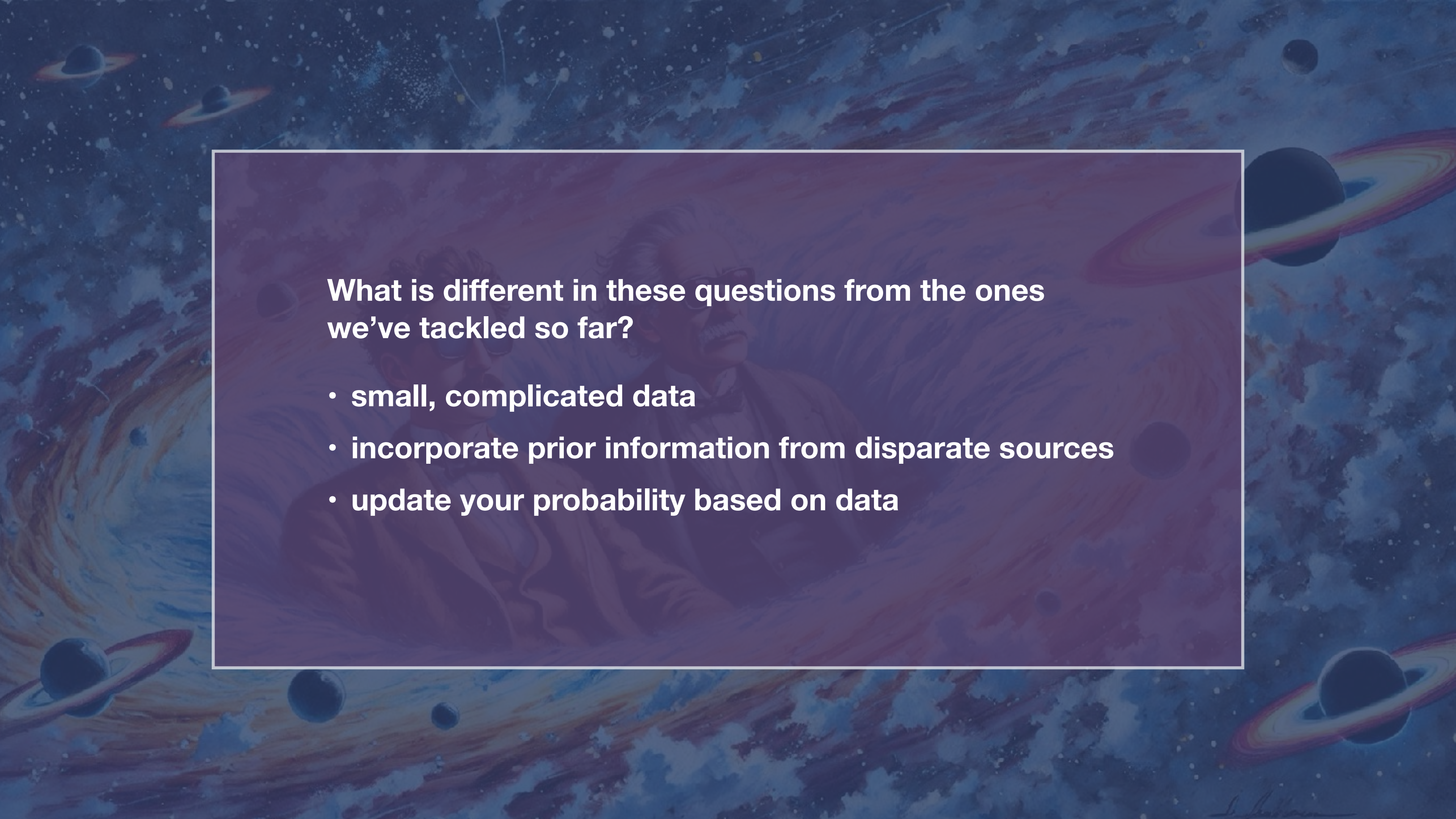
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- **small, complicated data**
- **incorporate prior information from disparate sources**
- **update your probability based on data**



Is it a random variable?

<https://app.wooclap.com/STATPHYSASTRO25>



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IS A RANDOM VARIABLE!

STATISTICS BATTLE OF THE CENTURY

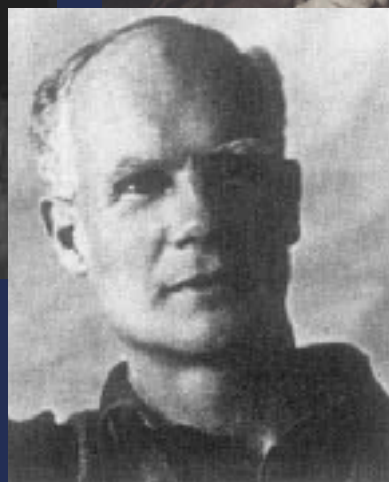


STATISTICS BATTLE OF THE CENTURY

Karl Pearson



Ronald A Fisher



Egon Pearson



Jerzy Neyman

+ most of 20th century
academic statisticians

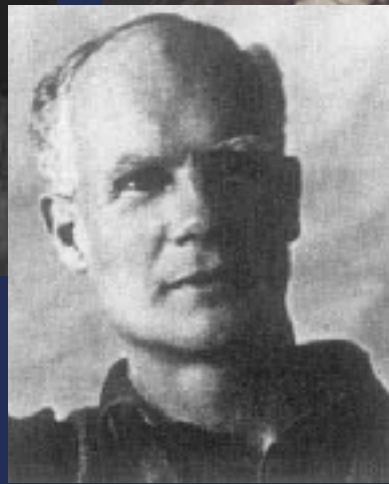
FREQUENTIST

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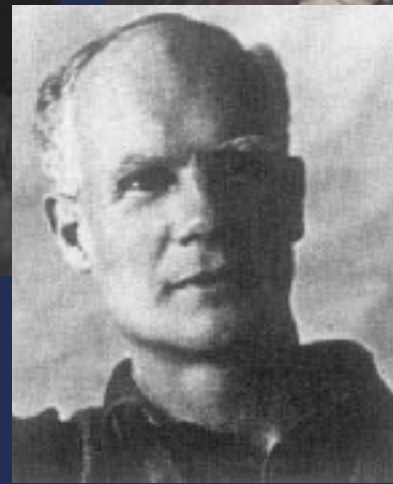
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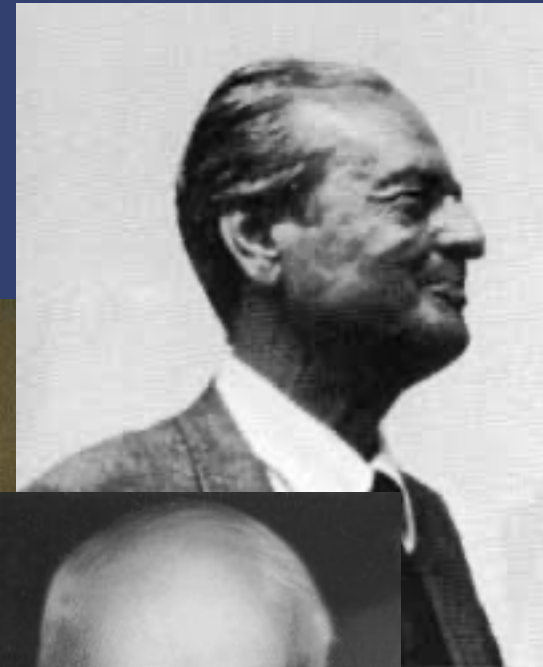
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FREQUENTIST

Thomas Bayes



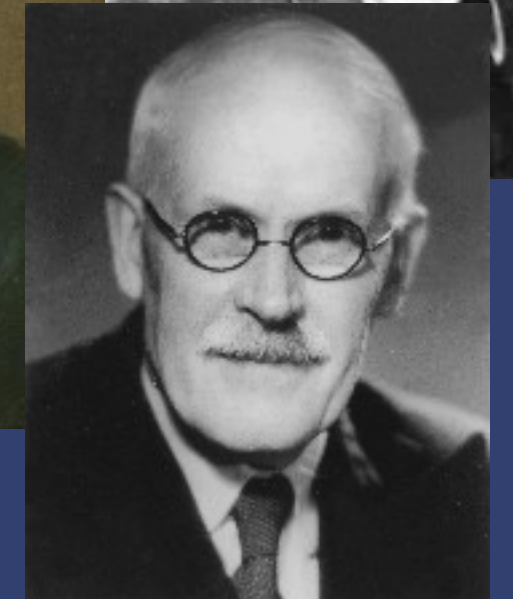
Bruno de Finetti



Alan Turing



Pierre Simon
Maquis de Laplace



Harold Jeffreys

VS

BAYESIAN

FREQUENTIST



*except the mathematical assumptions of your statistical tests

BAYESIAN



FREQUENTIST

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- probability defined as a long-running frequency derived from many repeated, identical trials (“proportion”)

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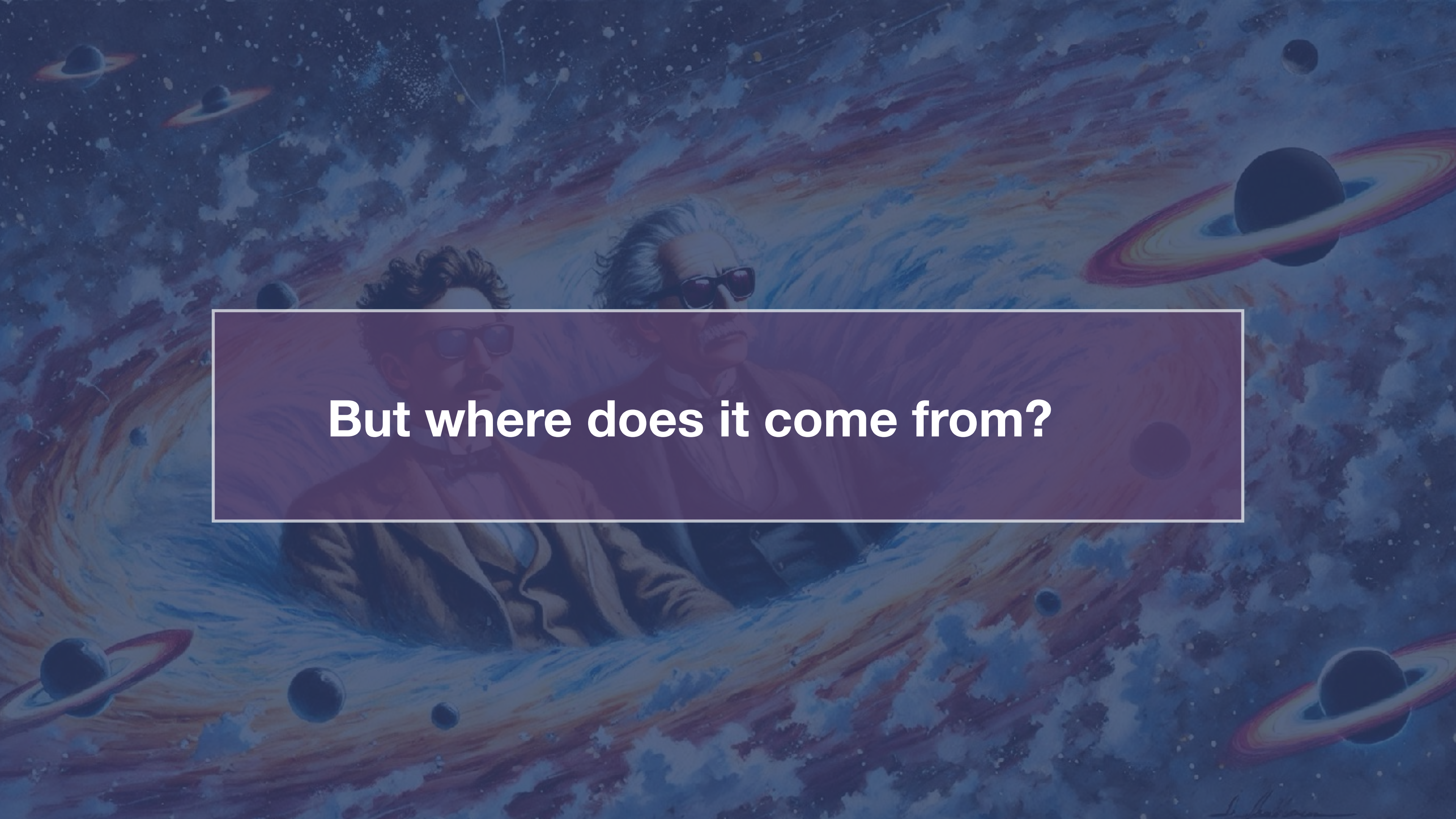


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- Our belief in an outcome is a combination of our prior beliefs in that outcome, and the information supplied by the data.
- Enables direct computation of probabilities of hypotheses

A painting depicting two prominent physicists, Albert Einstein and Stephen Hawking, floating in a cosmic space. Einstein, on the right, is shown with his characteristic wild white hair and glasses, wearing a dark suit. Hawking, on the left, has dark hair and a mustache, wearing glasses and a brown jacket. They are surrounded by a vibrant, swirling nebula in shades of blue, purple, and orange. Several planets with prominent rings, resembling Saturn, are scattered throughout the scene. The overall style is artistic and evocative, suggesting a deep connection between these scientists and the universe they studied.

But where does it come from?

A surreal scene featuring four white goats with brown horns and patterned sweaters. Two goats stand in the foreground, one on the left and one on the right. Two other goats are positioned behind three open, light blue doors in the background. The central door reveals a classic car with a checkered floor and a license plate that reads 'VIII 82 E?'. The room has wood-paneled walls and framed portraits of men. A semi-transparent purple box with white text is centered over the image.

(Conditional) Probabilities: Recap

Probability

Note: there exists more than one formalization of probability.

**Most common:
Kolmogorov formulation
and Cox formulation**

Probability is a (real-valued) set function that assigns to each event A in the sample space S a number $P(A)$, called the probability of the event A , such that the following hold:

1. The probability of any event must be nonnegative, that is, $P(A) \geq 0$.
2. The probability of the sample space is 1, that is, $P(S) = 1$.
3. Given mutually exclusive events $A_1, A_2, A_3, \dots$ that is, where $A_i \cap A_j = \emptyset$, for $i \neq j$,

- the probability of a finite union of the events is the sum of the probabilities of the individual events, that is:

$$P(A_1 \cup A_2 \cup \dots \cup A_k) = P(A_1) + P(A_2) + \dots + P(A_k)$$

More Probability

Probability of the Null:

If A is contained in B , then

Probability an event A is between 0 and 1

Probability of the complement of A is

Probability of A or B is



Notation

Probability of S

Probability of A

Probability of A and B

Probability of A or B

Probability of A given B

Probability of not A



Conditional Probabilities

- $P(\text{Astronomy}) = ?$
- $P(2 \mid \text{GRAPPA}) = ?$
- $P(\text{TP} \mid 3) = ?$
- Is $P(1 \mid \text{Astro}) = P(\text{Astro} \mid 1)$?
- $\sum P(\text{Other} \mid \text{year}) = ?$

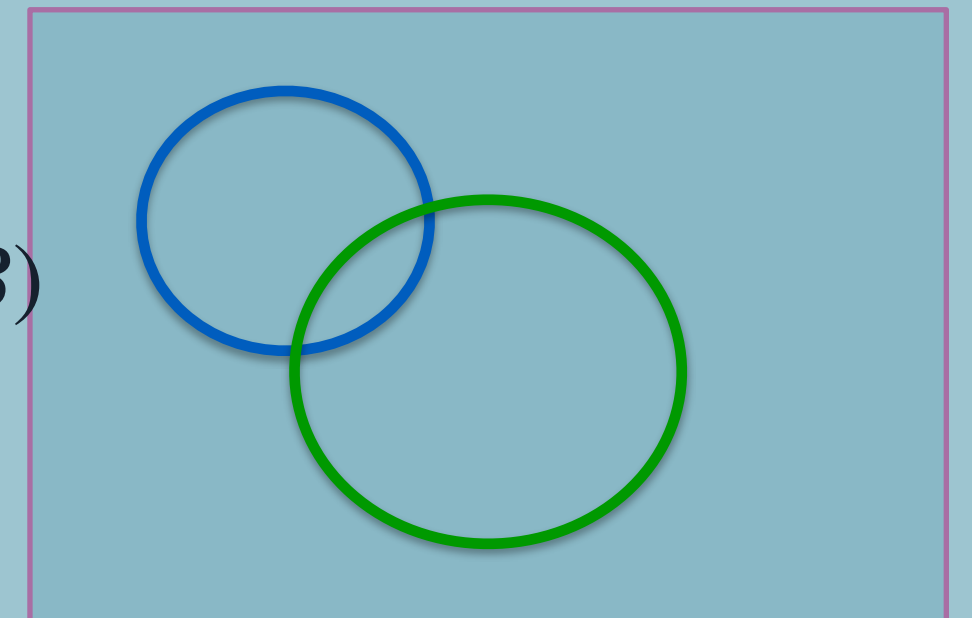
Suppose that this class has the following distribution of students:

Year/Track	Astronomy	GRAPPA	Theoretical Physics	Other	Total
1	19	9	2	3	33
2	3	2	1	0	6
3	1	0	1	2	4
Total	23	11	4	5	43

The probability of an event given some other event:

$$P(B \mid A) = \frac{P(A, B)}{P(A)}$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



Important: $p(\text{🌧️} | \text{☁️}) \neq p(\text{☁️} | \text{🌧️})$

Bayes Theorem

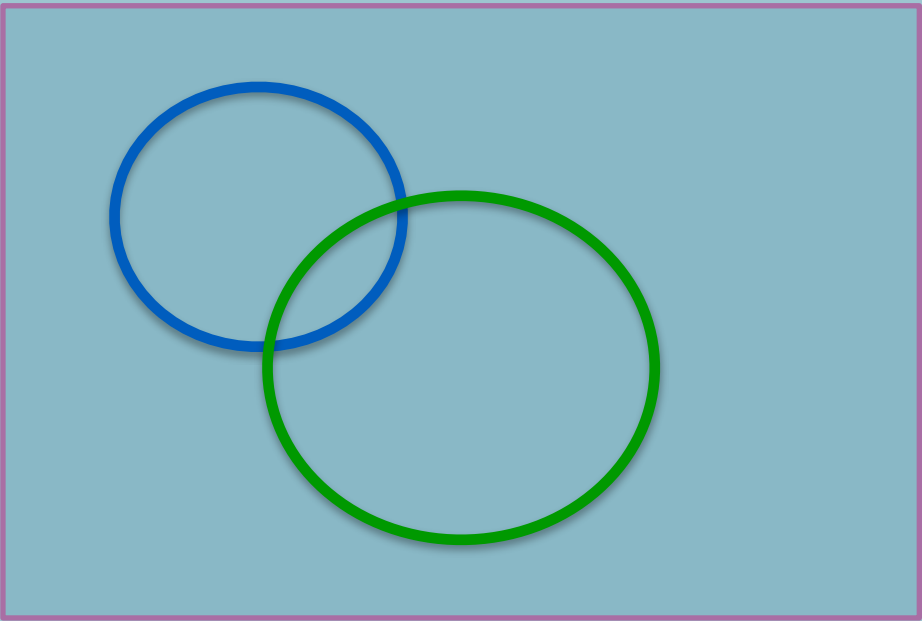
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The probability of an event given some other event:

$$P(B | A) = \frac{P(A, B)}{P(A)}$$

What is $P(A | B)$?



Questions

$$\int p(x | y) dx =$$

$$\int p(x | y) dy =$$



Questions

$$\int p(x | y) dx = 1$$

$$\int p(x | y) dy =$$



Questions

$$\int p(x | y) dx = 1$$

$$\int p(x | y) dy = \text{👎}$$



Multiplication Rule

$$p(x | y) = p(x, y) / p(y)$$



Multiplication Rule

$$p(x | y) = p(x, y) / p(y)$$

$$p(x, y) = p(x | y)p(y)$$



Multiplication Rule

$$p(x | y) = p(x, y) / p(y)$$

$$p(x, y) = p(x | y)p(y)$$

$$p(x) = \int_{-\infty}^{\infty} p(x, y) dy = \int_{-\infty}^{\infty} p(x | y)p(y) dy$$



$$p(x|y) = p(x, y)/p(y)$$

$$p(x, y) = p(x|y)p(y)$$

$$p(y|x) = ?$$



Marginalisation

$$p(x | y) = p(x, y) / p(y)$$

$$p(x, y) = p(x | y)p(y)$$

$$p(y | x) = ?$$



$$p(x, y) = p(x | y)p(y)$$

$$p(x, y) = p(y | x)p(x)$$



Bayes' Theorem

$$p(x, y) = p(x | y)p(y)$$

$$p(x, y) = p(y | x)p(x)$$

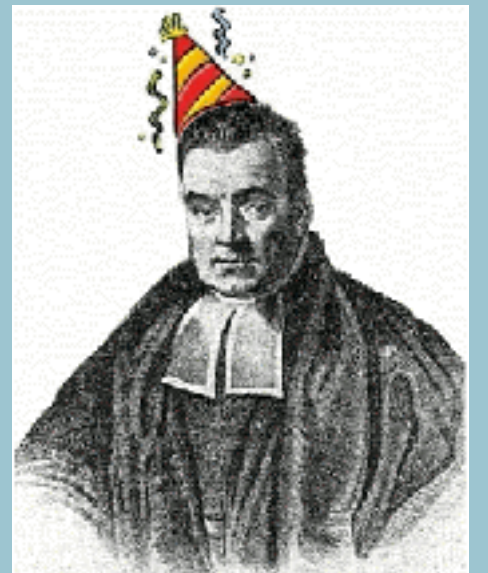


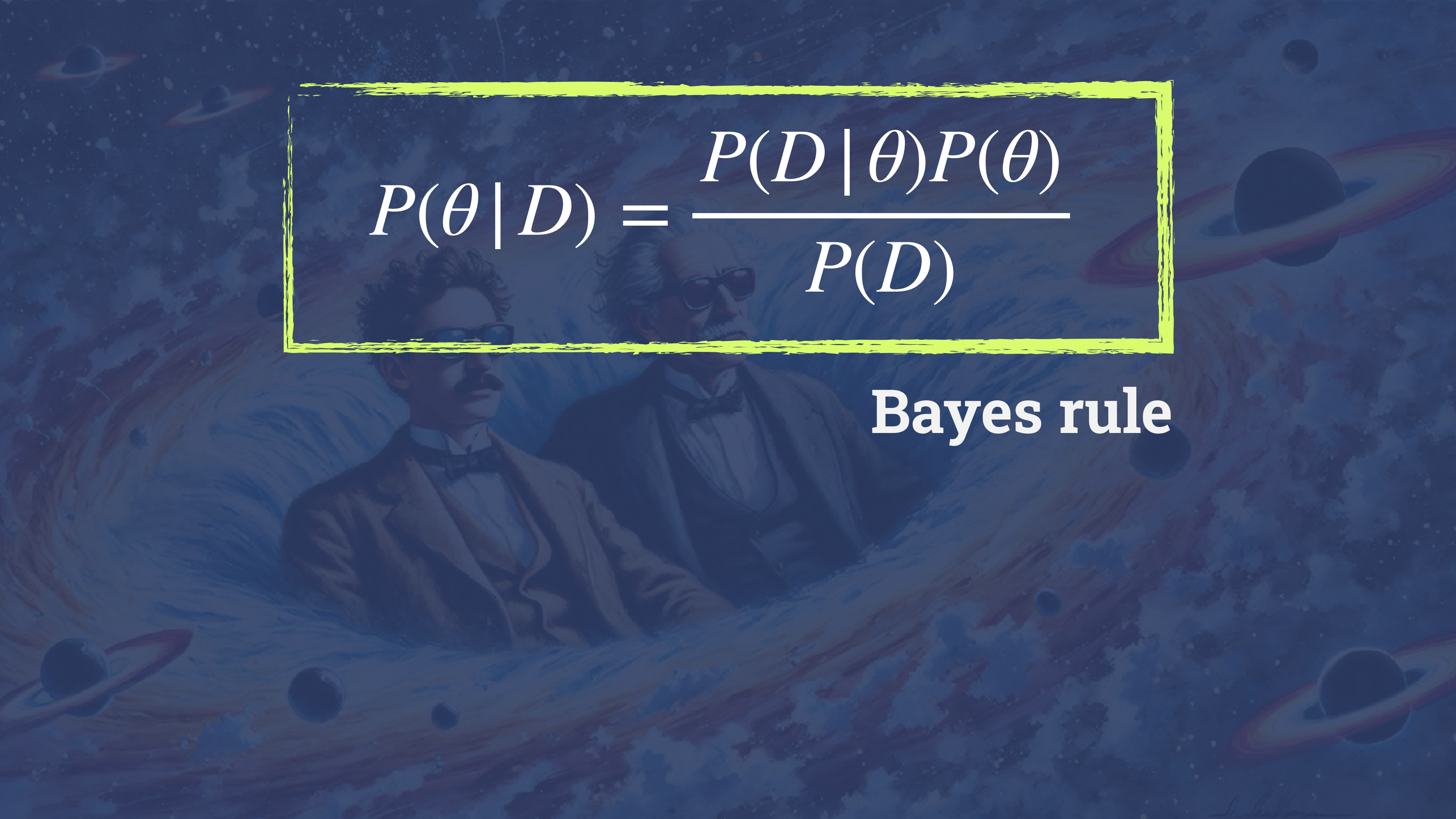
Bayes' Theorem

$$p(x, y) = p(x | y)p(y)$$

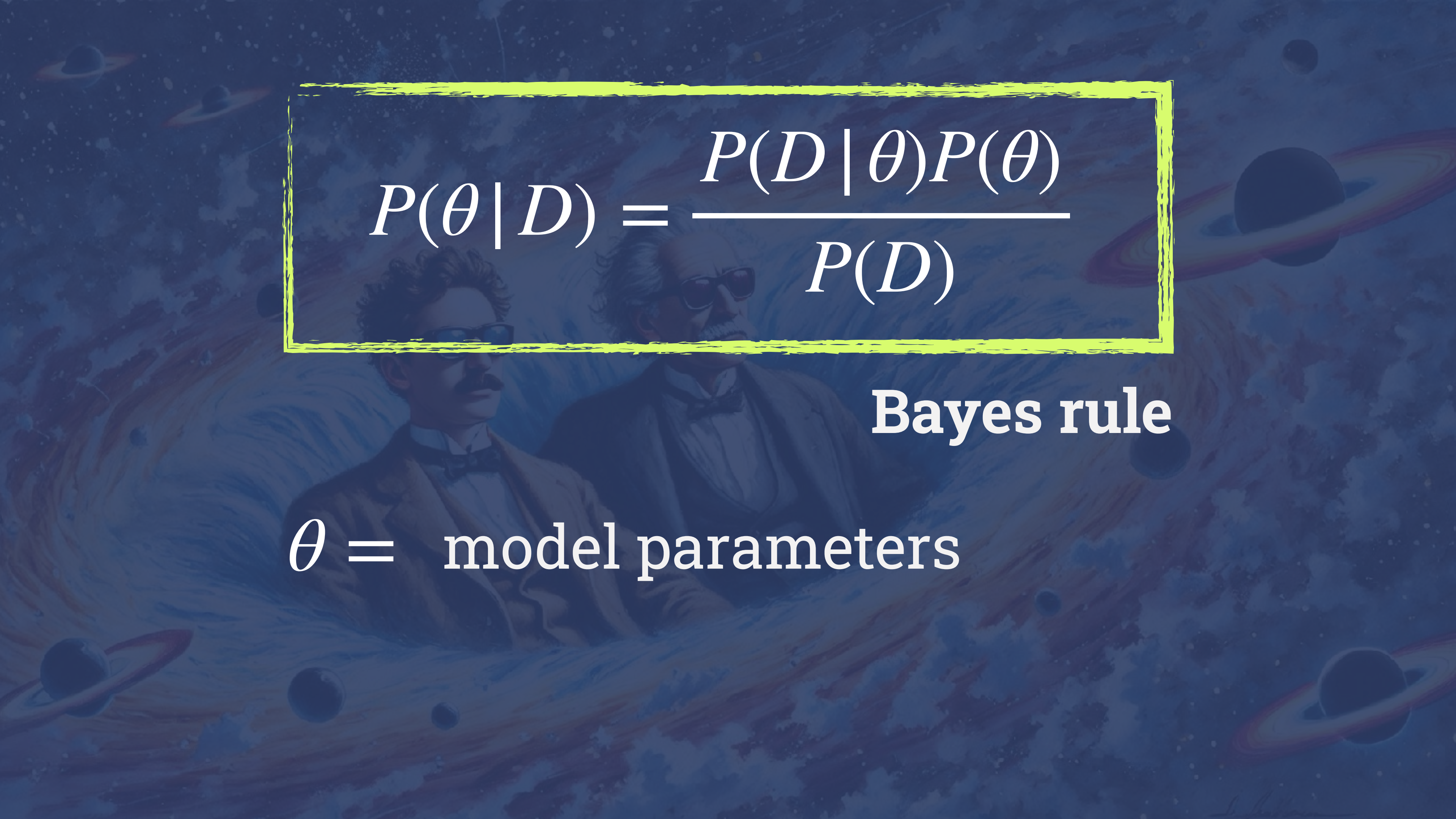
$$p(x, y) = p(y | x)p(x)$$

$$p(x | y)p(y) = p(y | x)p(x)$$




$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

Bayes rule

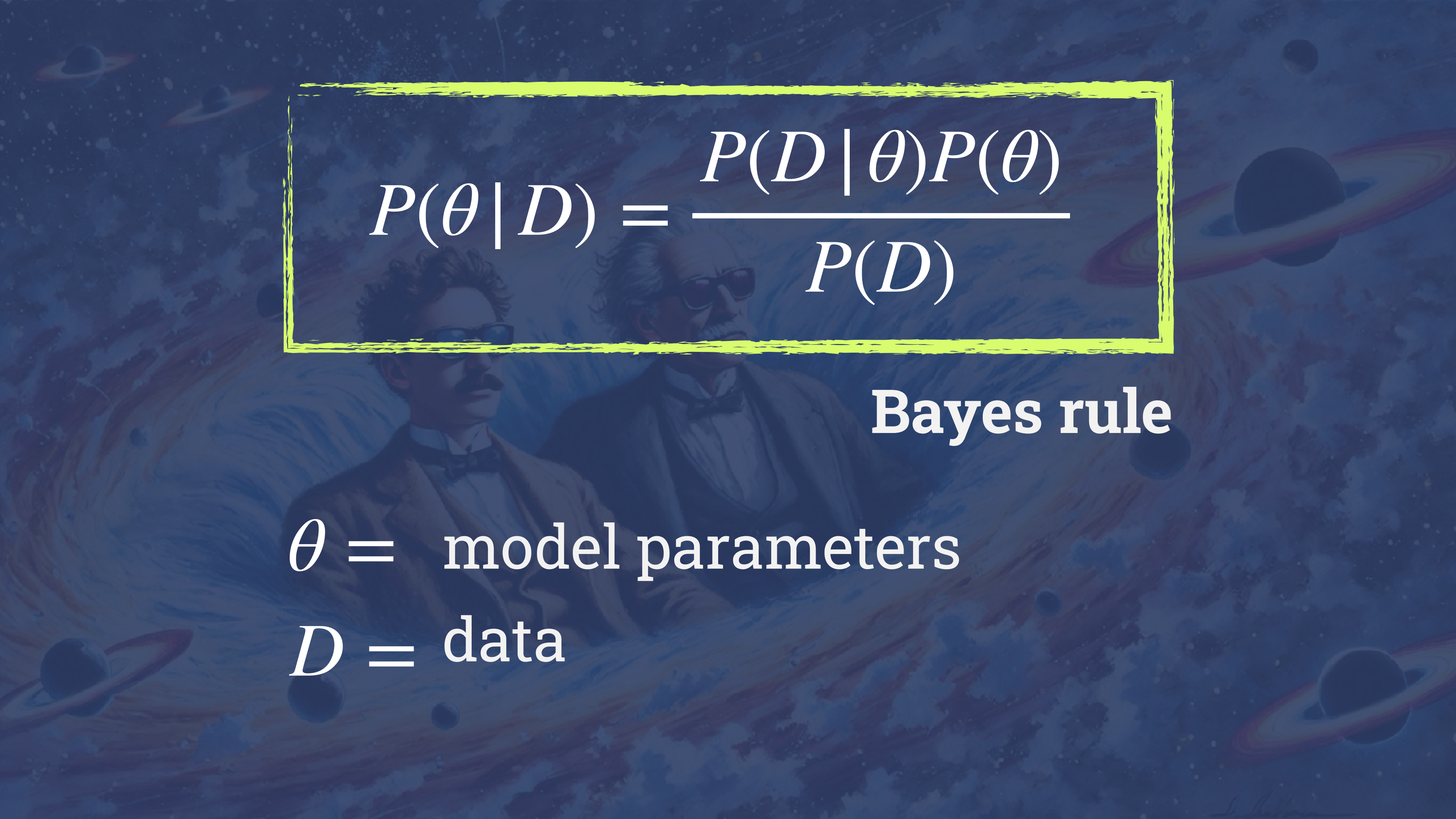


A cosmic background featuring a deep blue space filled with swirling galaxies, distant stars, and several ringed planets, including Saturn, in various orientations. The overall tone is dark and mysterious.

$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

Bayes rule

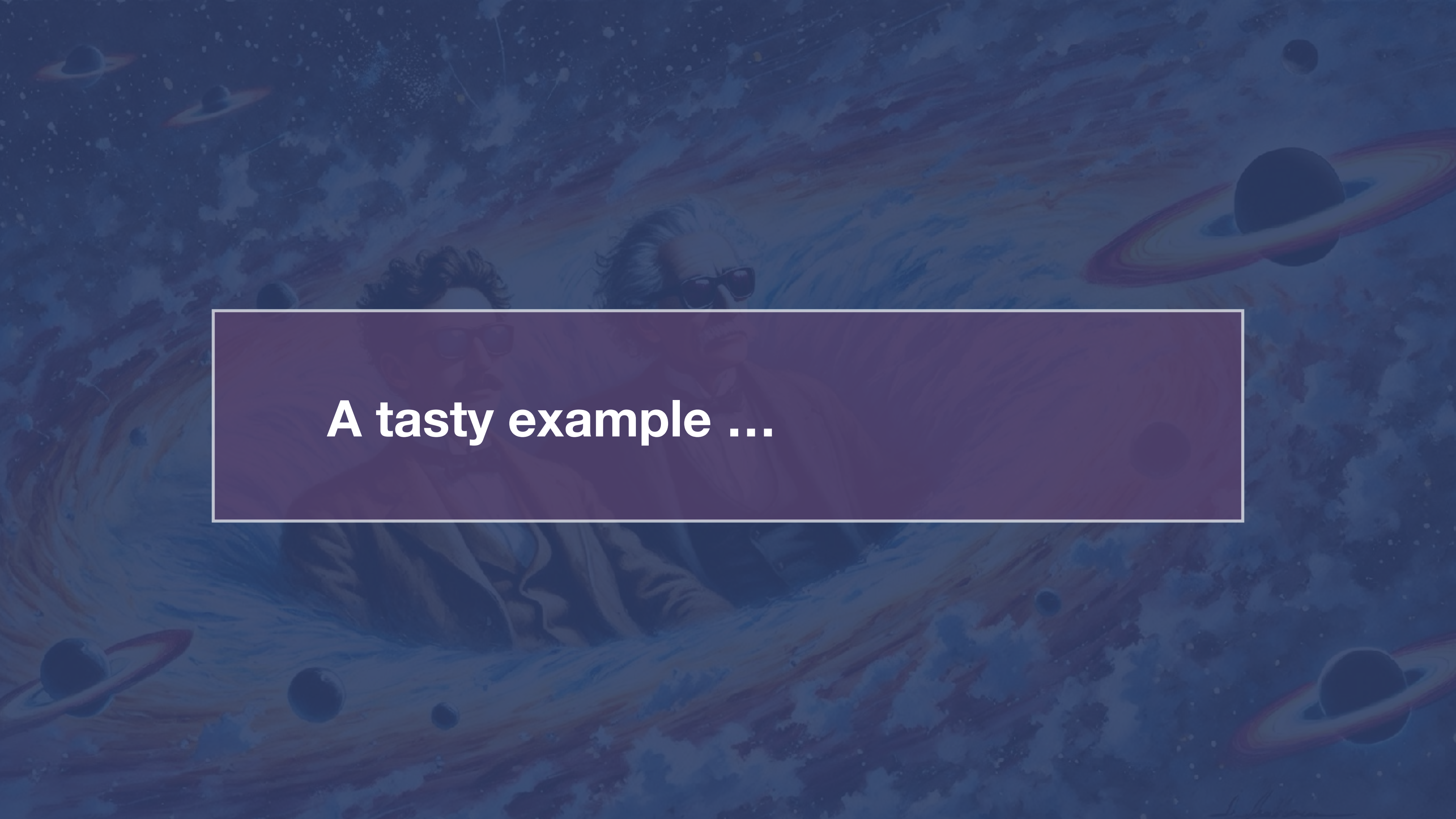
θ = model parameters


$$P(\theta | D) = \frac{P(D | \theta)P(\theta)}{P(D)}$$

Bayes rule

θ = model parameters

D = data

The background of the slide is a deep blue and purple cosmic scene. It features several galaxies with prominent spiral arms, some appearing as bright, glowing structures. Scattered throughout the space are various celestial bodies, including planets with rings (resembling Saturn) and smaller, dark spheres. The overall effect is a sense of vastness and wonder in the universe.

A tasty example ...









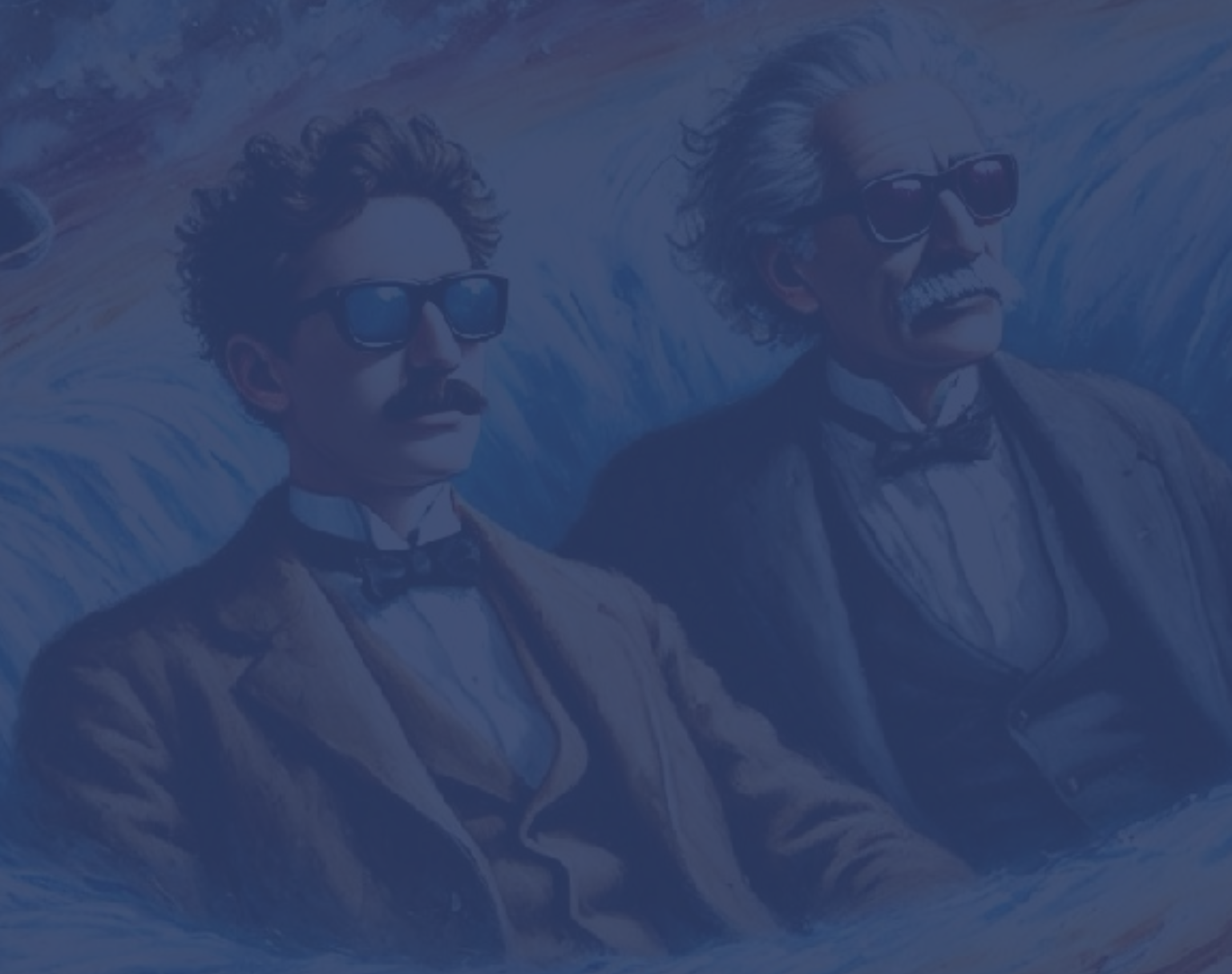
posterior

likelihood

prior

$$P(\theta | D) \propto P(D | \theta) P(\theta)$$

Find the posterior distribution for the percentage of **blue** m&ms made at the factory

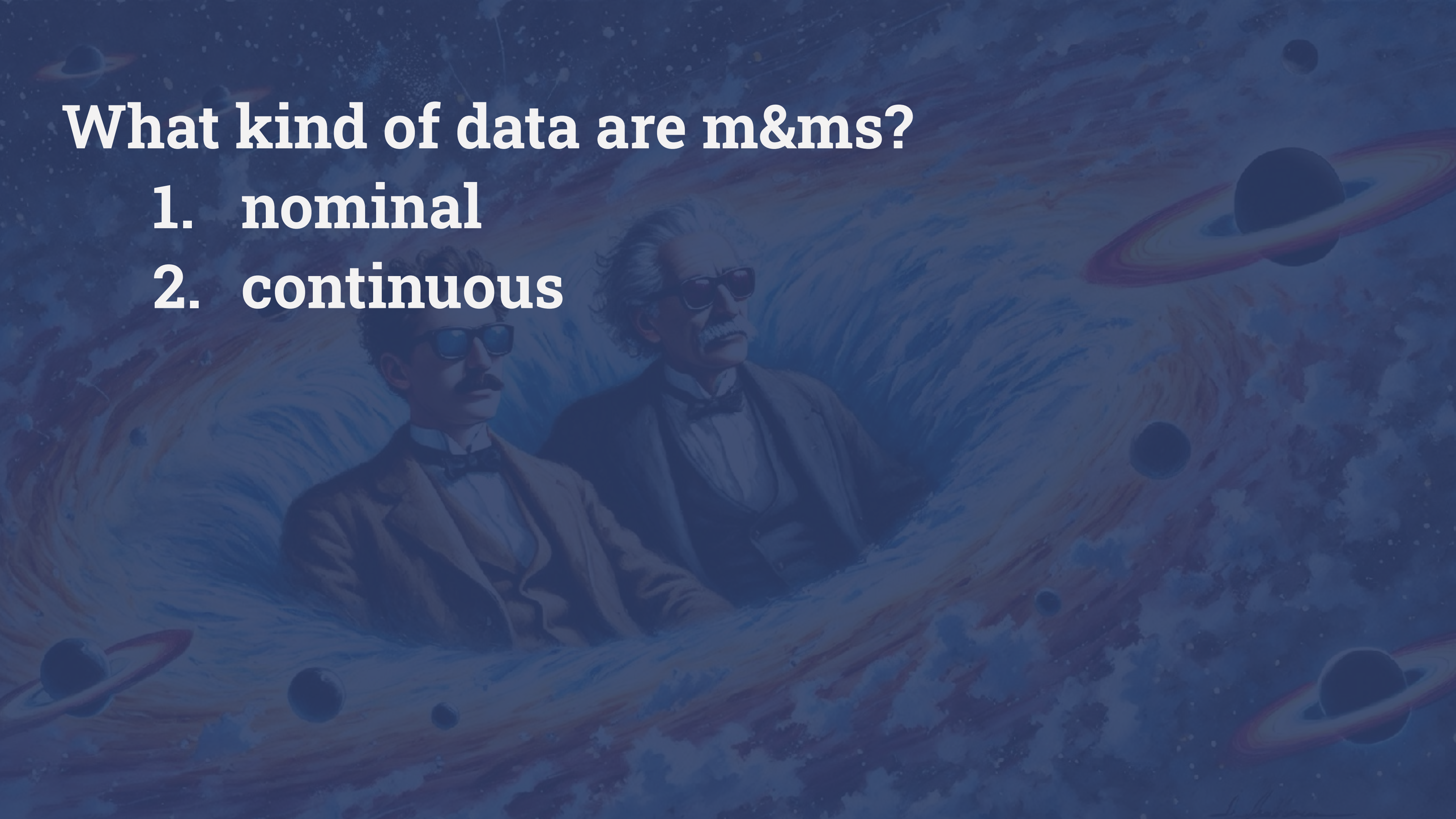


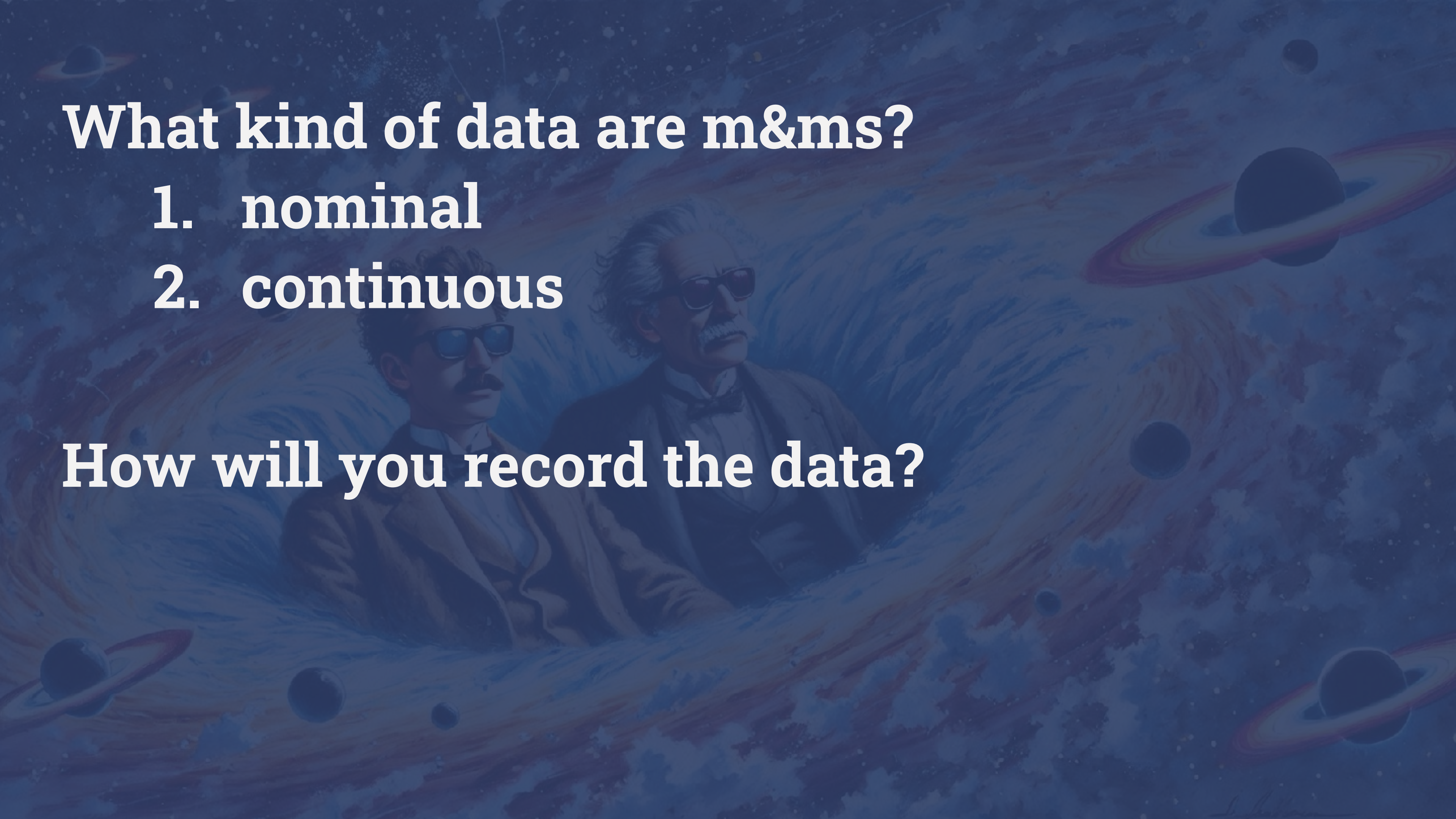
Question: Why do **you** think there are sometimes more or fewer blue M&Ms in the package?



What kind of data are m&ms?

1. nominal
2. continuous

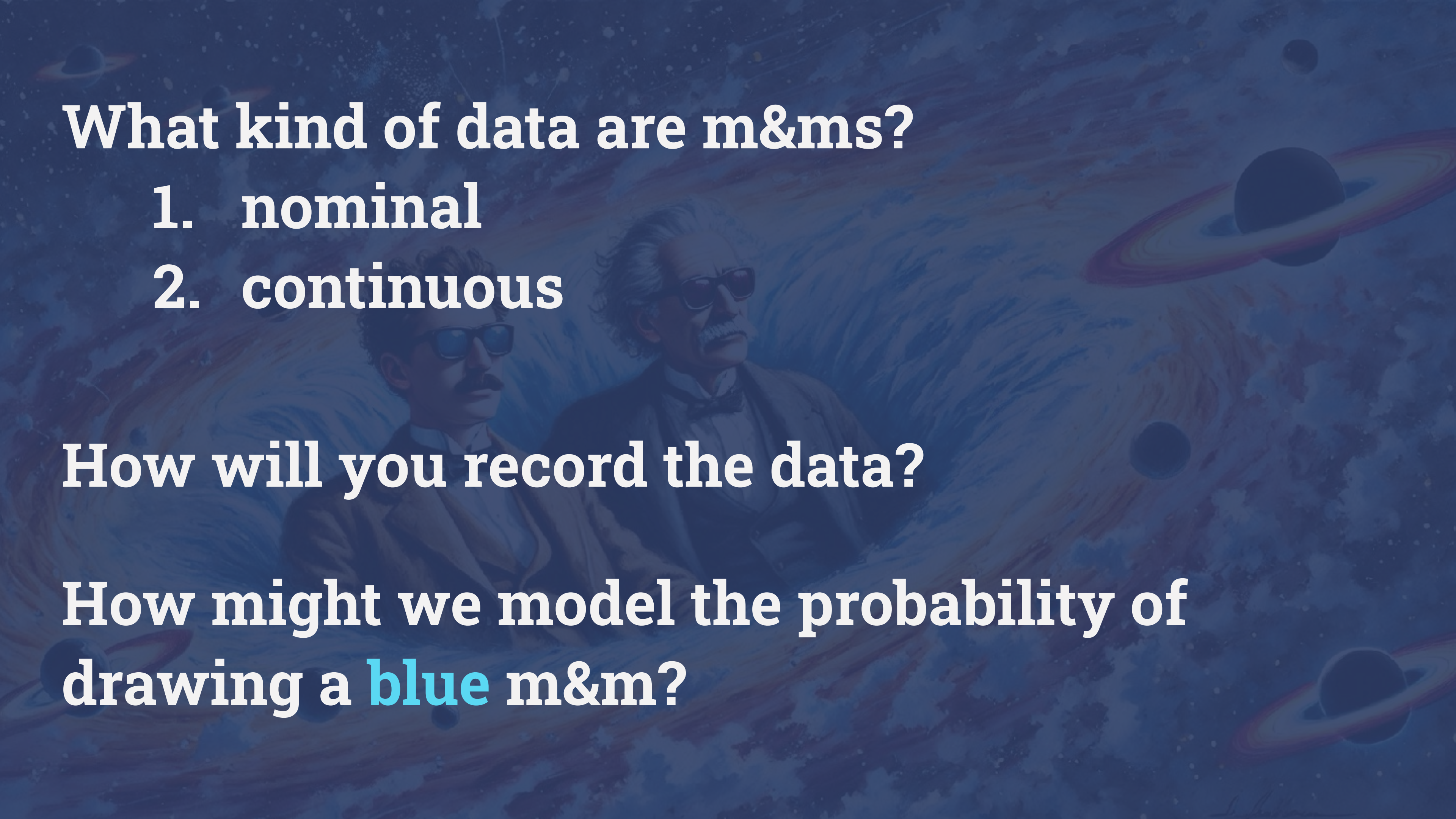


The background of the slide is a deep blue space scene filled with stars, nebulae, and several ringed planets. In the center, there is a faint, artistic illustration of two men, Albert Einstein and Stephen Hawking, wearing dark suits and bow ties. They are depicted from the chest up, looking towards the right. The overall aesthetic is scientific and intellectual.

What kind of data are m&ms?

1. nominal
2. continuous

How will you record the data?

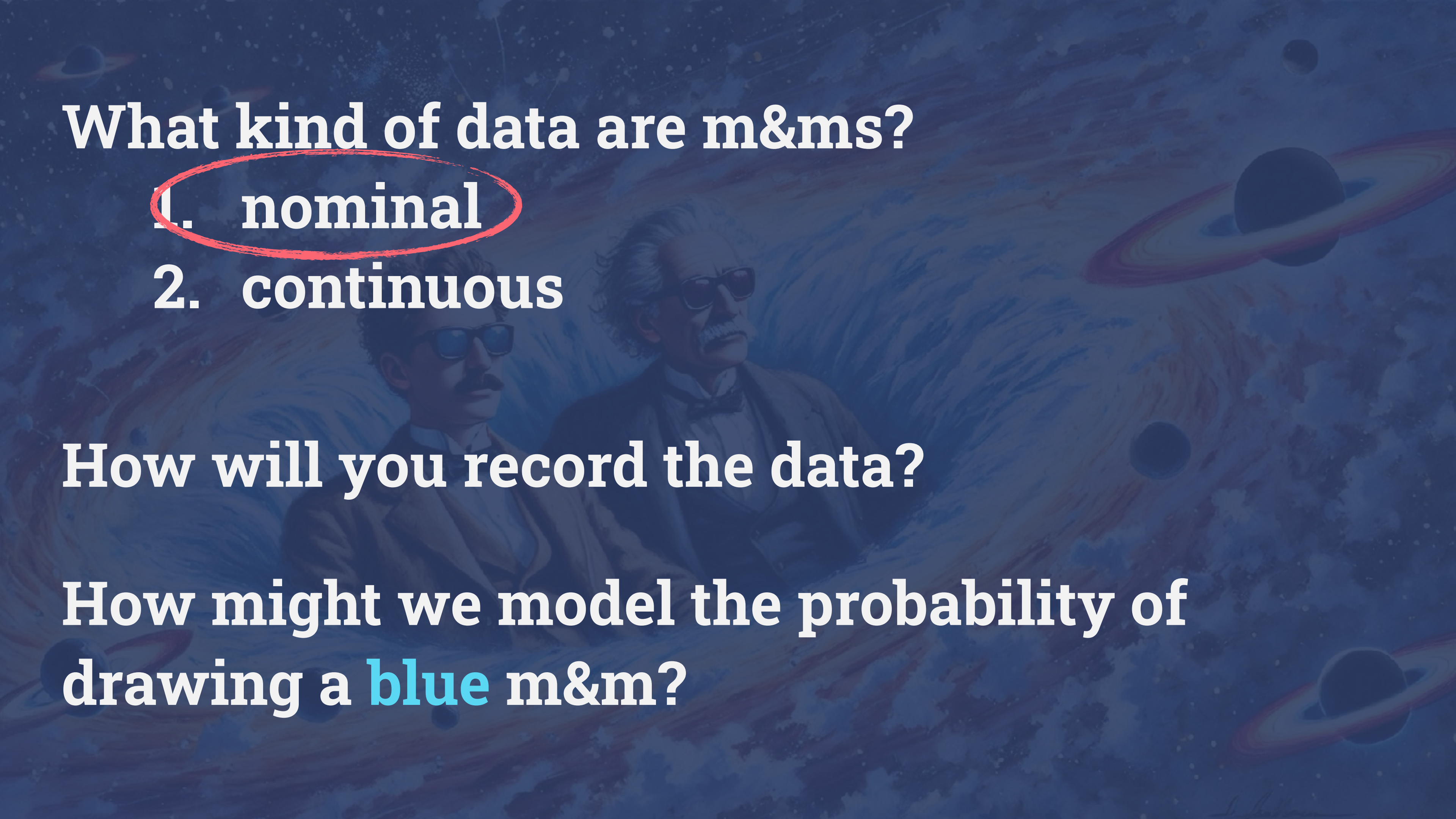
The background of the slide is a dark blue, textured space scene. It features faint, colorful nebulae and several planets, including one with prominent rings. In the center, there is a faint, semi-transparent illustration of two men. On the right is Albert Einstein, with his characteristic wild hair and mustache, wearing a suit and bow tie. On the left is a younger man with a mustache and glasses, also in a suit. They appear to be looking towards the right.

What kind of data are m&ms?

1. nominal
2. continuous

How will you record the data?

How might we model the probability of drawing a blue m&m?

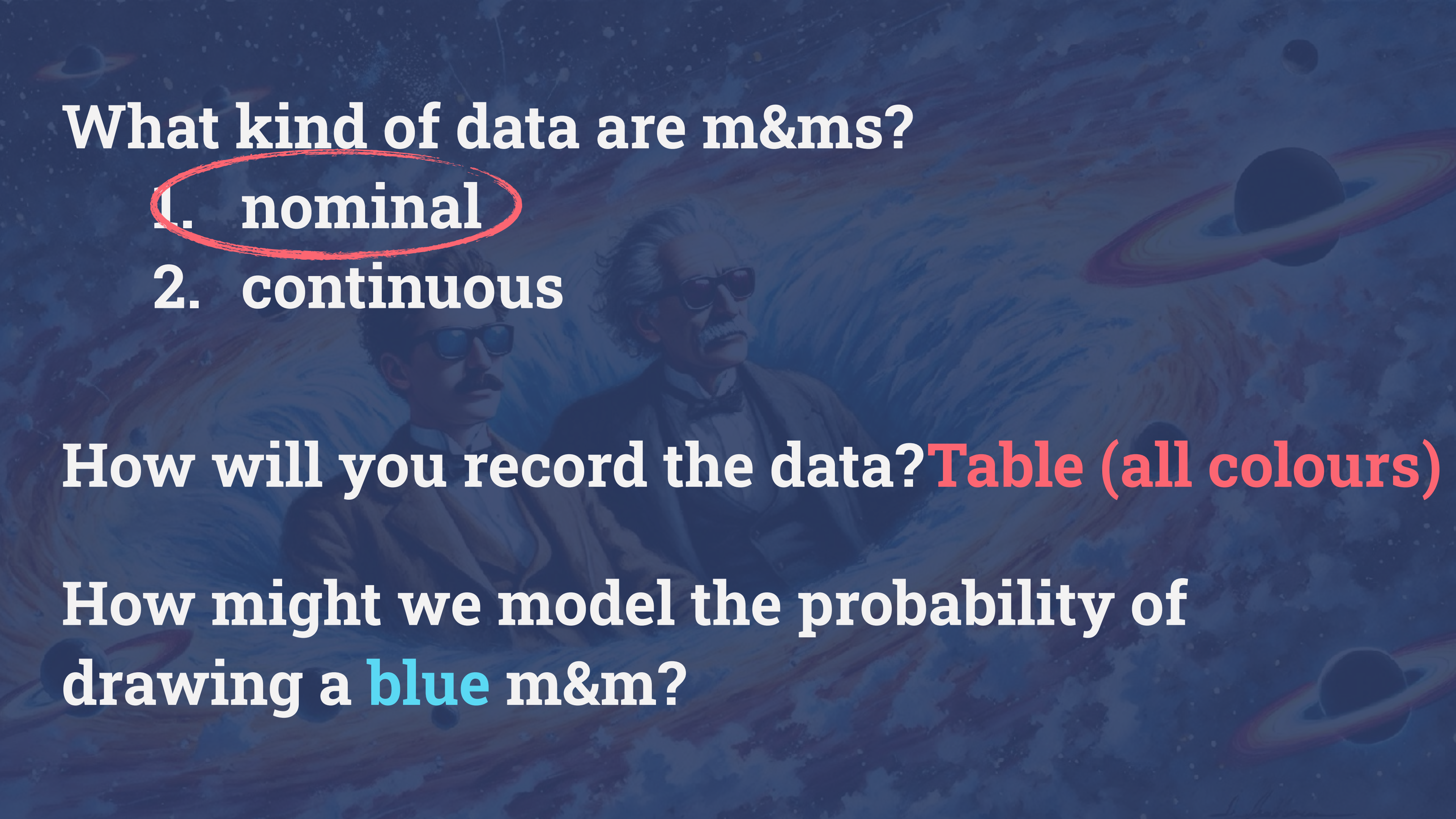


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How might we model the probability of drawing a blue m&m?



What kind of data are m&ms?

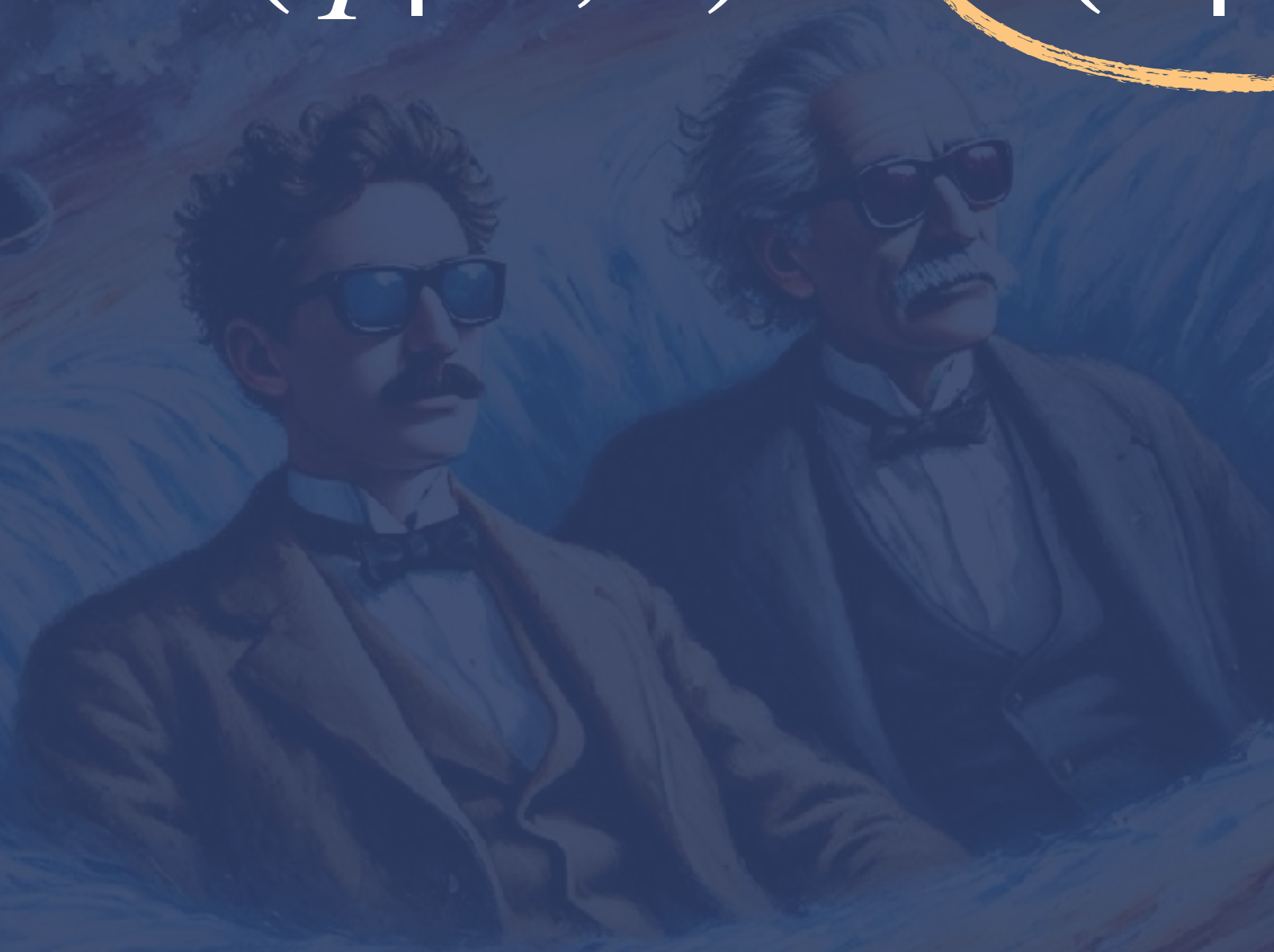
1. nominal
2. continuous

How will you record the data? **Table (all colours)**

How might we model the probability of drawing a **blue** m&m?

likelihood

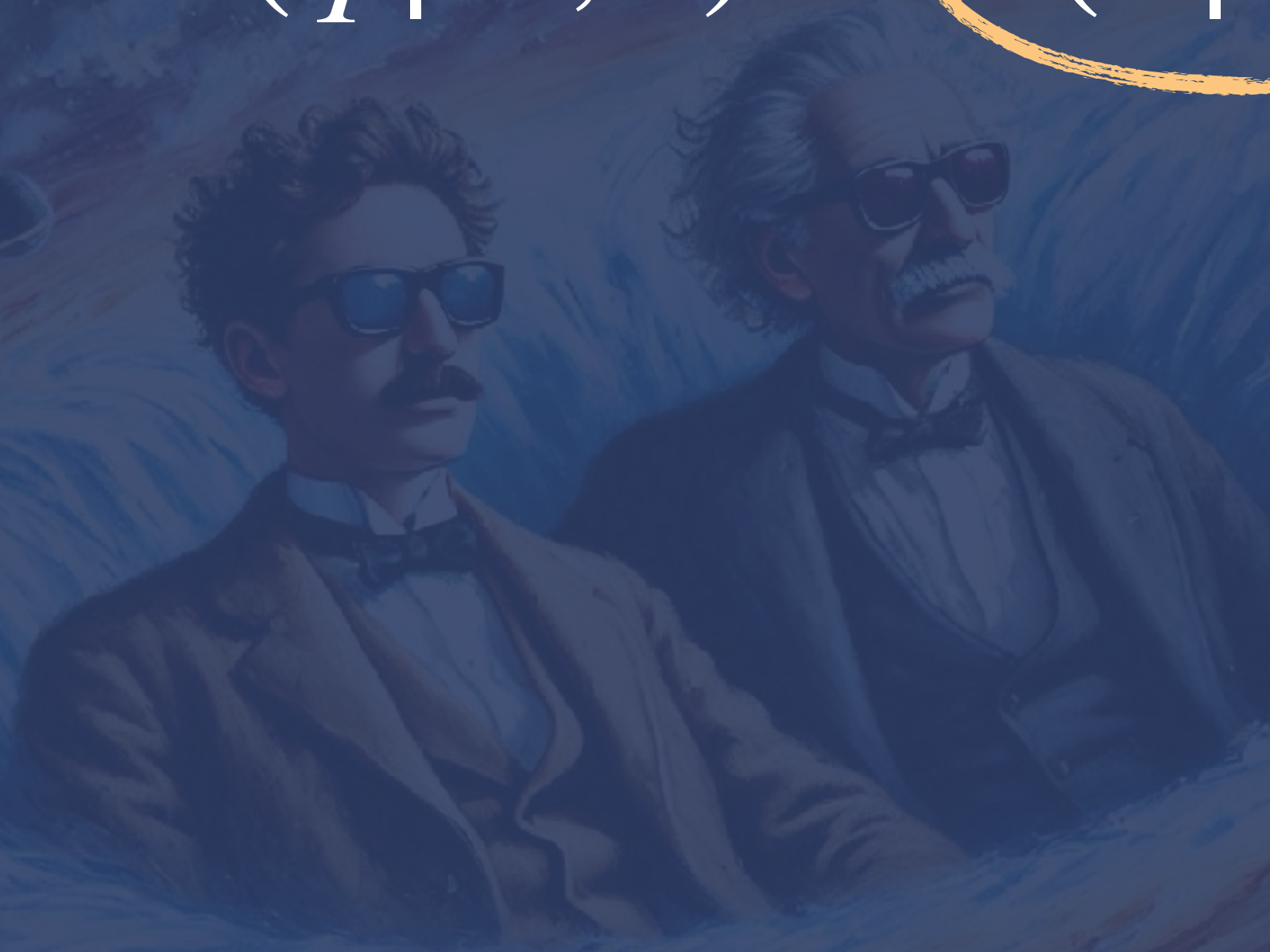
$$P(q | N, k) \propto P(k | N, q) P(q)$$



likelihood

$$P(q | N, k) \propto P(k | N, q) P(q)$$

$k =$



likelihood

$$P(q | N, k) \propto P(k | N, q)P(q)$$

k = number of successes (blue m&ms)

likelihood

$$P(q | N, k) \propto P(k | N, q) P(q)$$

k = number of successes (blue m&ms)

$N - k$ =

likelihood

$$P(q | N, k) \propto P(k | N, q)P(q)$$

k = number of successes (blue m&ms)

$N - k$ = number of failures (not-blue m&ms)

likelihood

$$P(q | N, k) \propto P(k | N, q)P(q)$$

k = number of successes (blue m&ms)

$N - k$ = number of failures (not-blue m&ms)

q =

likelihood

$$P(q | N, k) \propto P(k | N, q)P(q)$$

k = number of successes (blue m&ms)

$N - k$ = number of failures (not-blue m&ms)

q = probability of drawing a blue m&m

likelihood

$$P(q | N, k) \propto P(k | N, q)P(q)$$

k = number of successes (blue m&ms)

$N - k$ = number of failures (not-blue m&ms)

q = probability of drawing a blue m&m

$$P(\theta | D) \propto P(D | \theta)P(\theta) \quad \text{with} \quad \begin{array}{l} D = k \\ \theta = q \end{array}$$

The Binomial Distribution

Observed fraction of
blue M&Ms in a
single bag

True fraction of blue
M&Ms produced at
the factory



$$p(D \mid \theta) = p(k \mid N, q) \sim \text{Binom}(N, k, q)$$

“The probability of observing k blue M&Ms out of N
given that the true fraction of M&Ms is q ”

The Binomial Distribution

Observed fraction of
blue M&Ms in a
single bag

True fraction of blue
M&Ms produced at
the factory



$$p(D \mid \theta) = p(k \mid N, q) \sim \text{Binom}(N, k, q)$$
$$= \binom{N}{k} q^k (1 - q)^{(N-k)}$$

“The probability of observing **k** blue M&Ms out of **N**
given that the true fraction of M&Ms is **q**”

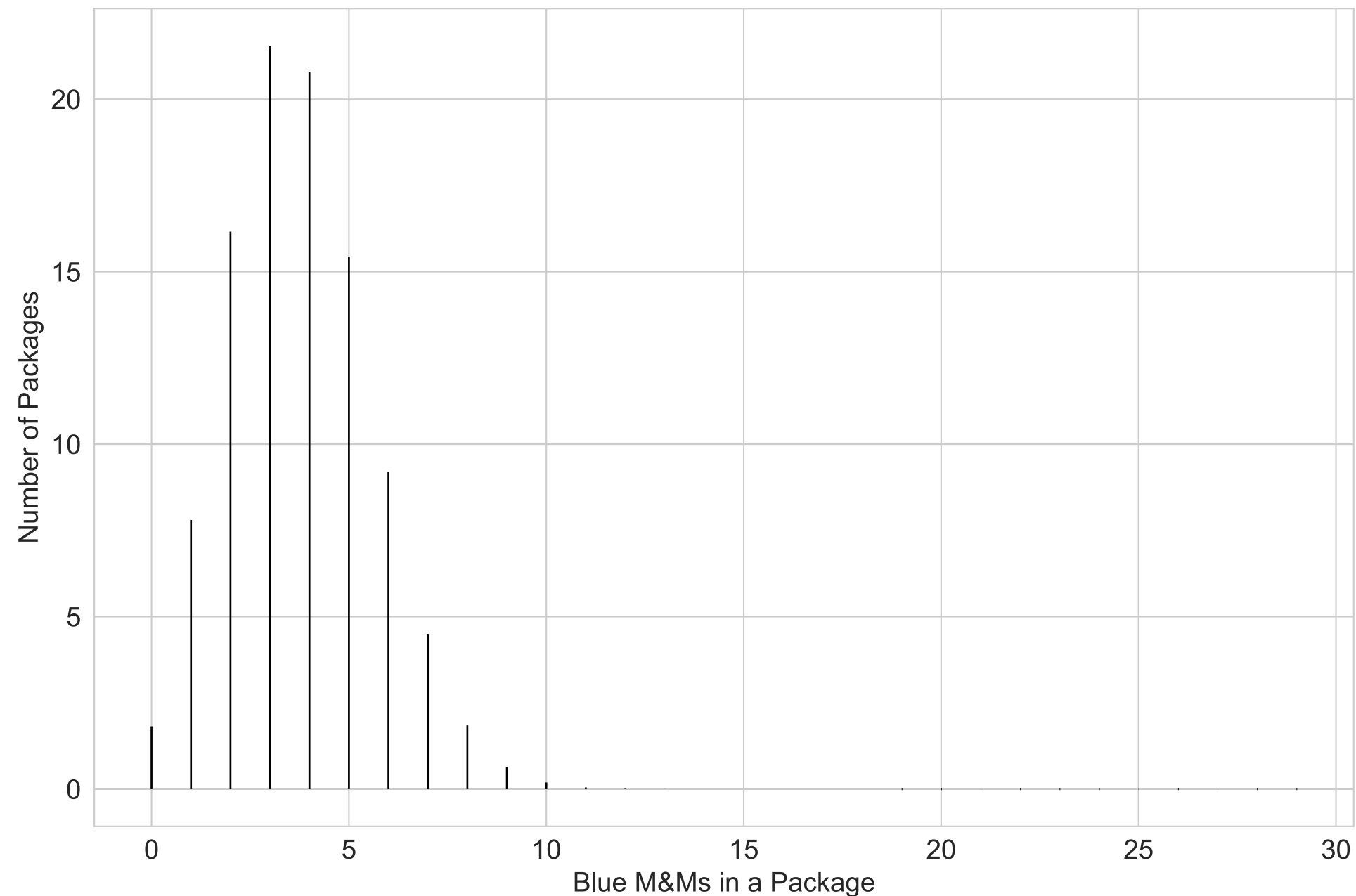


What do you think the fraction of blue M&Ms produced at the factor should be?



First Assumption: The M&Ms are mixed together at the factory and then distributed into packages.

Second Assumption: All colours are equally distributed

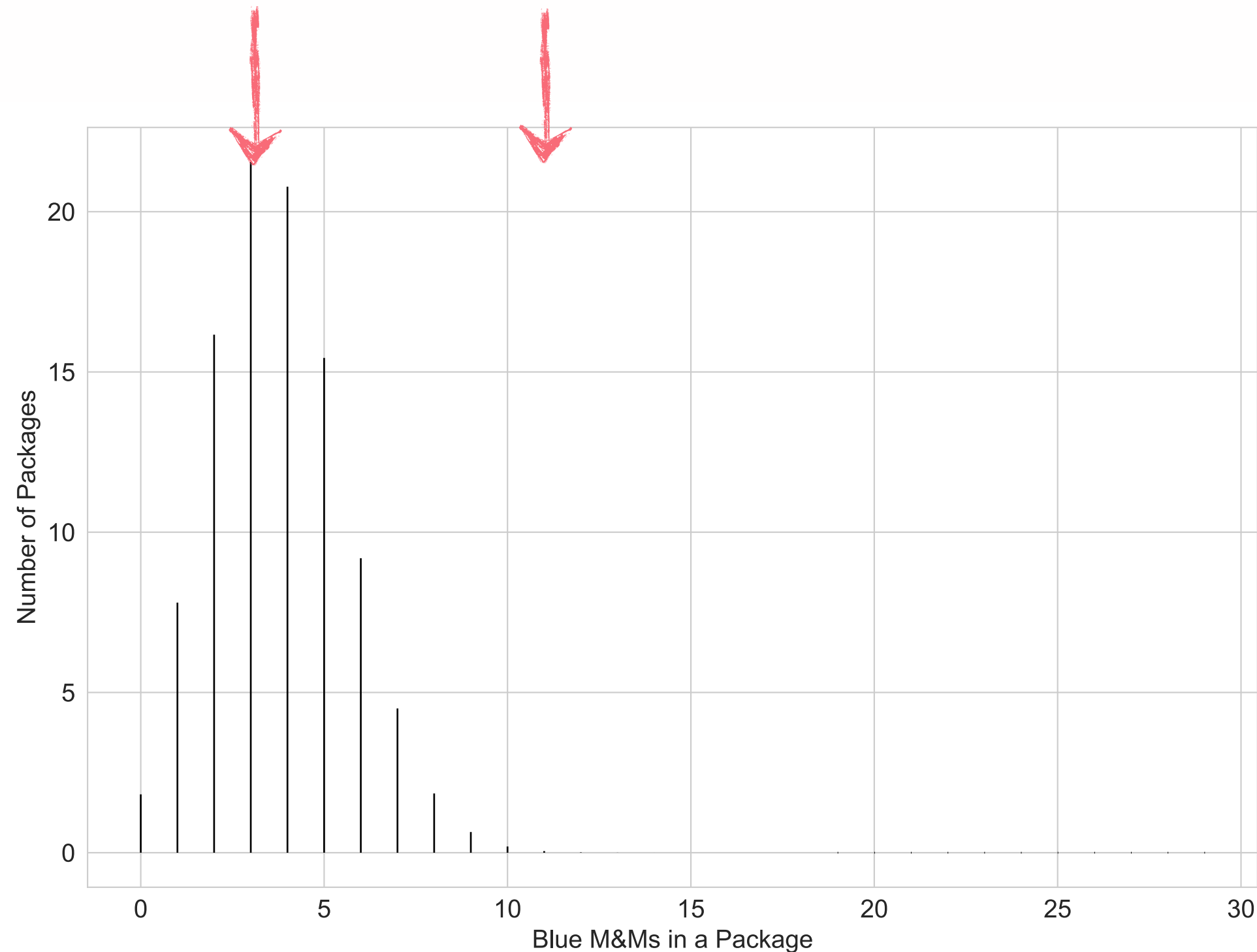


Binomial distribution

Blue Fraction: 0.125

First Assumption: The M&Ms are mixed together at the factory and then distributed into packages.

Second Assumption: All colours are equally distributed

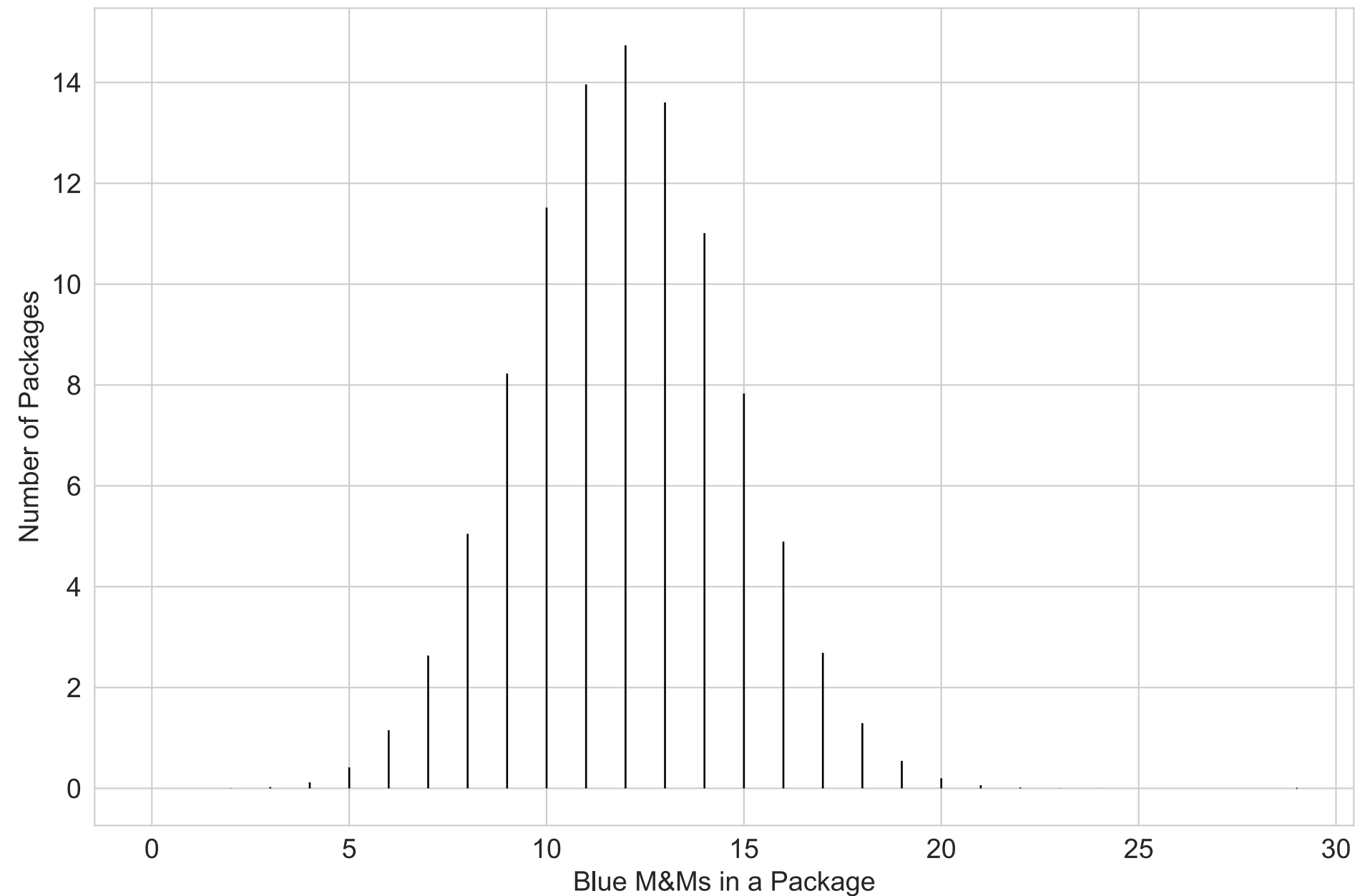


Binomial distribution

Blue Fraction: 0.125

First Assumption: The M&Ms are mixed together at the factory and then distributed into packages.

~~Second Assumption: All colours are equally distributed~~

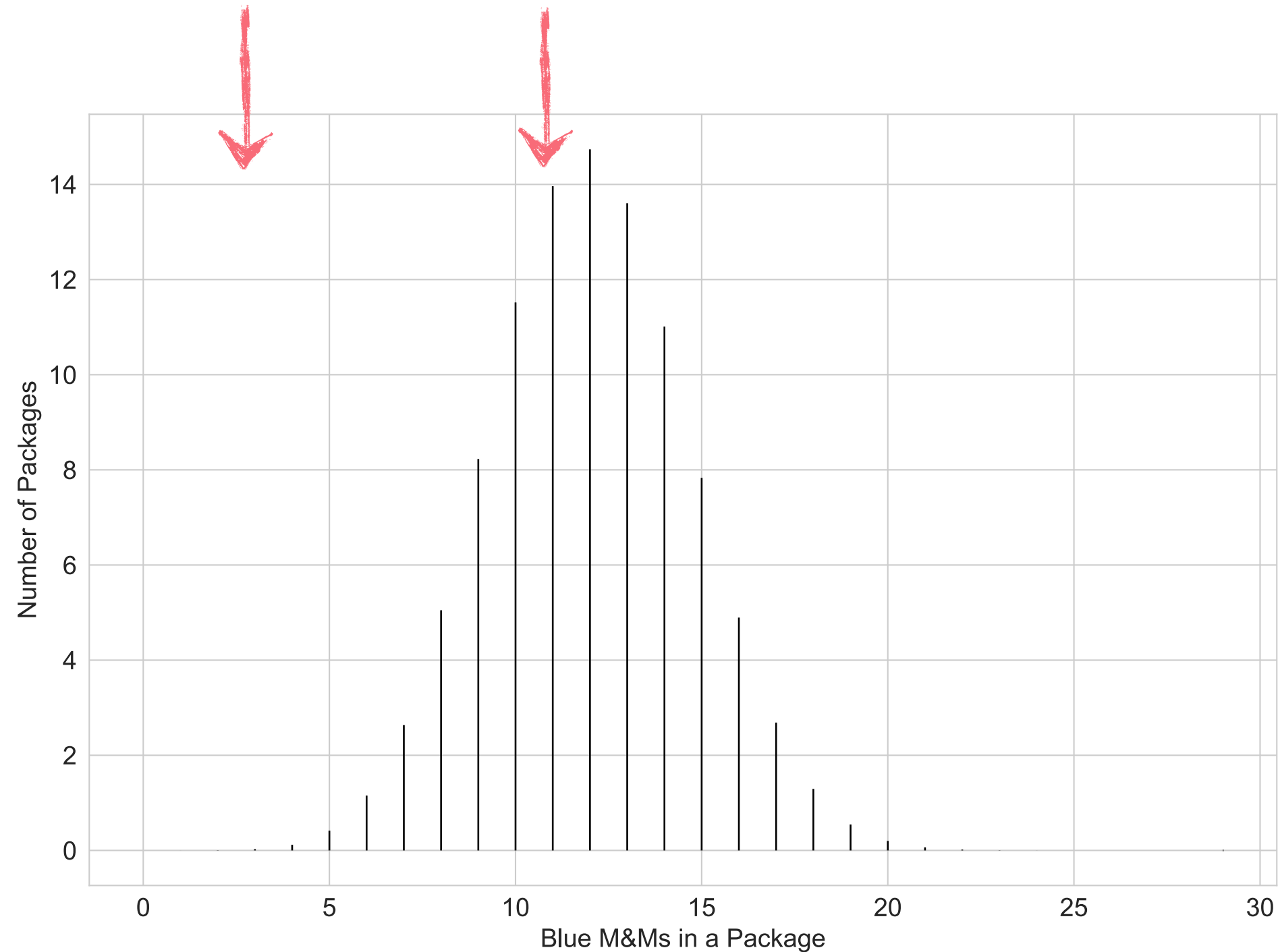


Binomial distribution

Blue Fraction: 0.4

First Assumption: The M&Ms are mixed together at the factory and then distributed into packages.

~~Second Assumption: All colours are equally distributed~~



Binomial distribution

Blue Fraction: 0.4

Question: Can we use this somehow
to estimate the fraction of blue
M&Ms?

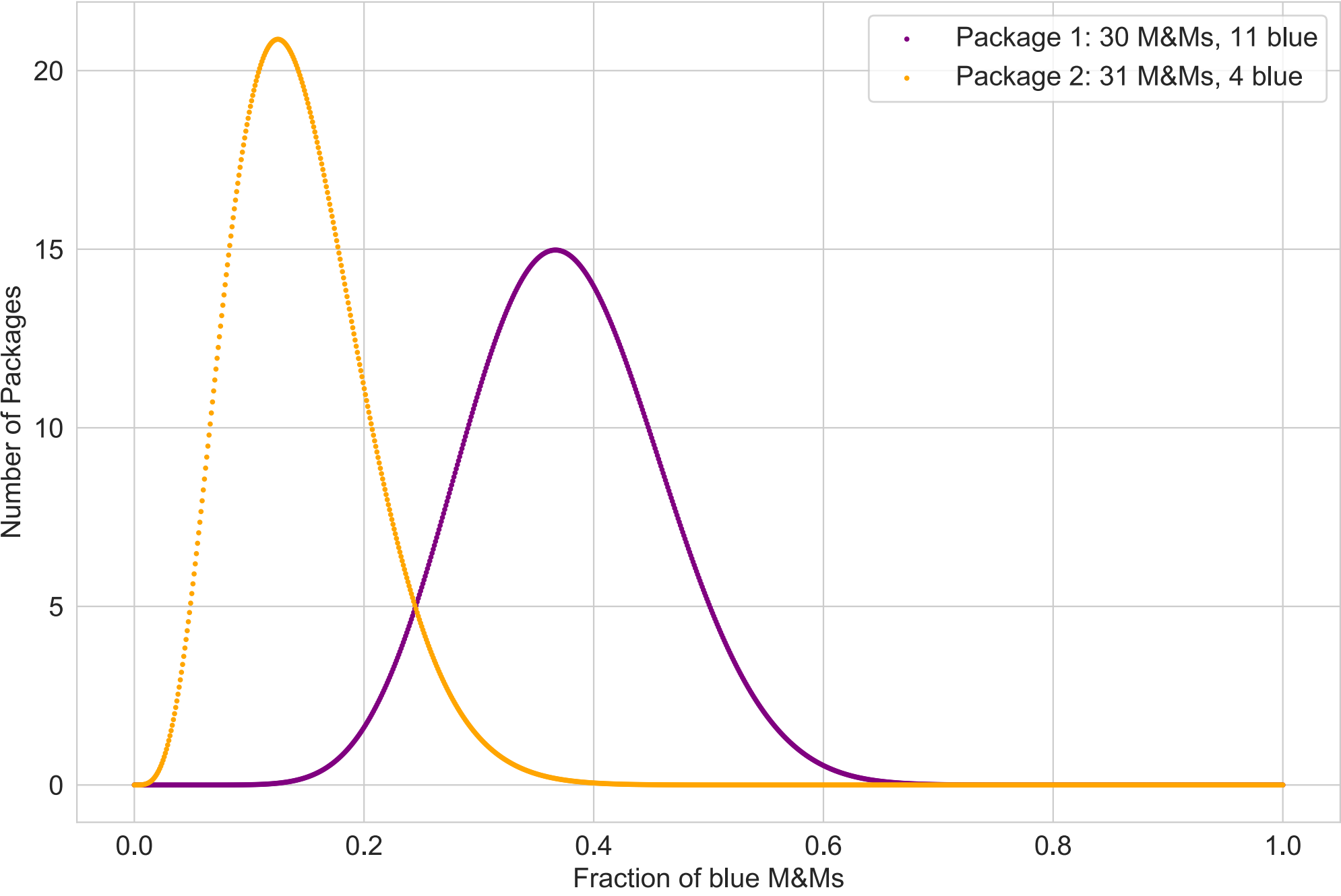


https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

Exercises 1, 2, 3

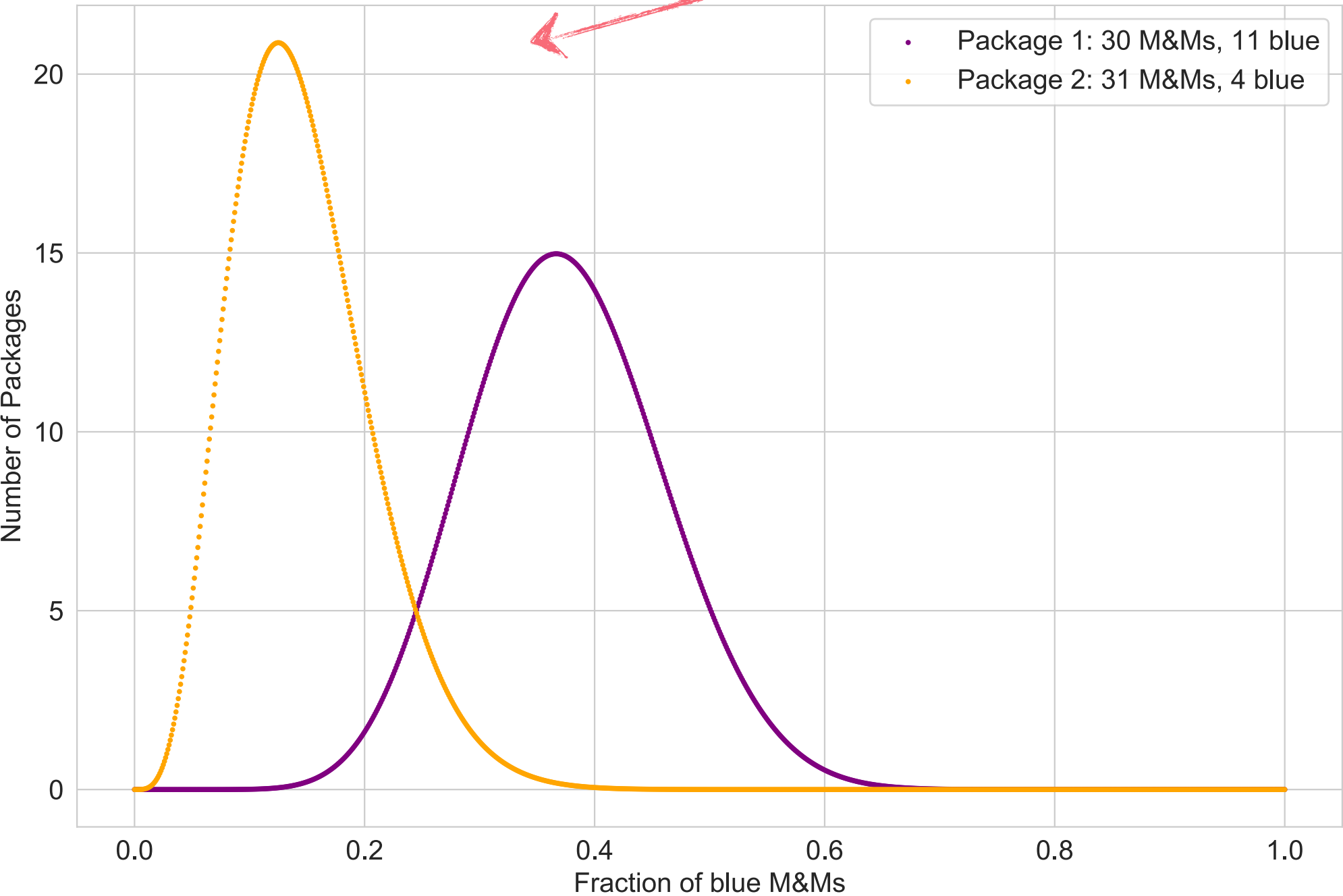


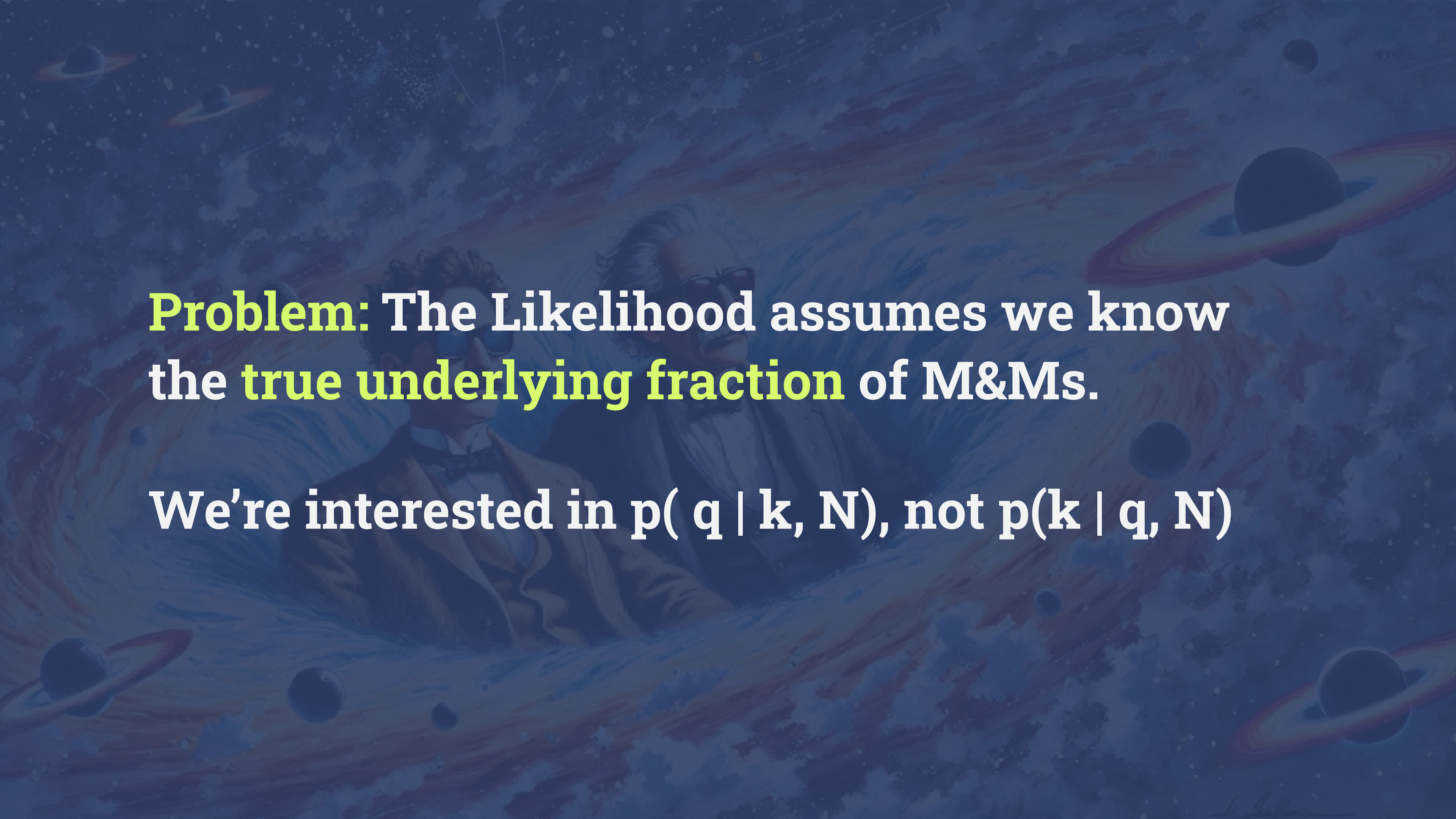
The Likelihood Function



The Likelihood Function

use optimization routines to find the maximum



The background is a dark blue space-themed illustration. It features Albert Einstein and a man in a tuxedo and bow tie floating in the center. Surrounding them are various celestial bodies, including planets with rings (like Saturn) and smaller planets, all set against a backdrop of swirling nebulae and star trails.

Problem: The Likelihood assumes we know the **true underlying fraction** of M&Ms.

We're interested in $p(q | k, N)$, not $p(k | q, N)$

Conditional Probability Magic

$$\begin{array}{c} \text{posterior} \\ p(q \mid k, N) = \end{array} \frac{\begin{array}{cc} \text{likelihood} & \text{prior} \\ p(k \mid q, N) & p(q) \end{array}}{\begin{array}{c} p(k \mid N) \\ \text{Evidence/marginal} \\ \text{likelihood} \end{array}}$$

Bayes theorem



Conditional Probability Magic

$$\begin{array}{c} \text{posterior} \\ p(q \mid k, N) = \frac{\text{likelihood} \quad \text{prior} \\ p(k \mid q, N) \, p(q)}{p(k \mid N)} \\ \text{Evidence/marginal} \\ \text{likelihood} \end{array}$$

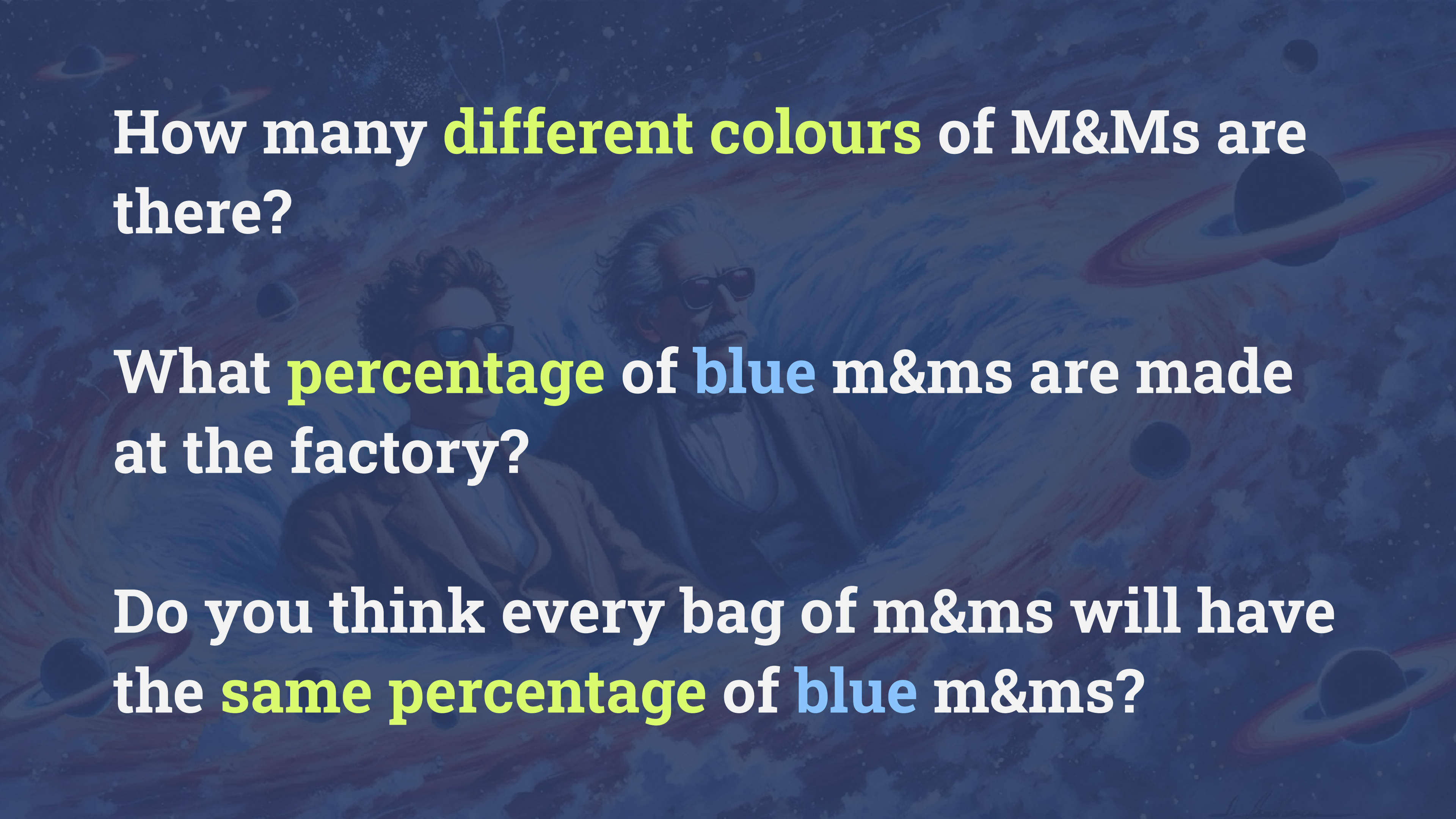
$$p(q \mid k, N) \propto p(k \mid q, N) \, p(q)$$

Bayes theorem



Question

What prior information do we have about $p(f)$?

The background is a dark blue space-themed illustration. It features faint, stylized depictions of galaxies, nebulae, and planets with rings. In the center, there are two figures: Albert Einstein on the right, wearing his characteristic wild hair and mustache, and Stephen Hawking on the left, wearing his signature dark-rimmed glasses. They appear to be floating or traveling through space.

How many **different colours** of M&Ms are there?

What **percentage** of **blue** m&ms are made at the factory?

Do you think every bag of m&ms will have the **same percentage** of **blue** m&ms?

Sketch your prior



percentage of blue m&ms

prior

$$p(q | k) \propto p(k | q)p(q)$$



prior

$$p(q | k) \propto p(k | q) p(q)$$

$$p(q) \propto q^{\alpha-1} (1-q)^{\beta-1}$$

prior

$$p(q | k) \propto p(k | q) p(q)$$

beta distribution

$$p(q) \propto q^{\alpha-1} (1-q)^{\beta-1}$$

prior

$$p(q | k) \propto p(k | q) p(q)$$

beta distribution

$$p(q) \propto q^{\alpha-1} (1-q)^{\beta-1}$$

“conjugate
prior”

prior

$$p(q | k) \propto p(k | q) p(q)$$

beta distribution

$$p(q) \propto q^{\alpha-1} (1-q)^{\beta-1}$$

“conjugate
prior”

$$\alpha = ?$$

prior

$$p(q | k) \propto p(k | q) p(q)$$

beta distribution

$$p(q) \propto q^{\alpha-1} (1-q)^{\beta-1}$$

“conjugate
prior”

$$\alpha = ? \quad \beta = ?$$

Write a function for the beta distribution and try out different values for alpha and beta

Some values to try:

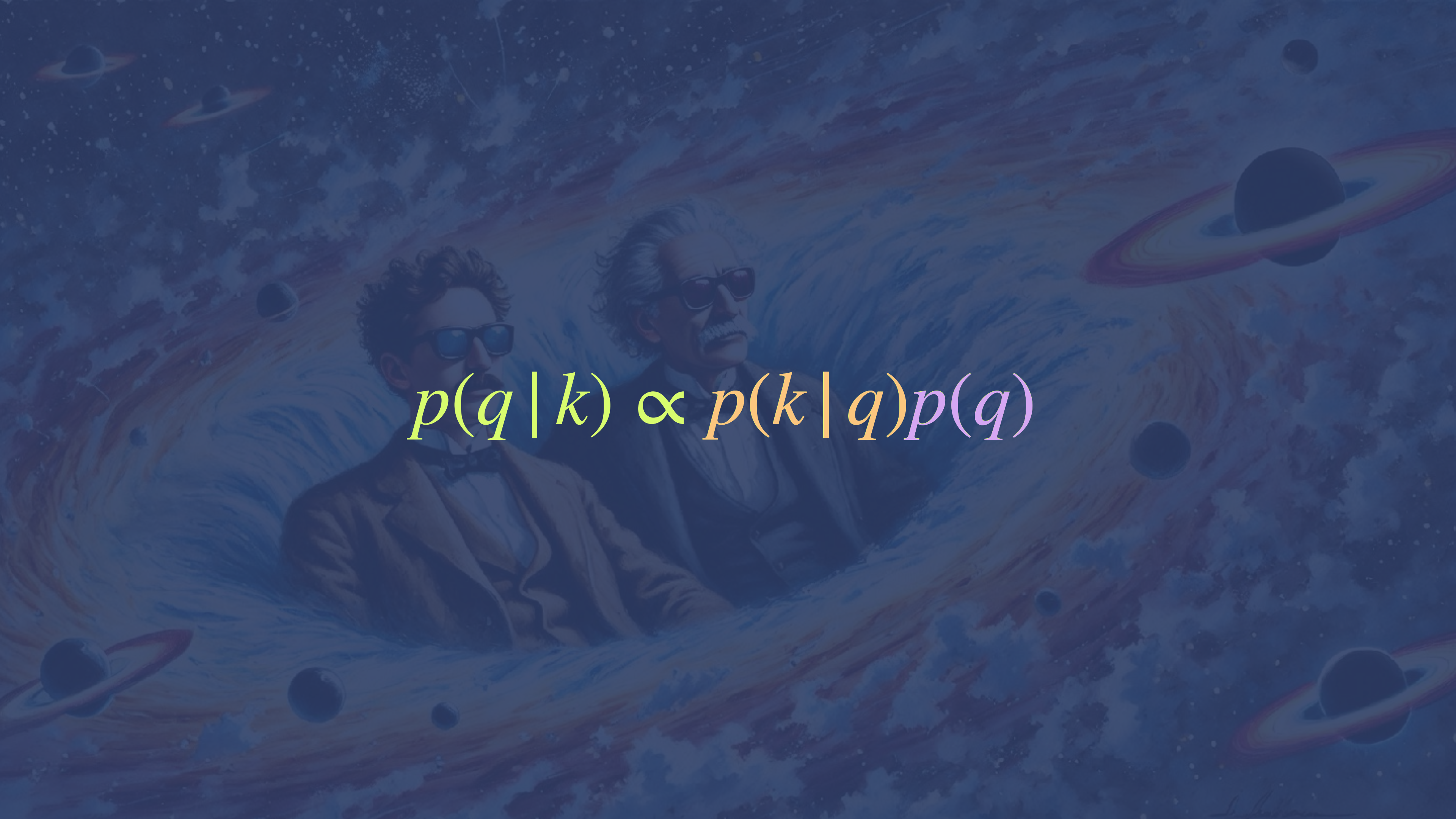
$$\alpha = \beta$$

$$\alpha = \beta = 1$$

α, β very large

α, β very small

https://github.com/philuttle/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb Exercise 5 + 6


$$p(q \mid k) \propto p(k \mid q)p(q)$$

likelihood

$$p(k | q, N) \propto q^k (1 - q)^{N-k}$$

$$p(q | k) \propto p(k | q)p(q)$$

likelihood

prior

$$p(k | q, N) \propto q^k (1 - q)^{N-k}$$

$$p(q) \propto q^{\alpha-1} (1 - q)^{\beta-1}$$

$$p(q | k) \propto p(k | q) p(q)$$

likelihood

prior

$$p(k | q, N) \propto q^k (1 - q)^{N-k}$$

$$p(q) \propto q^{\alpha-1} (1 - q)^{\beta-1}$$

$$p(q | k) \propto p(k | q) p(q)$$

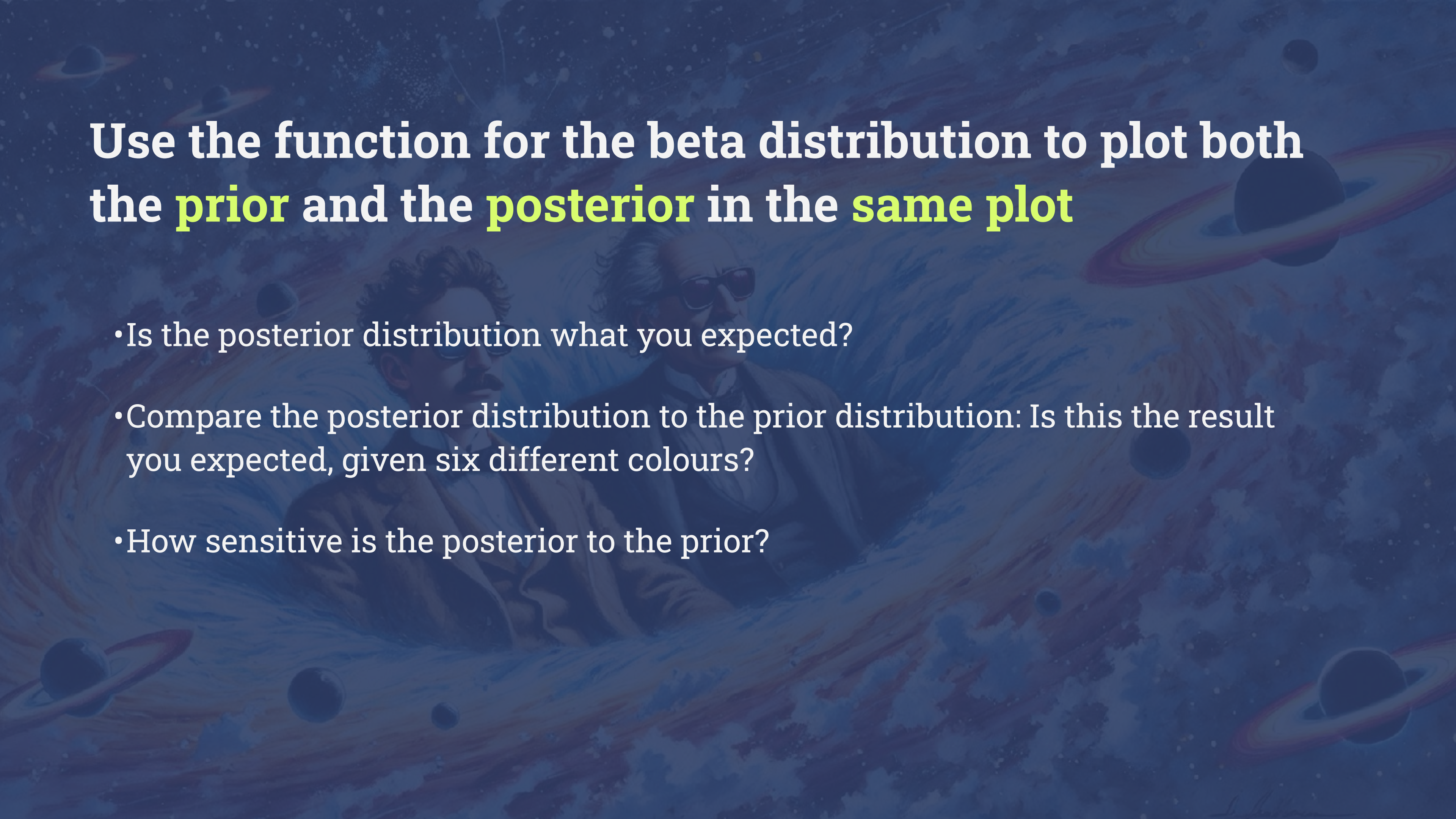
$$p(q | k, N) \propto q^{k+\alpha-1} (1 - q)^{N-k+\beta-1}$$

Gather some data!

record the counts for all colours and the total number of M&Ms:

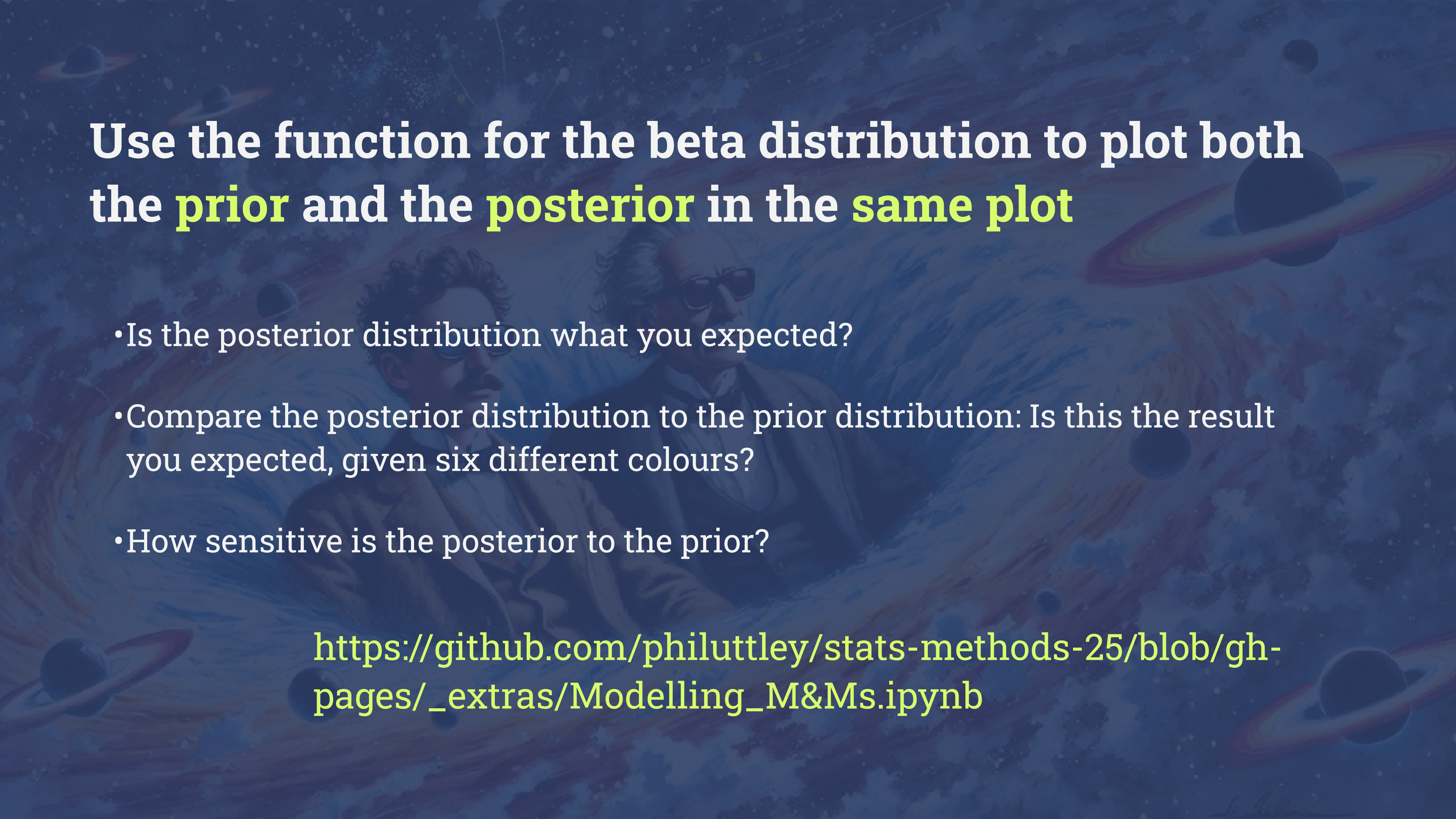
<https://forms.gle/Dvm1jpbUdTT6M1Vs8>





Use the function for the beta distribution to plot both the **prior** and the **posterior** in the **same plot**

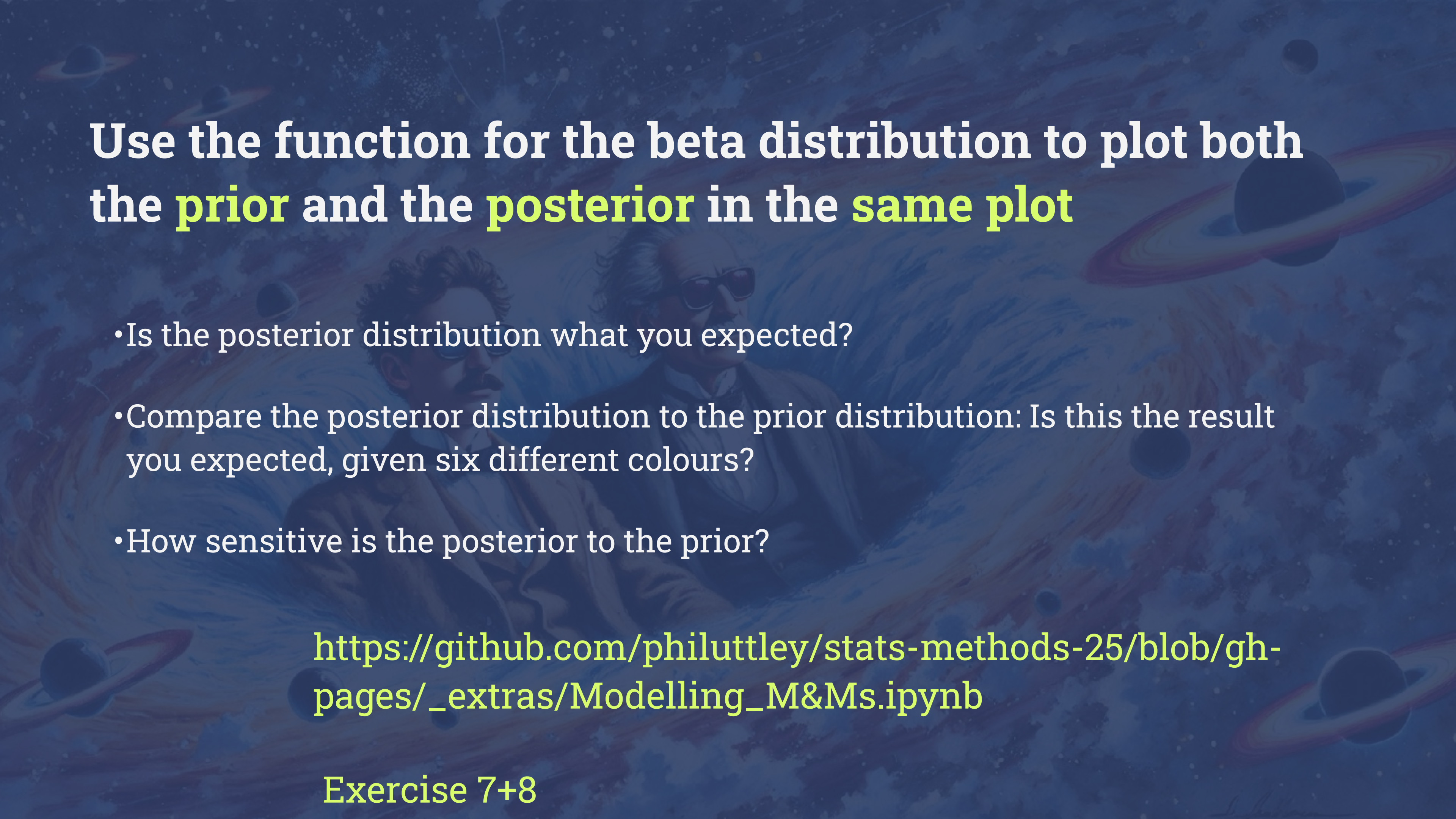
- Is the posterior distribution what you expected?
- Compare the posterior distribution to the prior distribution: Is this the result you expected, given six different colours?
- How sensitive is the posterior to the prior?



Use the function for the beta distribution to plot both the **prior** and the **posterior** in the **same plot**

- Is the posterior distribution what you expected?
- Compare the posterior distribution to the prior distribution: Is this the result you expected, given six different colours?
- How sensitive is the posterior to the prior?

https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

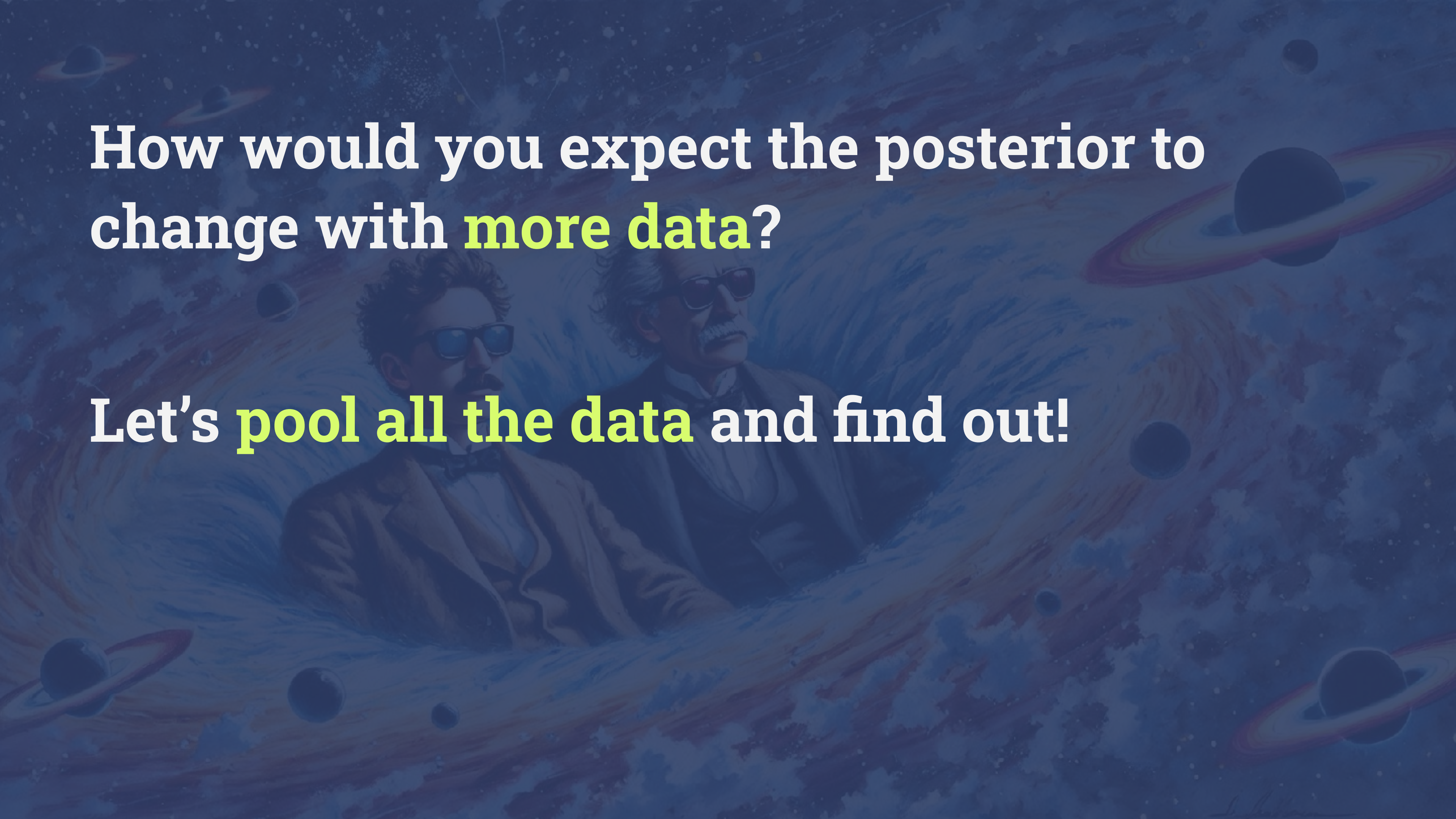


Use the function for the beta distribution to plot both the **prior** and the **posterior** in the **same plot**

- Is the posterior distribution what you expected?
- Compare the posterior distribution to the prior distribution: Is this the result you expected, given six different colours?
- How sensitive is the posterior to the prior?

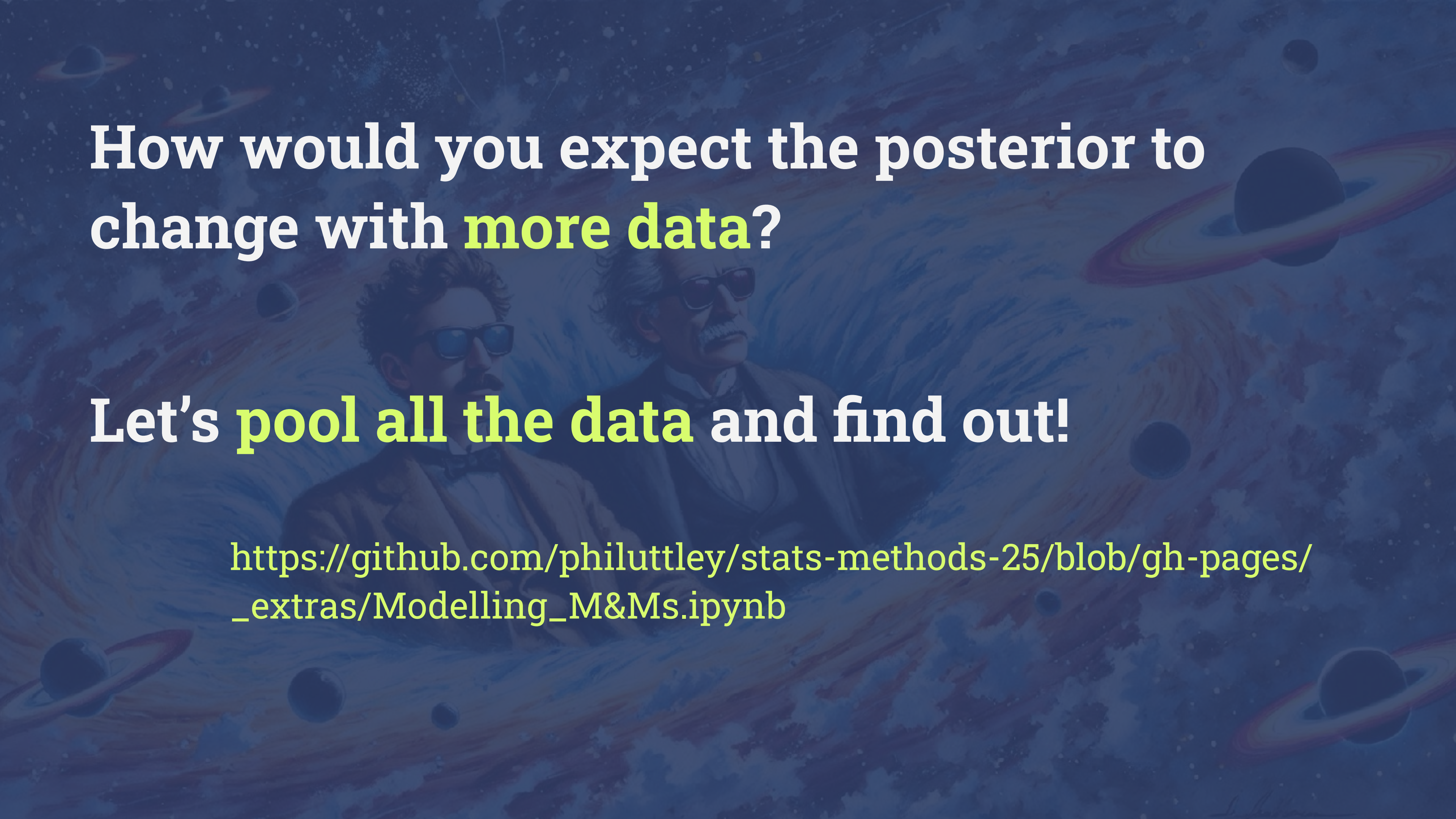
https://github.com/philuttley/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

Exercise 7+8



How would you expect the posterior to change with **more data**?

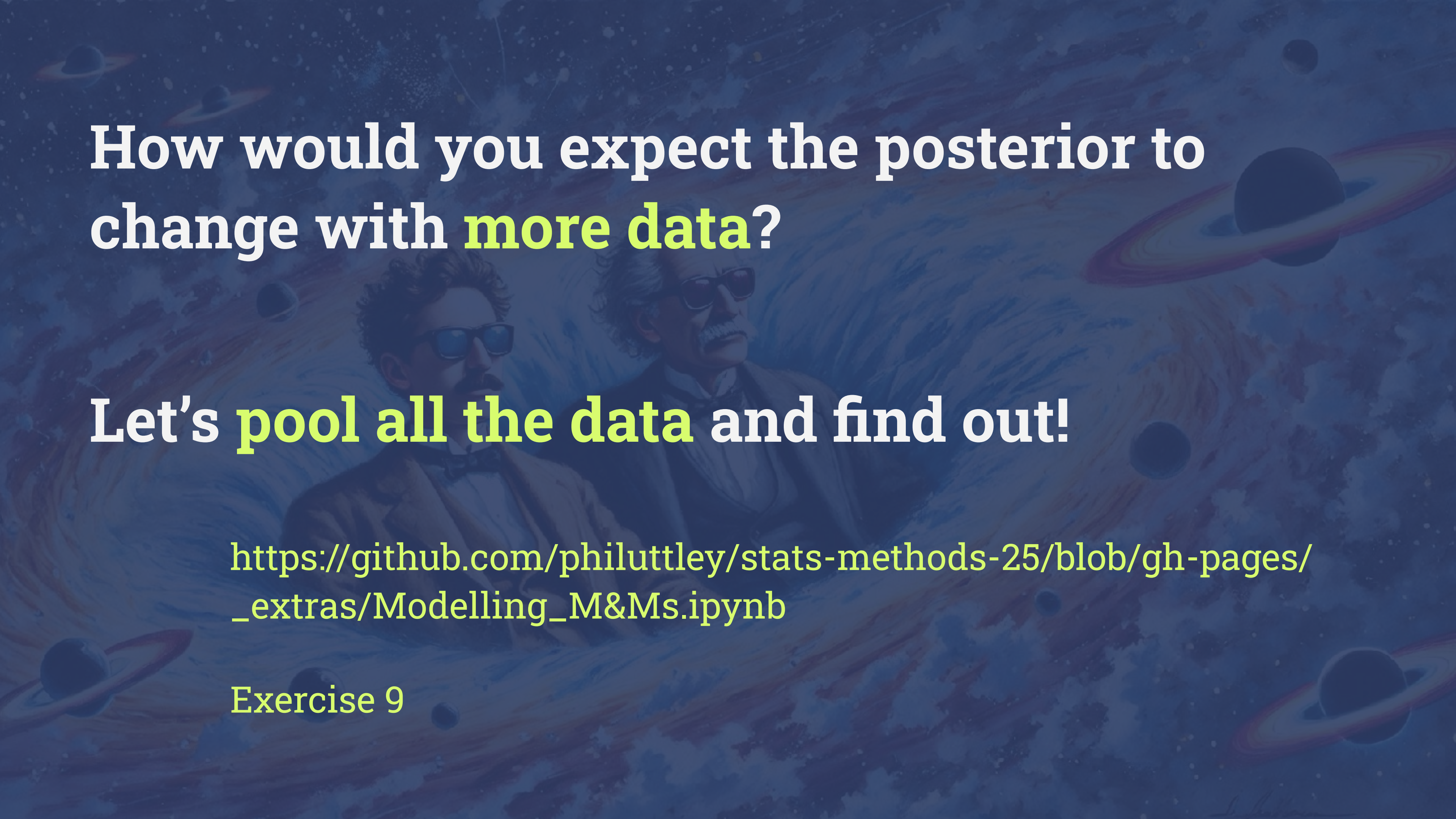
Let's **pool all the data** and find out!



How would you expect the posterior to change with **more data**?

Let's **pool all the data** and find out!

https://github.com/philuttle/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

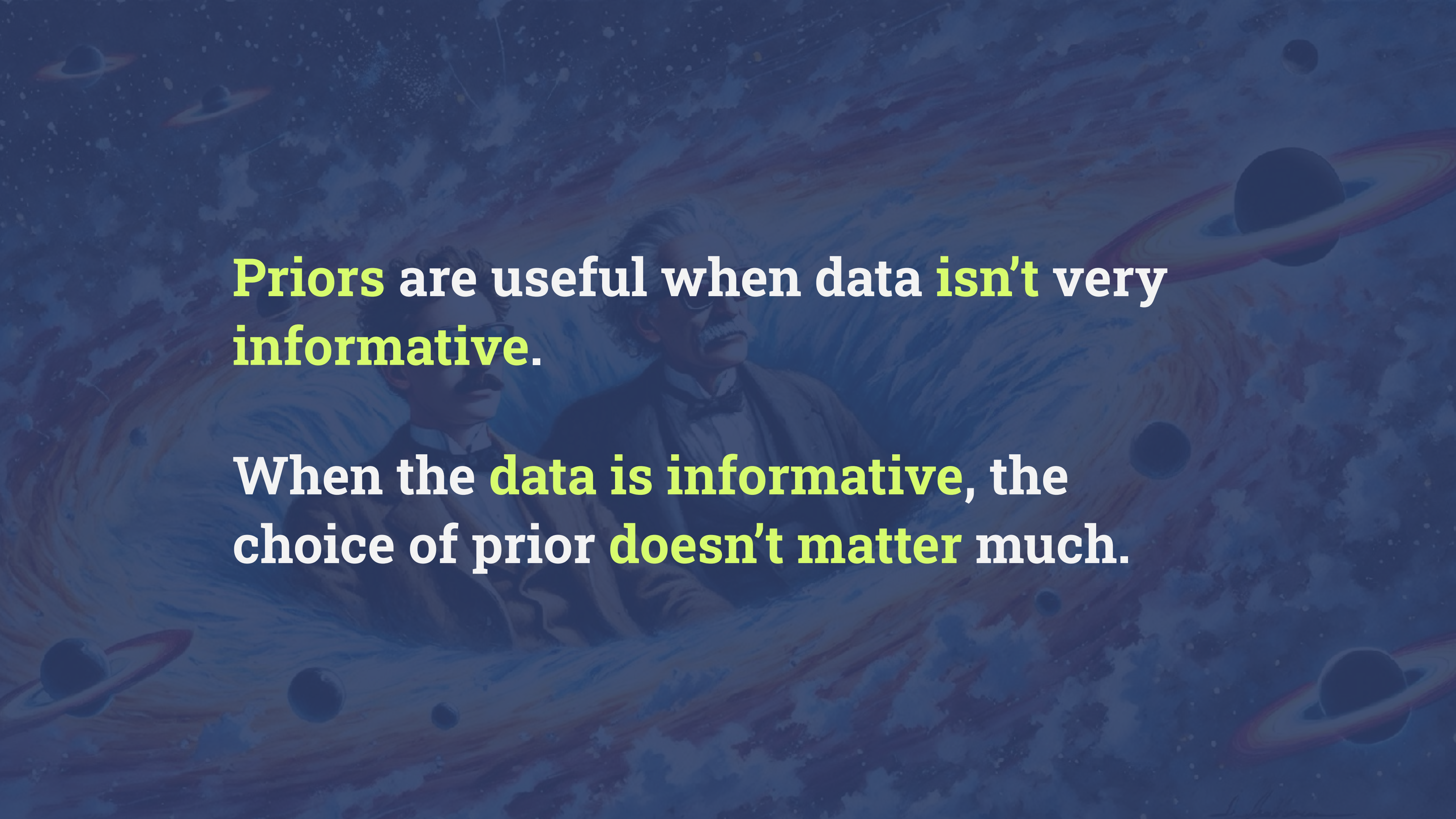
The background of the slide is a dark blue space-themed illustration. It features several galaxies with colorful spiral arms, several planets with rings (resembling Saturn), and two men in suits and sunglasses. The man on the left has dark hair and a mustache, while the man on the right has white hair and a mustache. They appear to be floating in space.

How would you expect the posterior to change with **more data**?

Let's **pool all the data** and find out!

https://github.com/philuttle/stats-methods-25/blob/gh-pages/_extras/Modelling_M&Ms.ipynb

Exercise 9



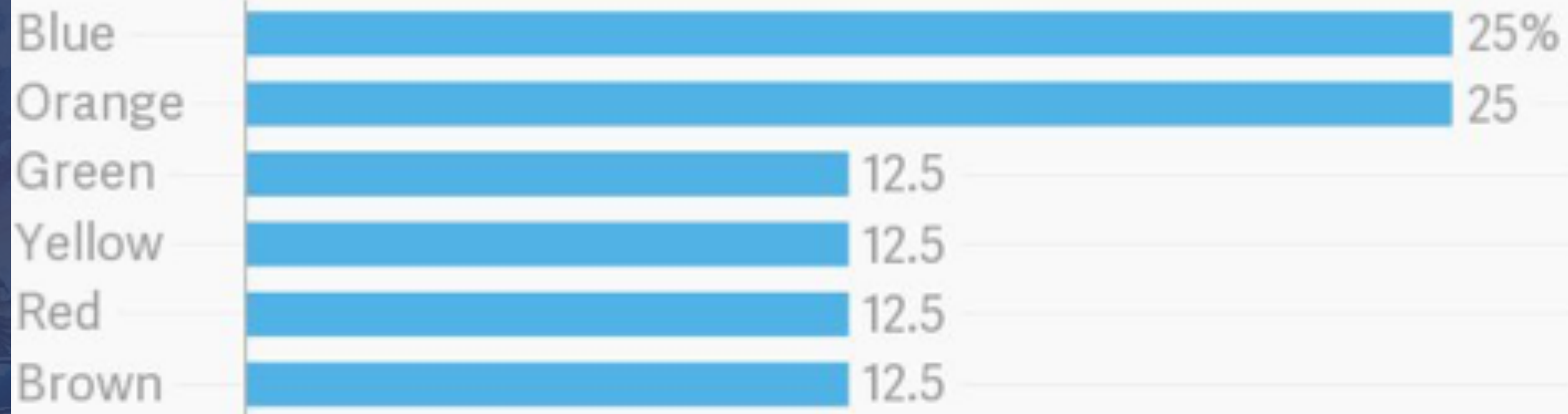
Priors are useful when data **isn't** very **informative**.

When the **data is informative**, the choice of prior **doesn't matter** much.

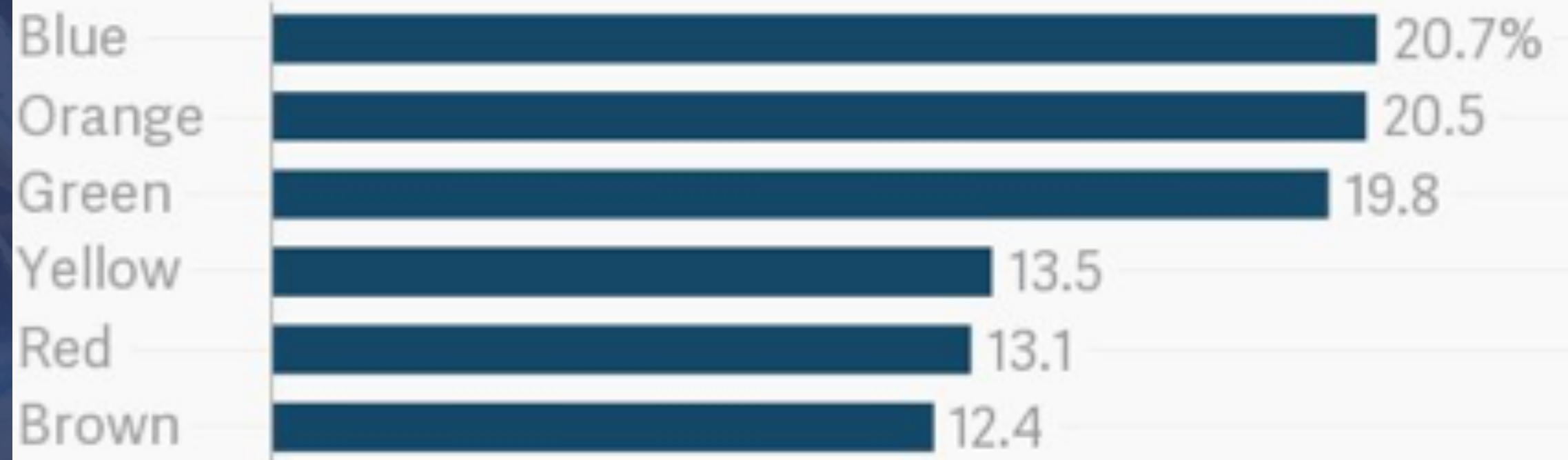
We kind of
know the
distributions of
US factories,
and different
factories make
different
distributions of
m&ms!

M&Ms color distribution, c. 2017

New Jersey factory



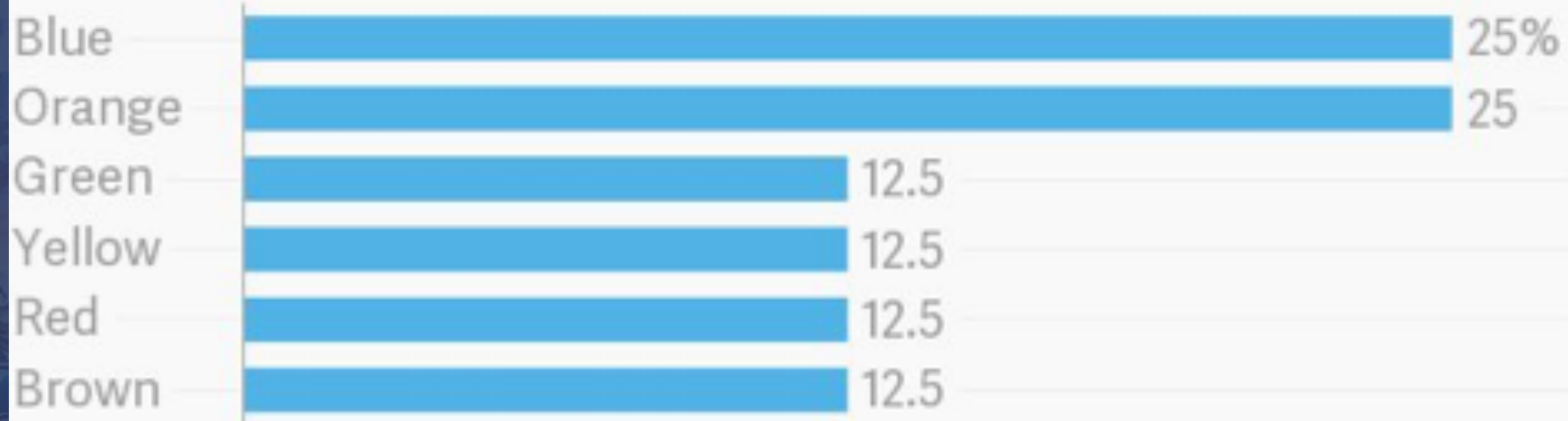
Tennessee factory



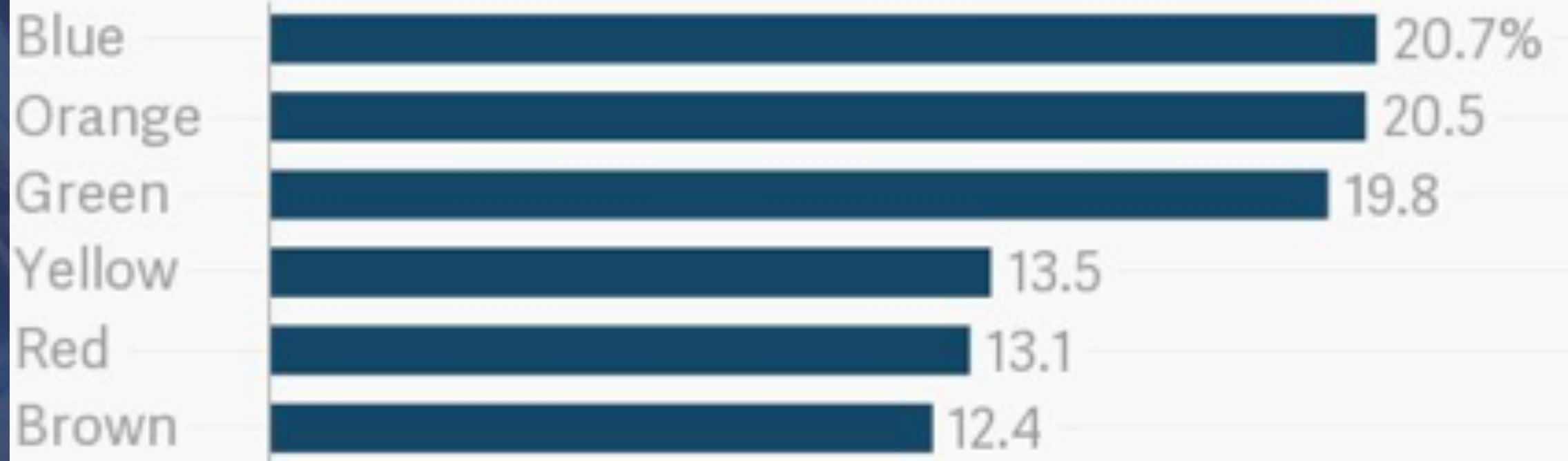
Do **European**
factories
produce
similar
proportions?

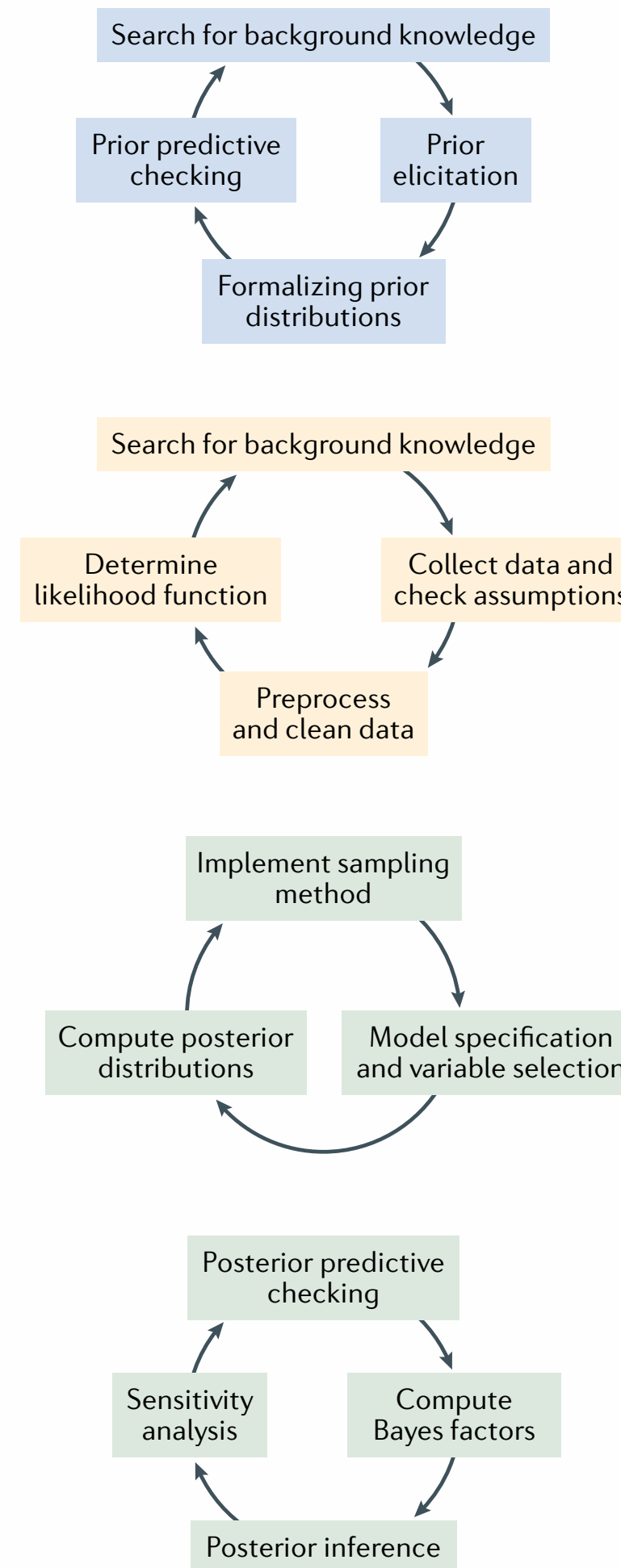
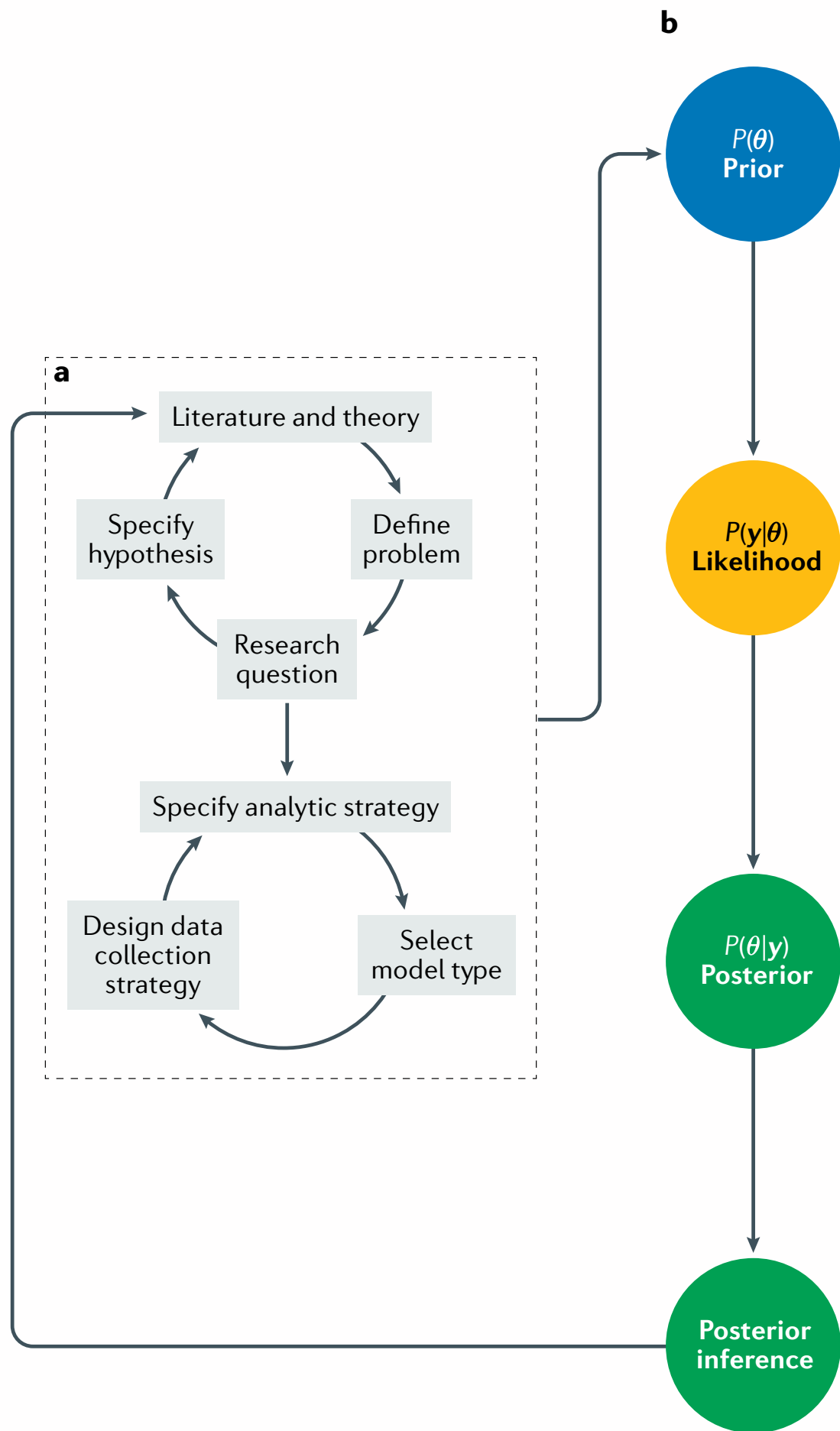
M&Ms color distribution, c. 2017

New Jersey factory



Tennessee factory



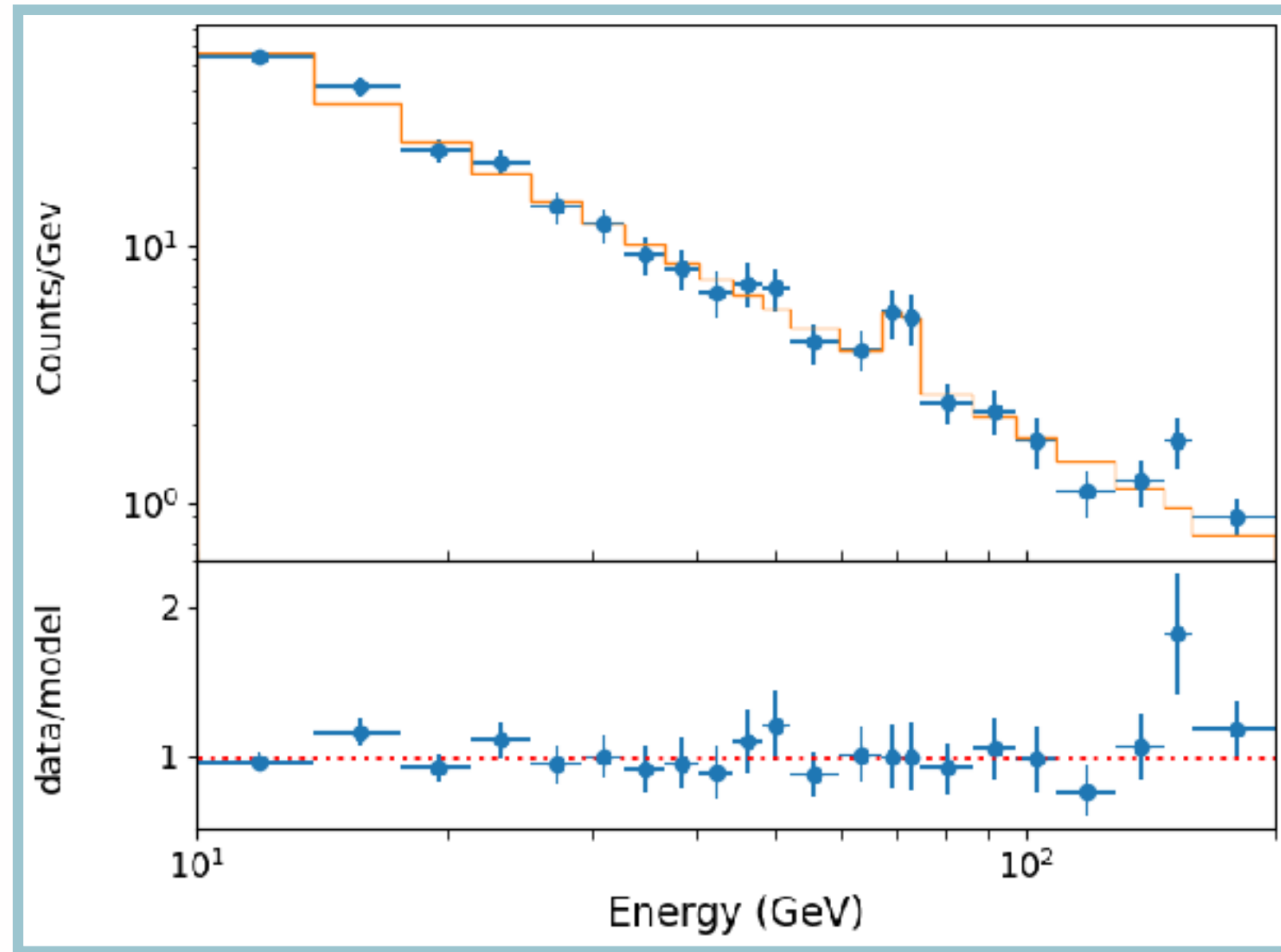


Have other people calculated the distribution of colours? What can we reason about $p(f)$?

Assumption that all colours are mixed together at the factory and then distributed into packages: **binomial likelihood**

What's the probability of a certain fraction of blue smarties given our observations, $p(f | b)$?

What about physics?



Data:

γ -ray spectrum

Model:

Power law + Gaussian line

Likelihood:

Poisson

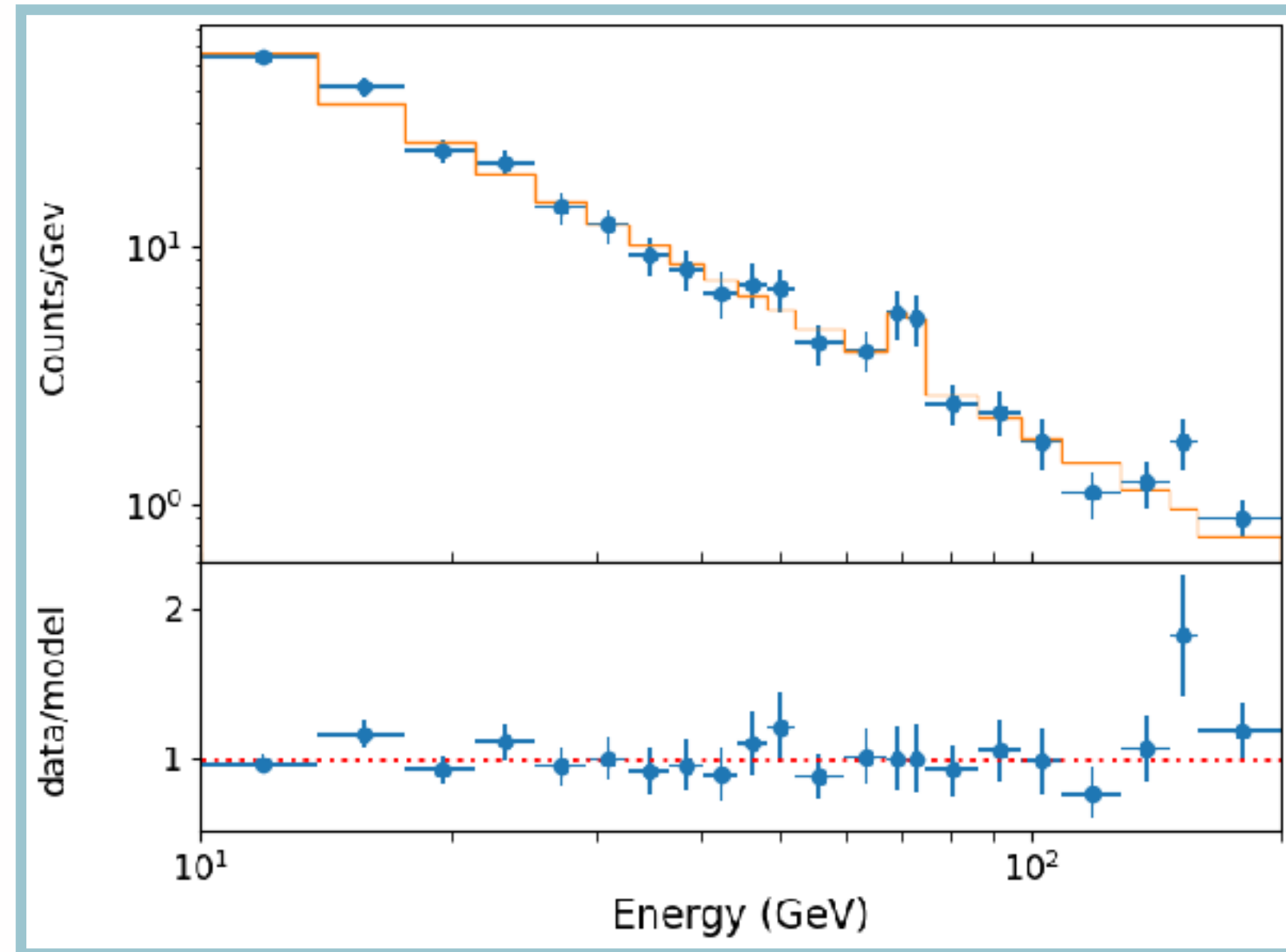
Parameters:

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}},$
 $\sigma_{\text{Line}} A_{\text{Line}}$

Priors:

???

What about physics?



Data:

Model:

Likelihood:

Parameters:

Priors:

X-ray spectrum

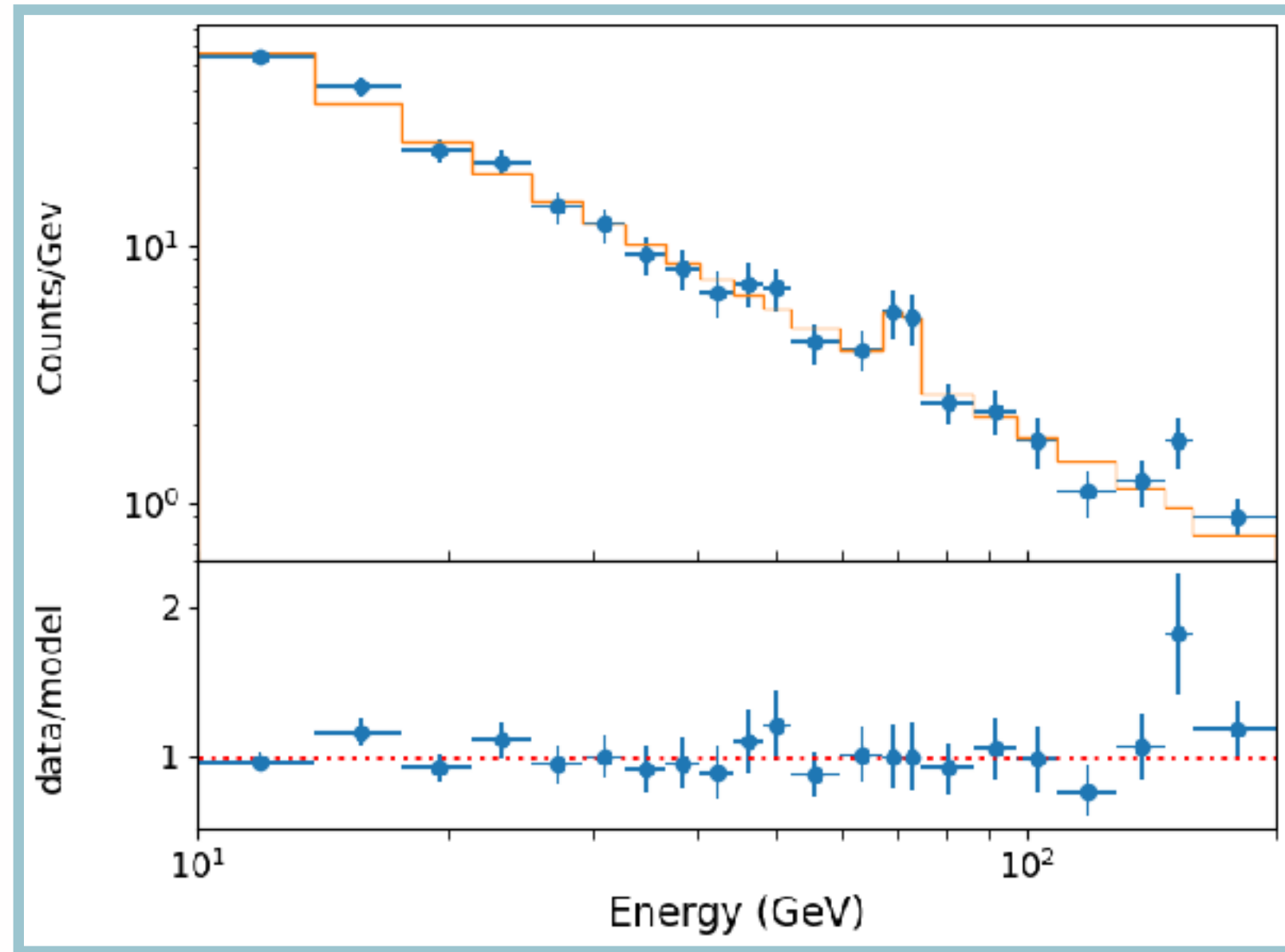
Power law + Gaussian line

Poisson

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}, A_{\text{Line}}$

???

What about physics?



Data:

Model:

Likelihood:

Parameters:

Priors:

X-ray spectrum

Power law + Gaussian line

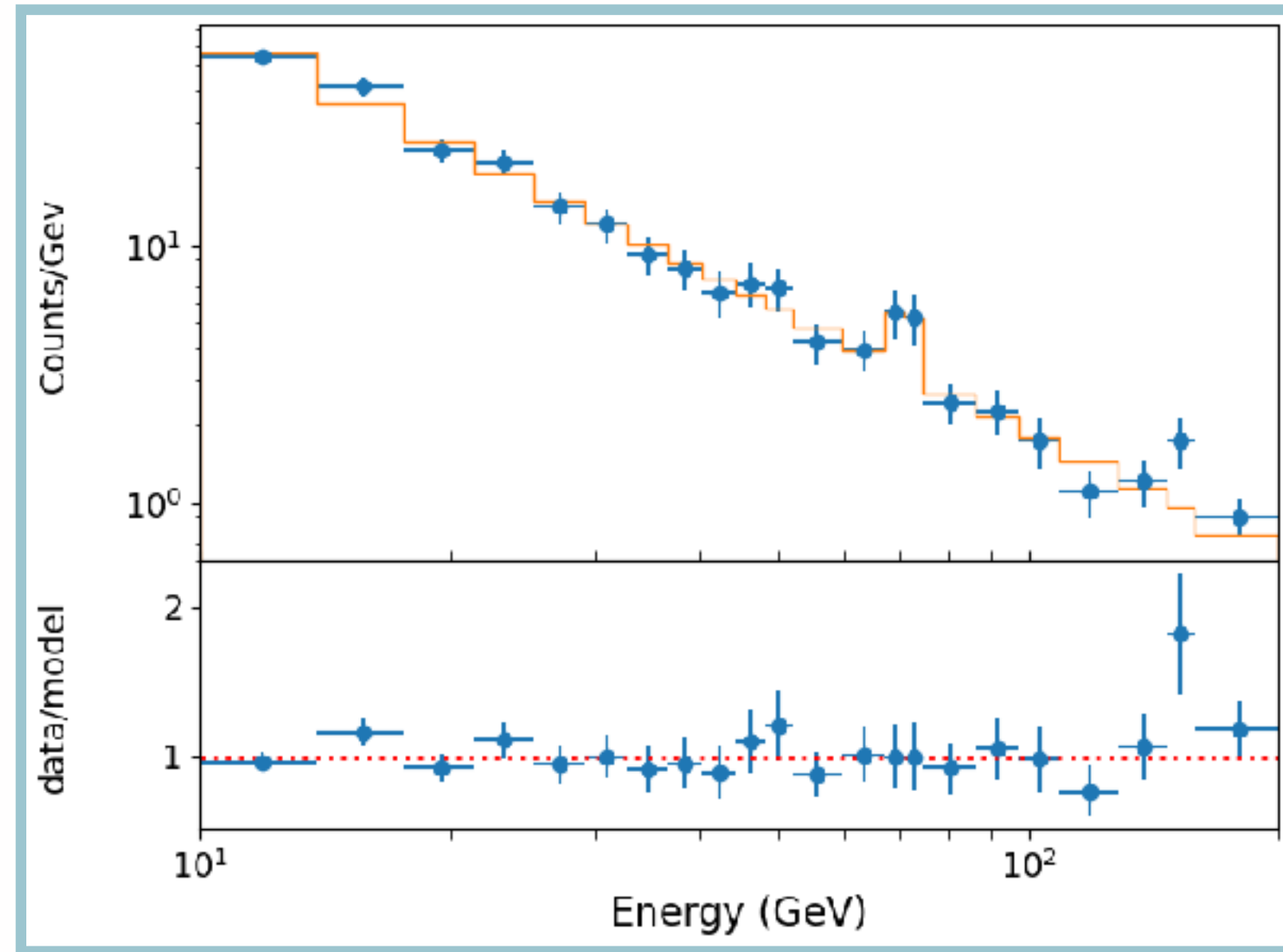
Poisson

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}, A_{\text{Line}}$

???

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}})}{p(D)}$$

What about physics?



Data:

X-ray spectrum

Model:

Power law + Gaussian line

Likelihood:

Poisson

Parameters:

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}, A_{\text{Line}}$

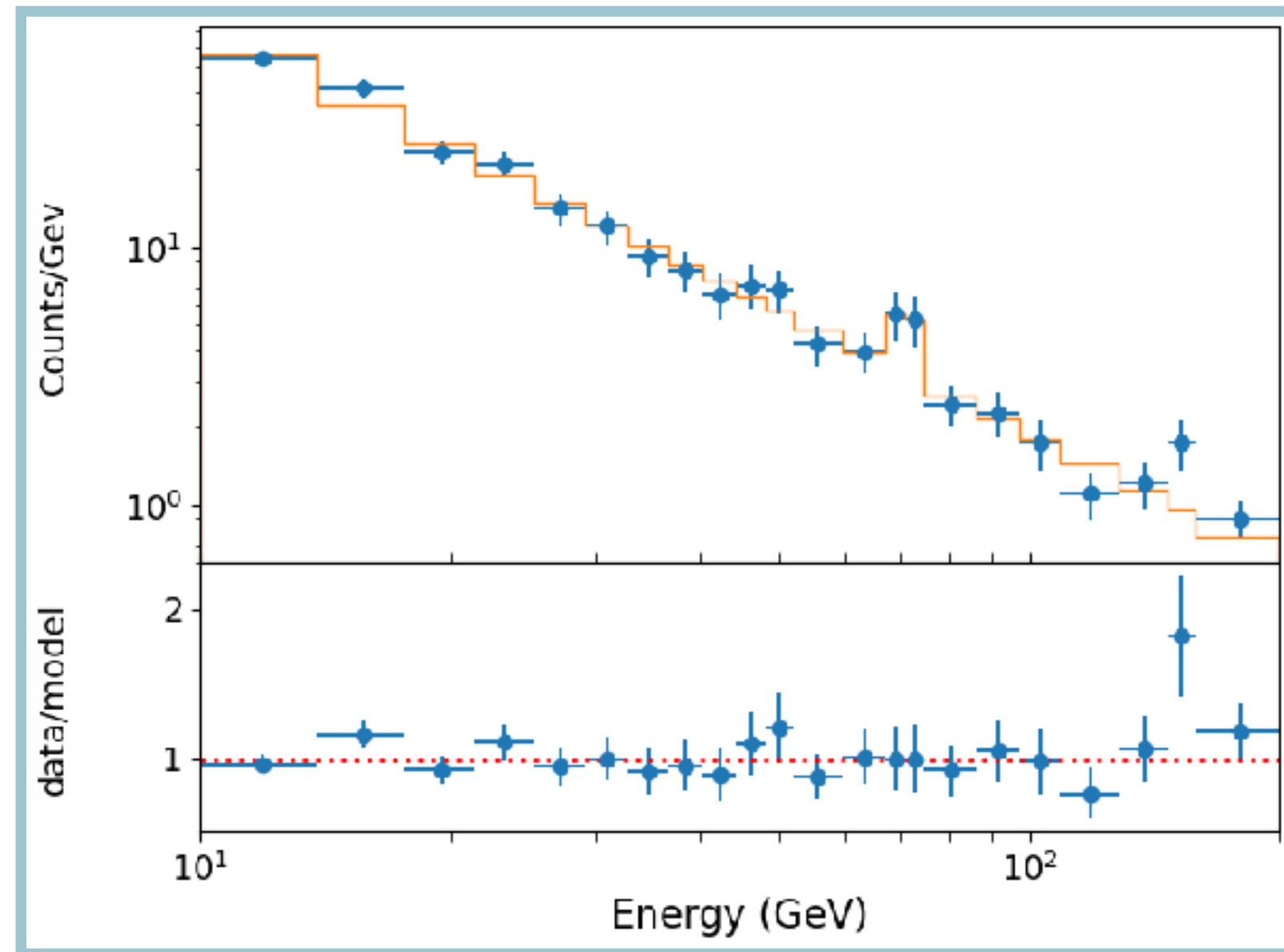
Priors:

???

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}})}{p(D)}$$

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}) p(\Gamma) p(A_{\text{Line}}) p(E_{\text{Line}}) p(\sigma_{\text{Line}})}{p(D)}$$

What about physics?



Data:

Model:

Likelihood:

Parameters:

Priors:

X-ray spectrum

Power law + Gaussian line

Poisson

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}, A_{\text{Line}}$

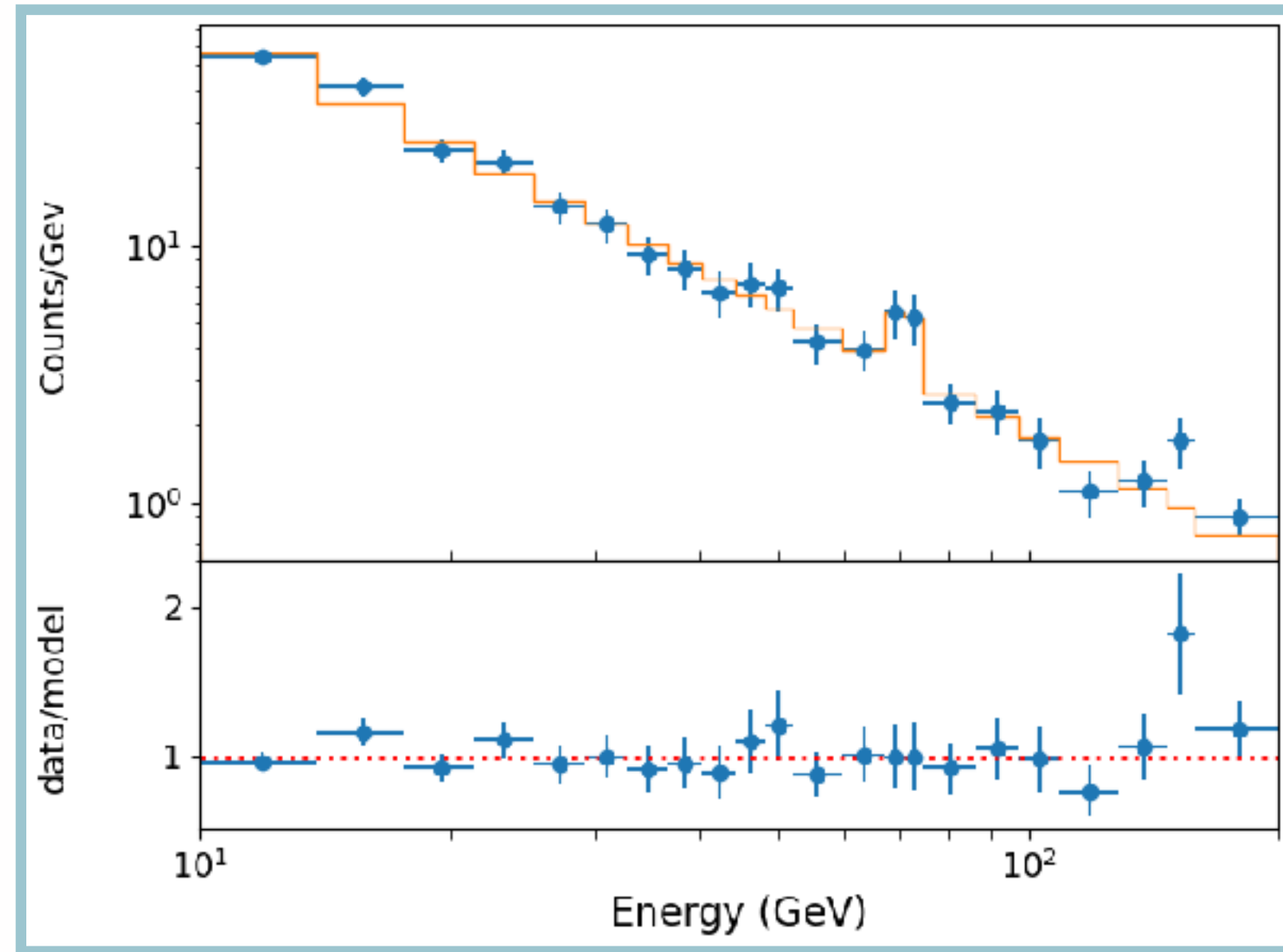
???

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}})}{p(D)}$$

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}) p(\Gamma) p(A_{\text{Line}}) p(E_{\text{Line}}) p(\sigma_{\text{Line}})}{p(D)}$$

**What assumption
have we made here?**

What about physics?



Data:

Model:

Likelihood:

Parameters:

Priors:

X-ray spectrum

Power law + Gaussian line

Poisson

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}, A_{\text{Line}}$

???

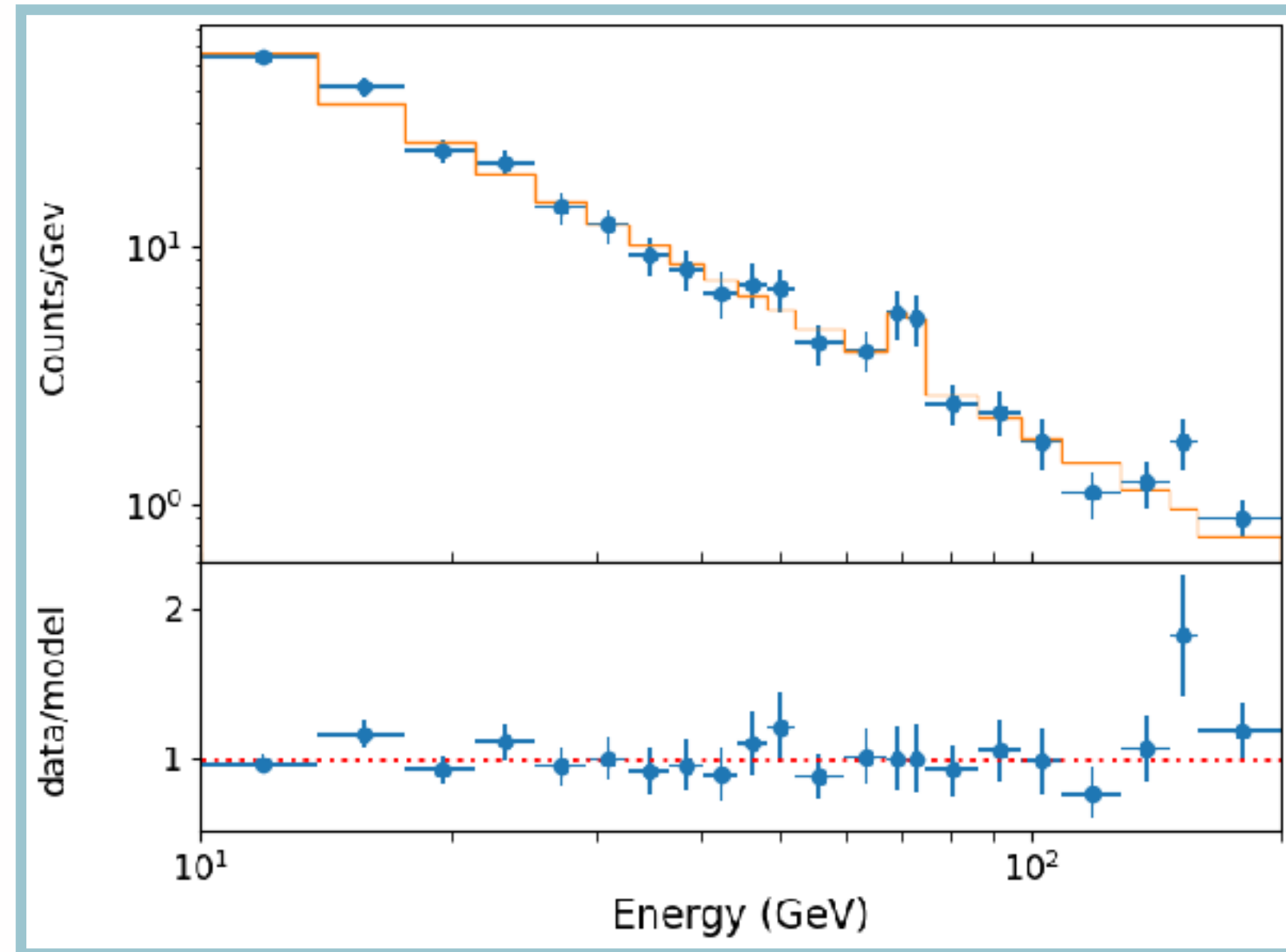
$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}})}{p(D)}$$

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}) p(\Gamma) p(A_{\text{Line}}) p(E_{\text{Line}}) p(\sigma_{\text{Line}})}{p(D)}$$

**What assumption
have we made here?**

**Independence of
priors!**

What about physics?



Data:

X-ray spectrum

Model:

Power law + Gaussian line

Likelihood:

Poisson

Parameters:

$A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}, A_{\text{Line}}$

Priors:

???

What priors should we choose?

$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}})}{p(D)}$$

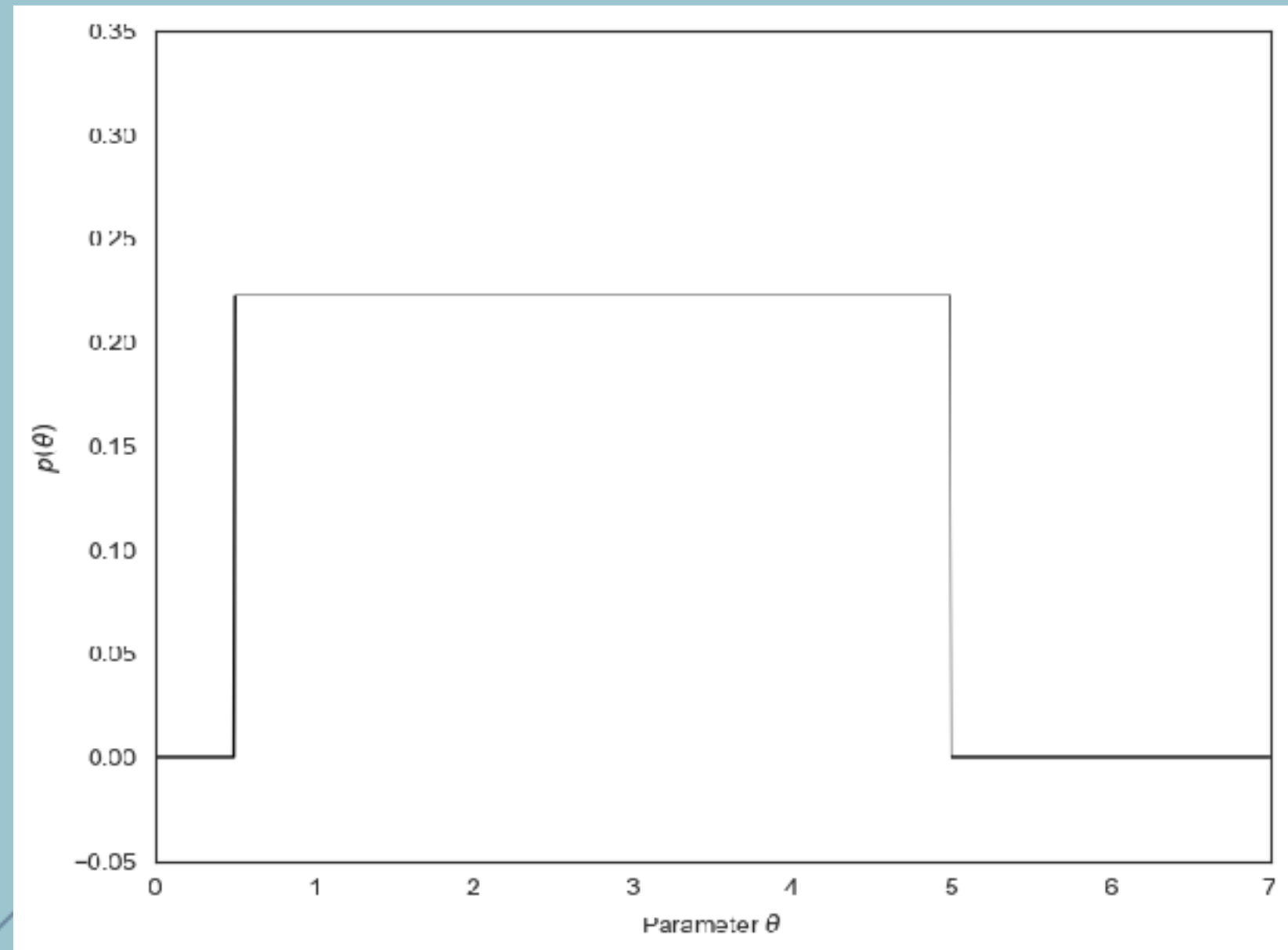
$$p(A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}} | D) = \frac{p(D | A_{\text{PL}}, \Gamma, A_{\text{Line}}, E_{\text{Line}}, \sigma_{\text{Line}}) p(A_{\text{PL}}) p(\Gamma) p(A_{\text{Line}}) p(E_{\text{Line}}) p(\sigma_{\text{Line}})}{p(D)}$$

**What assumption
have we made here?**

**Independence of
priors!**

Flat Priors:
 Γ

“I know nothing but I’m 100% sure
that my parameter is between $\theta_{\min}$ and
 $\theta_{\max}$ ”



Scale-free priors:

$A_{\text{PL}}, A_{\text{Line}}$

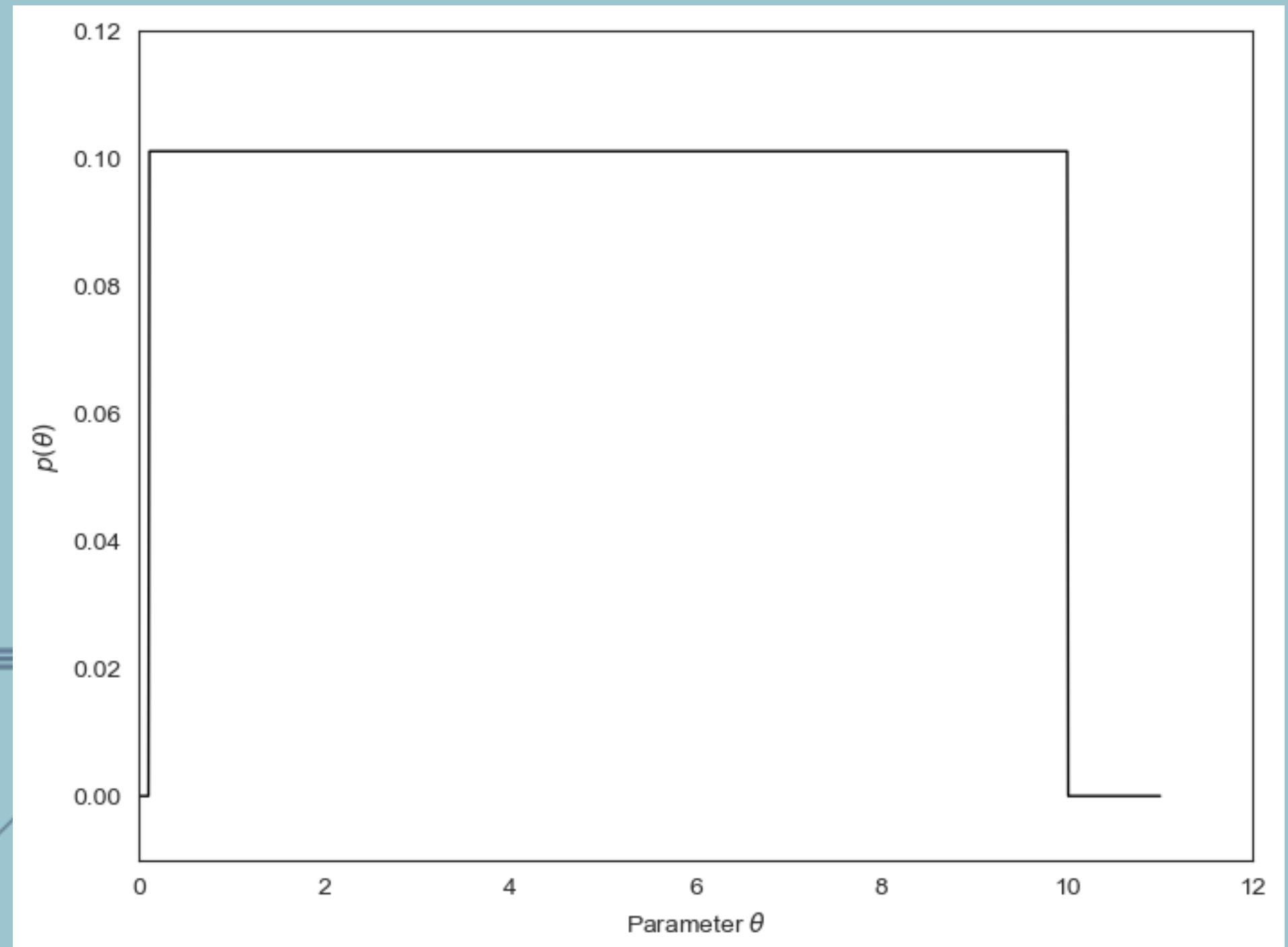
“I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10”



Scale-free priors:

A_{PL} , A_{Line}

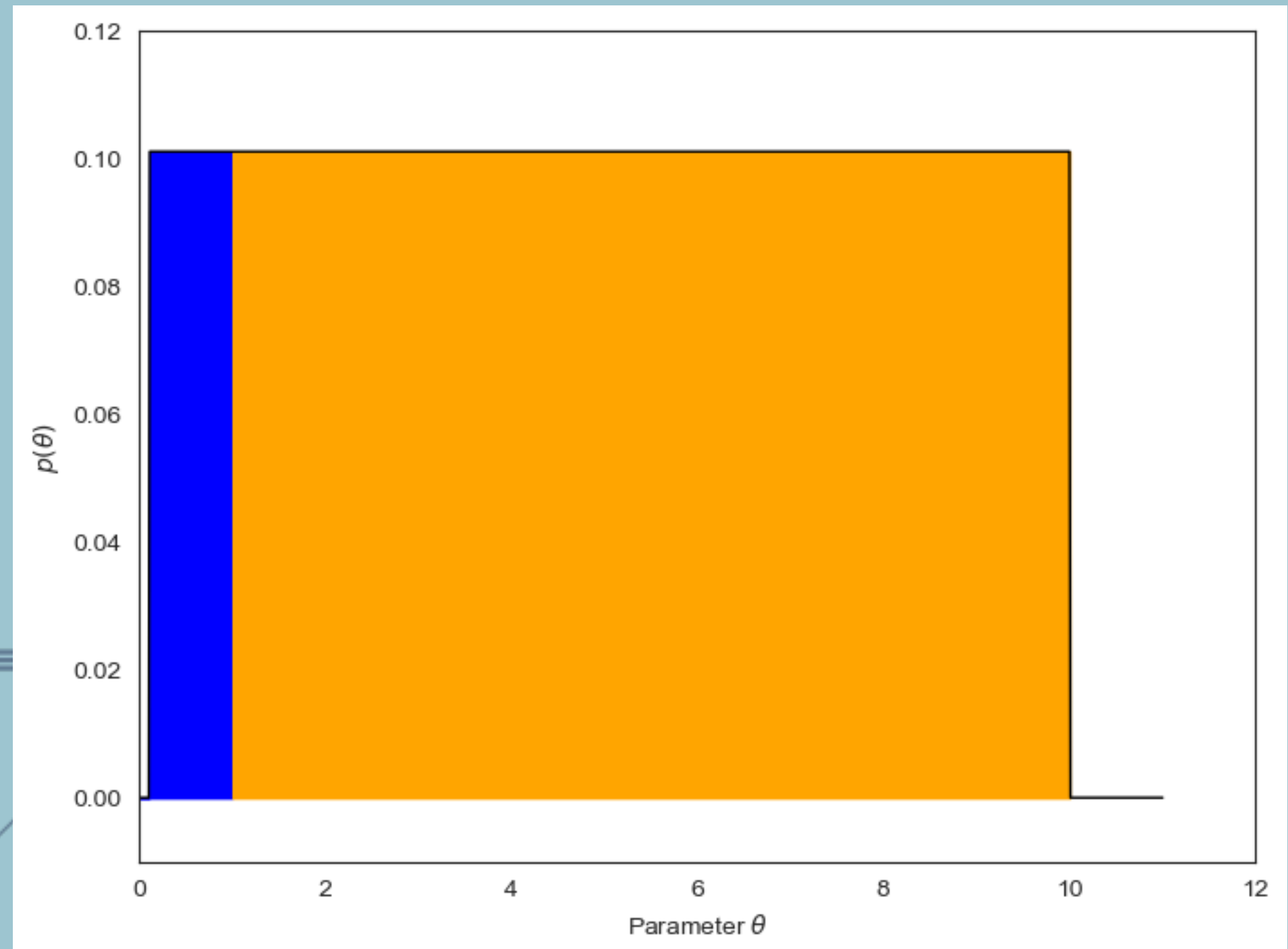
“I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10”



Scale-free priors:

A_{PL} , A_{Line}

“I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10”



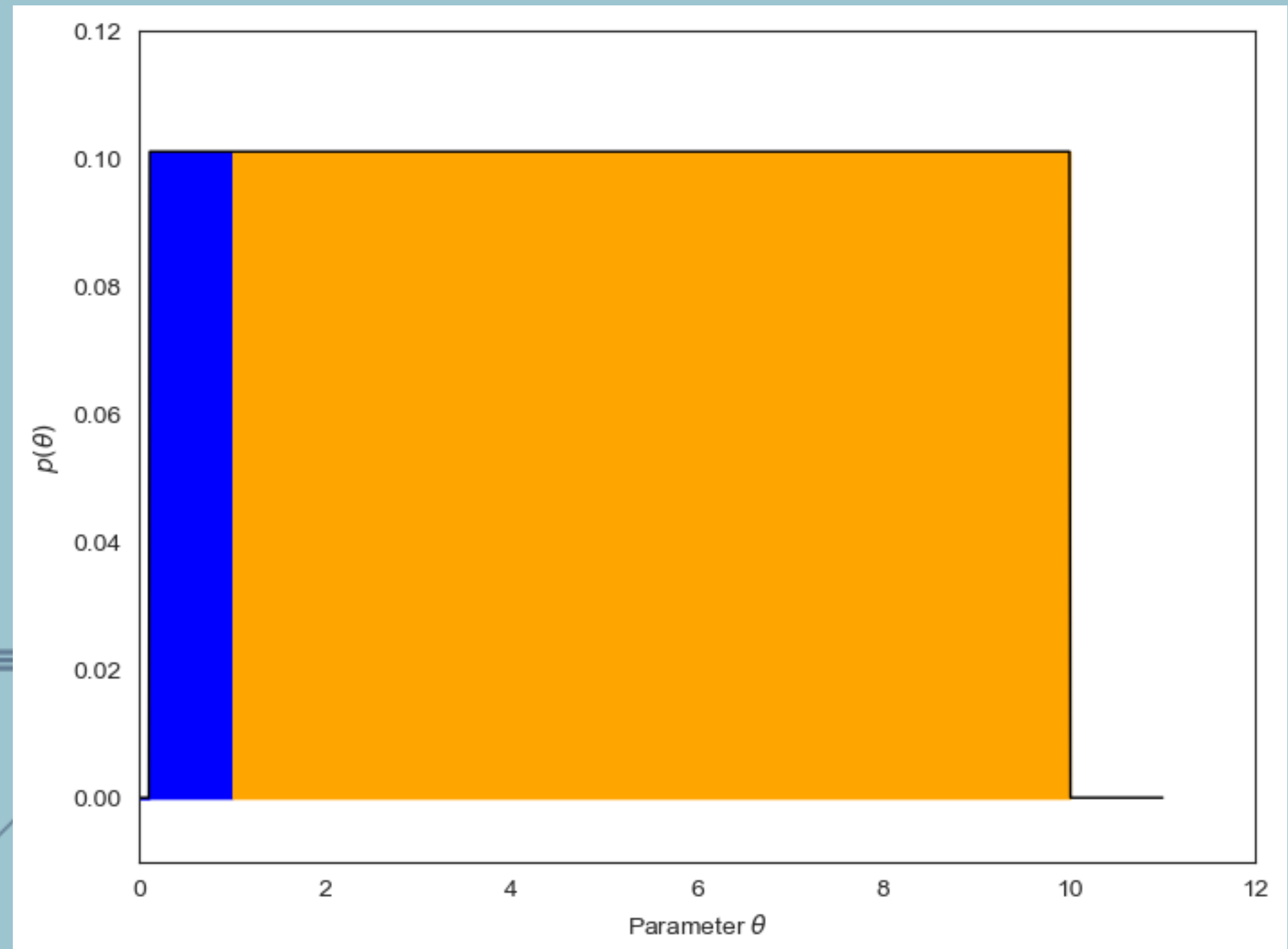
Scale-free priors:

$A_{\text{PL}}, A_{\text{Line}}$



$\log(A_{\text{PL}}), \log(A_{\text{Line}})$

“I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10”



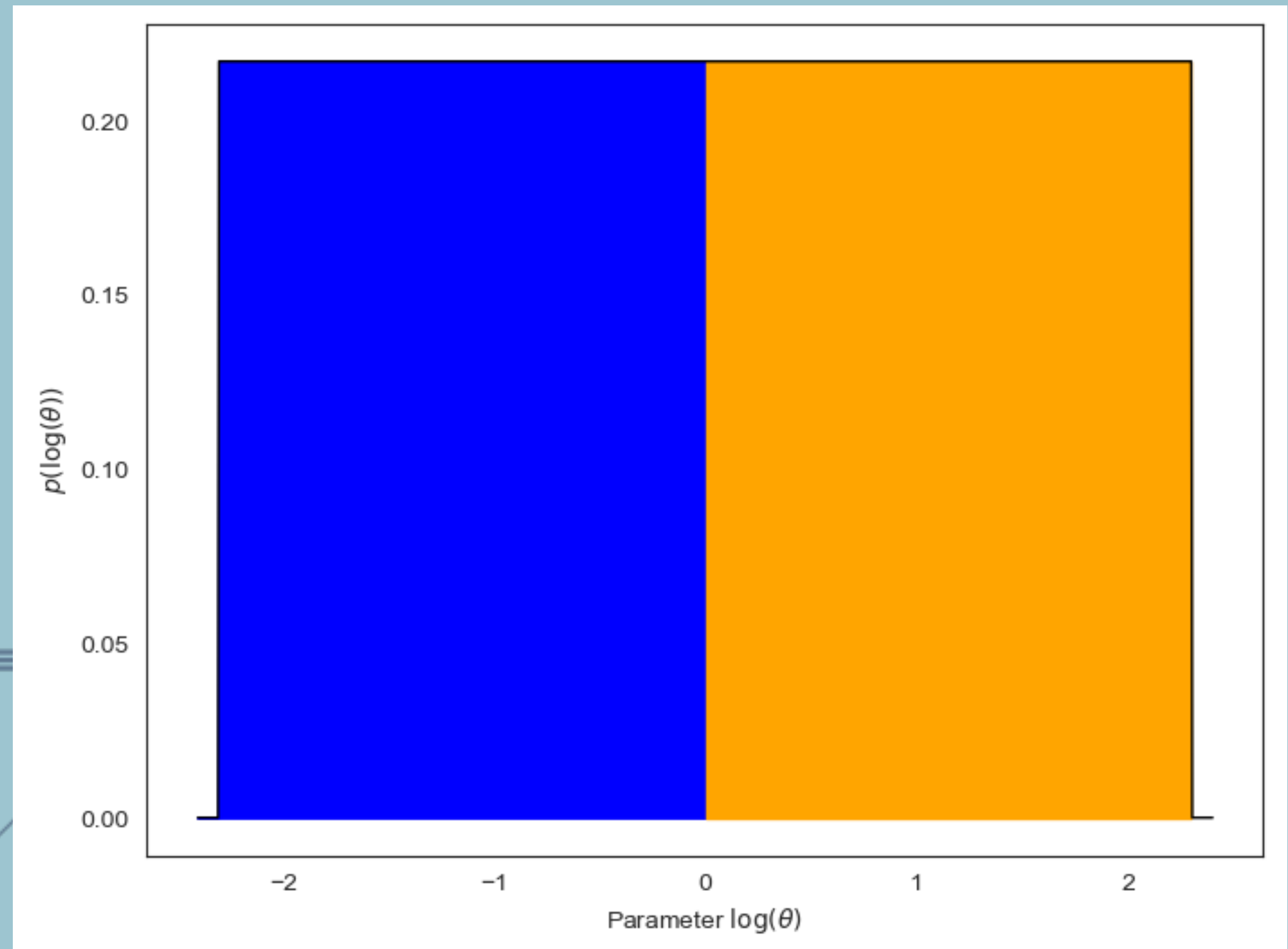
Scale-free priors:

$A_{\text{PL}}, A_{\text{Line}}$



$\log(A_{\text{PL}}), \log(A_{\text{Line}})$

“I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10”



Scale-free priors:

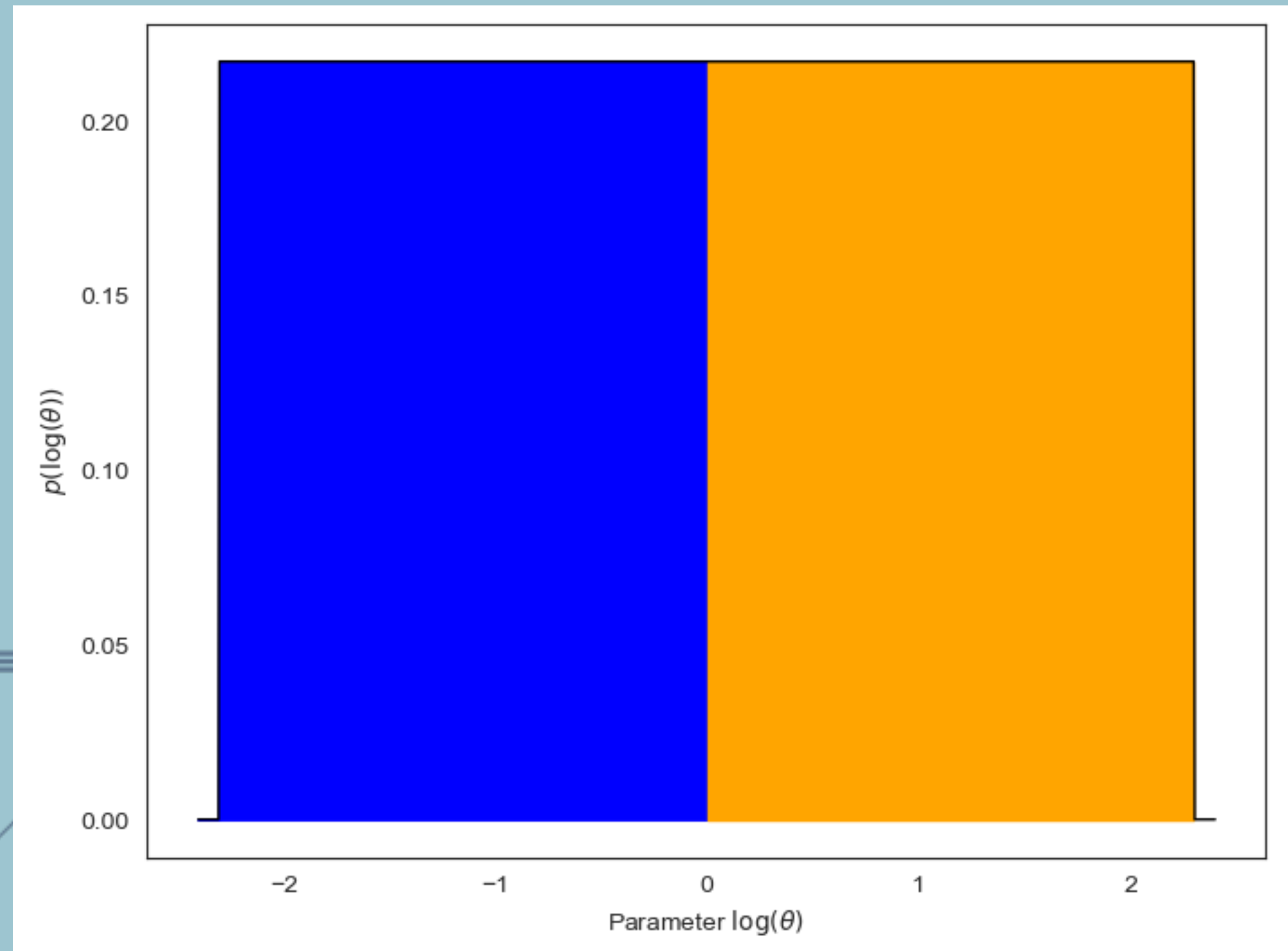
$A_{\text{PL}}, A_{\text{Line}}$



$\log(A_{\text{PL}}), \log(A_{\text{Line}})$

“I believe with equal prior probability that my parameter could be between 0.1 and 1 or between 1 and 10”

Also use this for
parameters that
cannot be
negative!

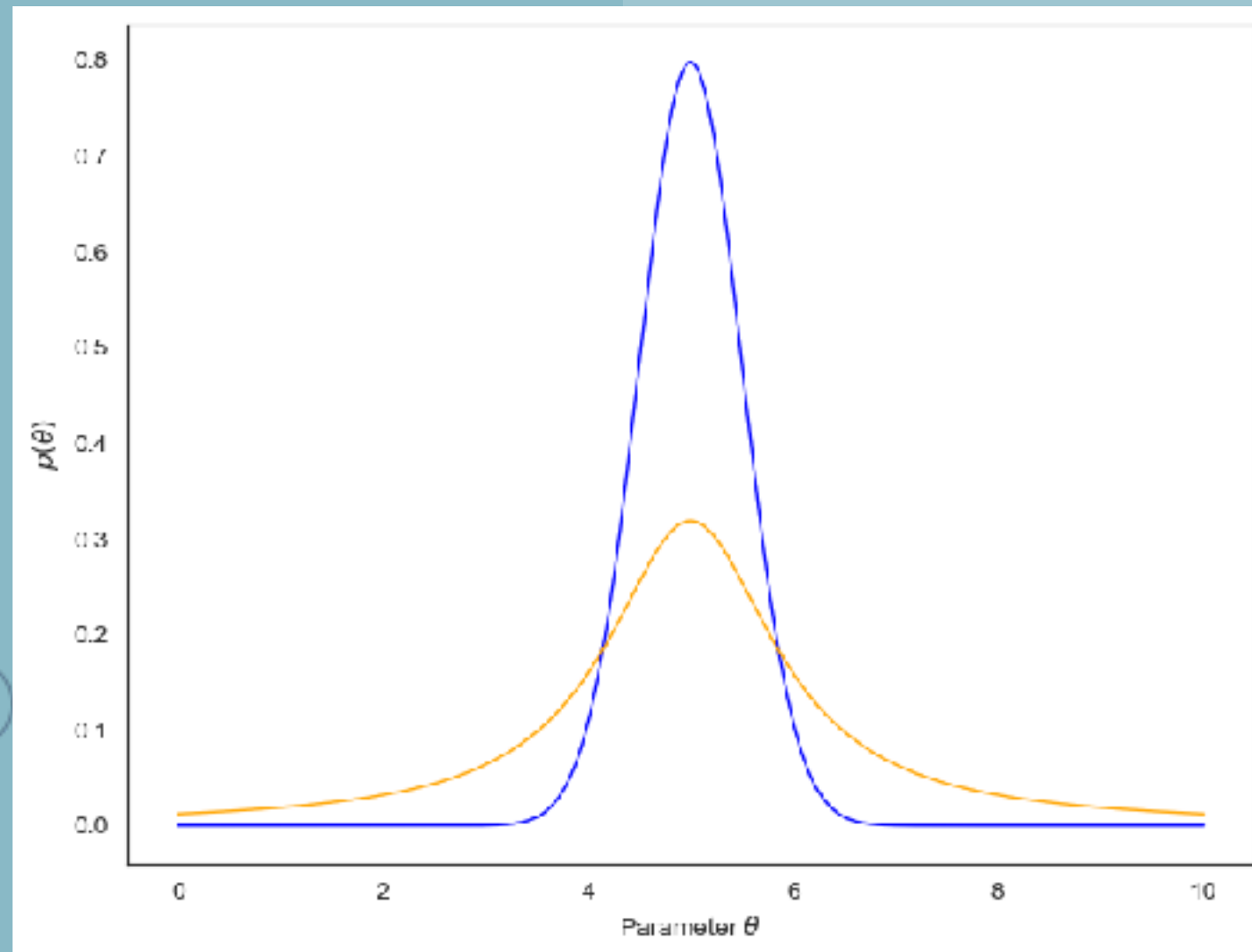


Other Distributions:

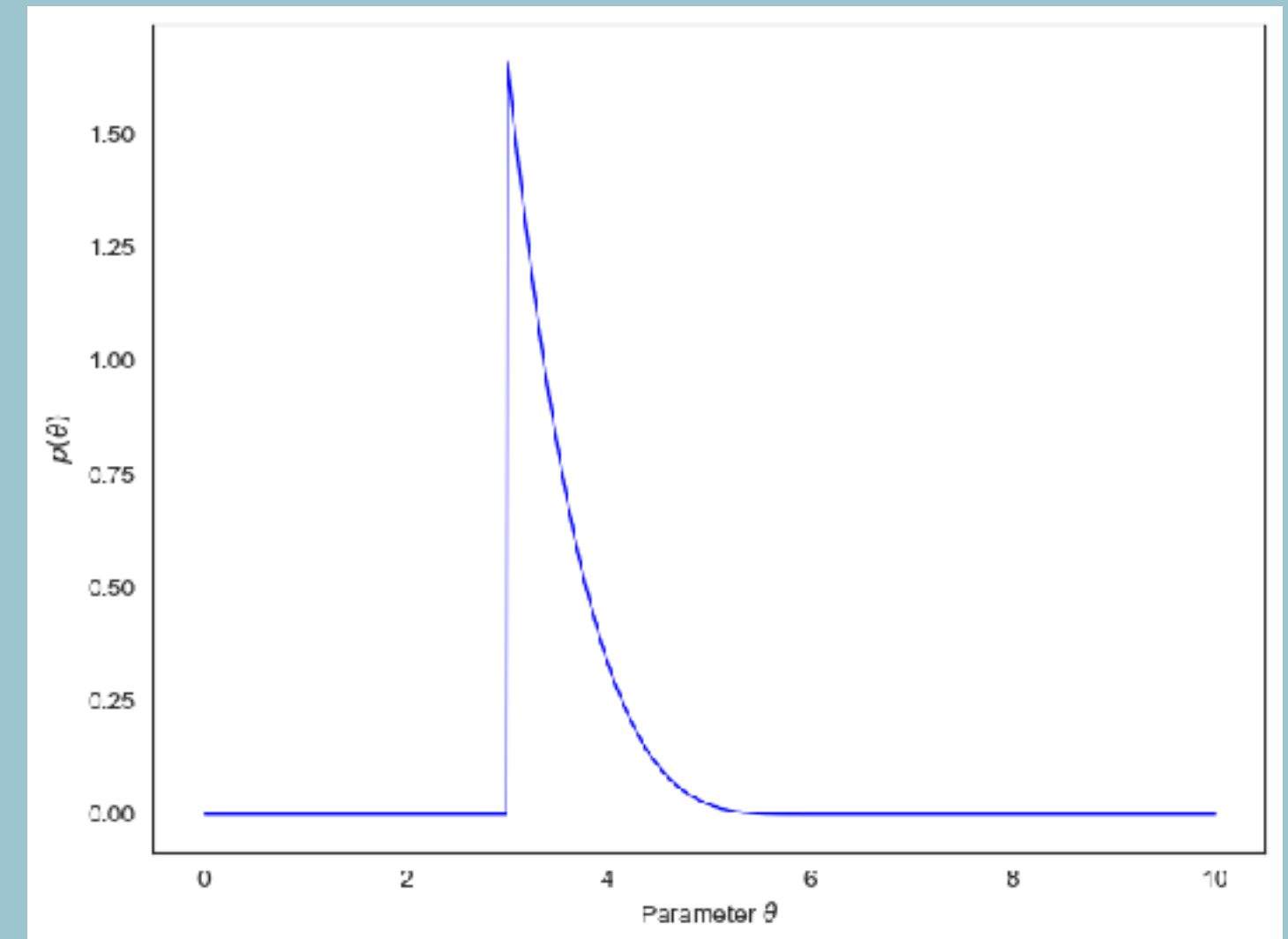
E_{line}

“I have prior information that says that my parameter θ should be around a value of $\hat{\theta}$, but I’m not sure.”

Gaussian and t-distribution



Beta-distribution



Reparametrizations

σ_{line}

“The possible width of a line generally scales with the line’s energy.”

$$p(E_{\text{line}}, \sigma_{\text{line}}) \neq p(E_{\text{line}})p(\sigma_{\text{line}})$$



Reparametrizations

σ_{line}

“The possible width of a line generally scales with the line’s energy.”

$$p(E_{\text{line}}, \sigma_{\text{line}}) \neq p(E_{\text{line}})p(\sigma_{\text{line}})$$



$$q = \frac{E_{\text{line}}}{\sigma_{\text{line}}}$$



Reparametrizations

σ_{line}

“The possible width of a line generally scales with the line’s energy.”

$$p(E_{\text{line}}, \sigma_{\text{line}}) \neq p(E_{\text{line}})p(\sigma_{\text{line}})$$

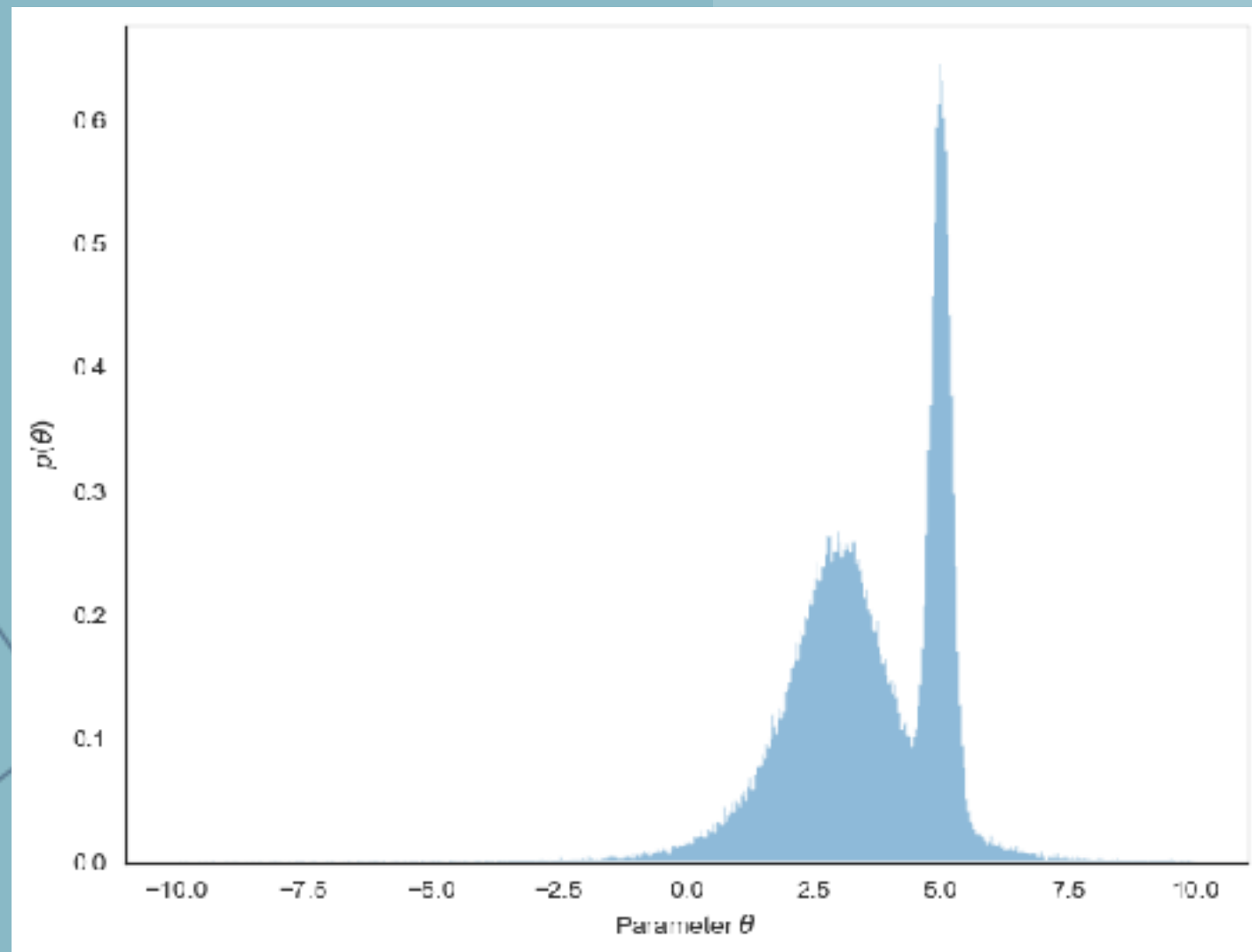
$$q = \frac{E_{\text{line}}}{\sigma_{\text{line}}}$$

$$p(E_{\text{line}}, q) = p(E_{\text{line}})p(q)$$



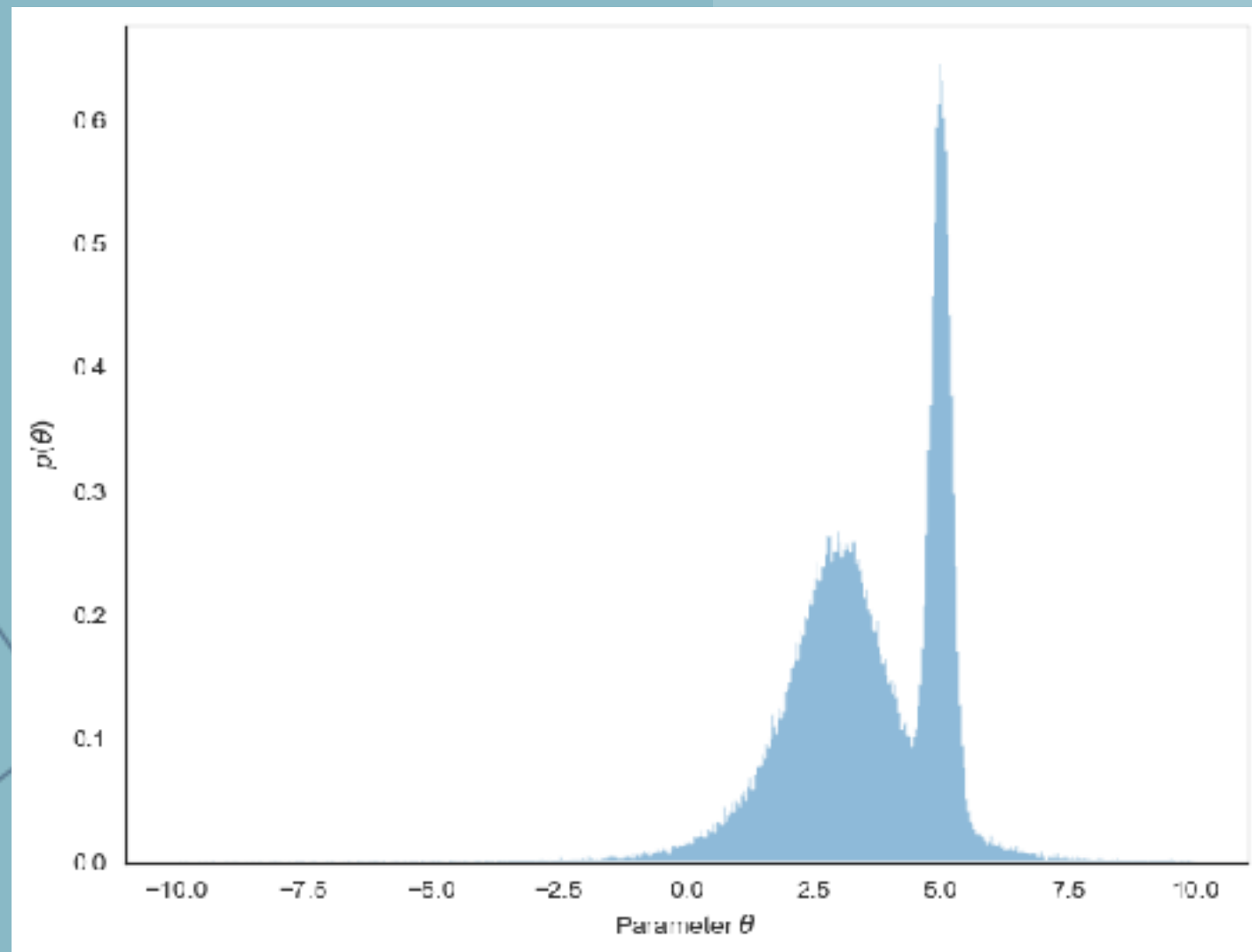
Empirical Distributions

“My prior is actually someone else’s
posterior”

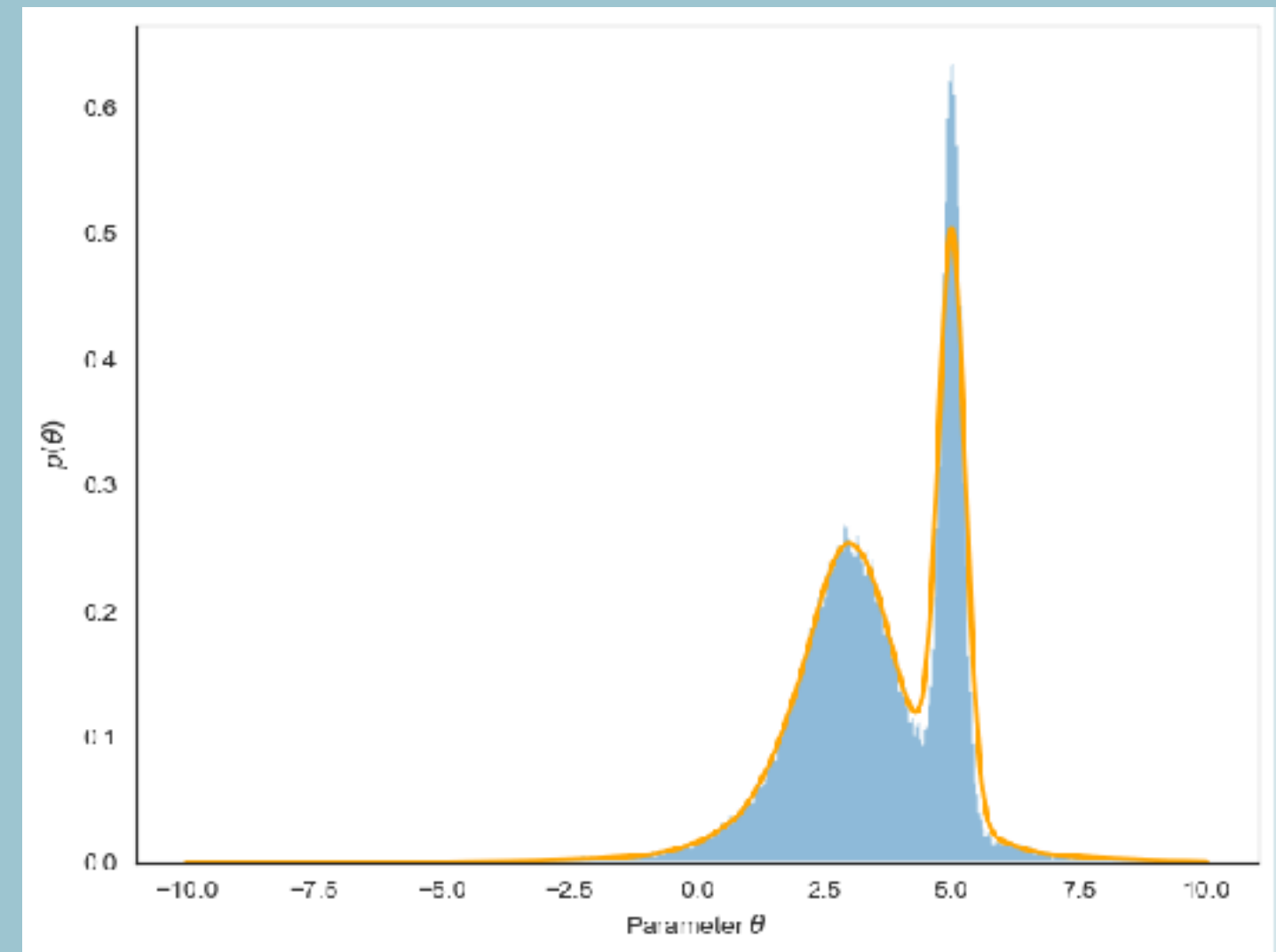


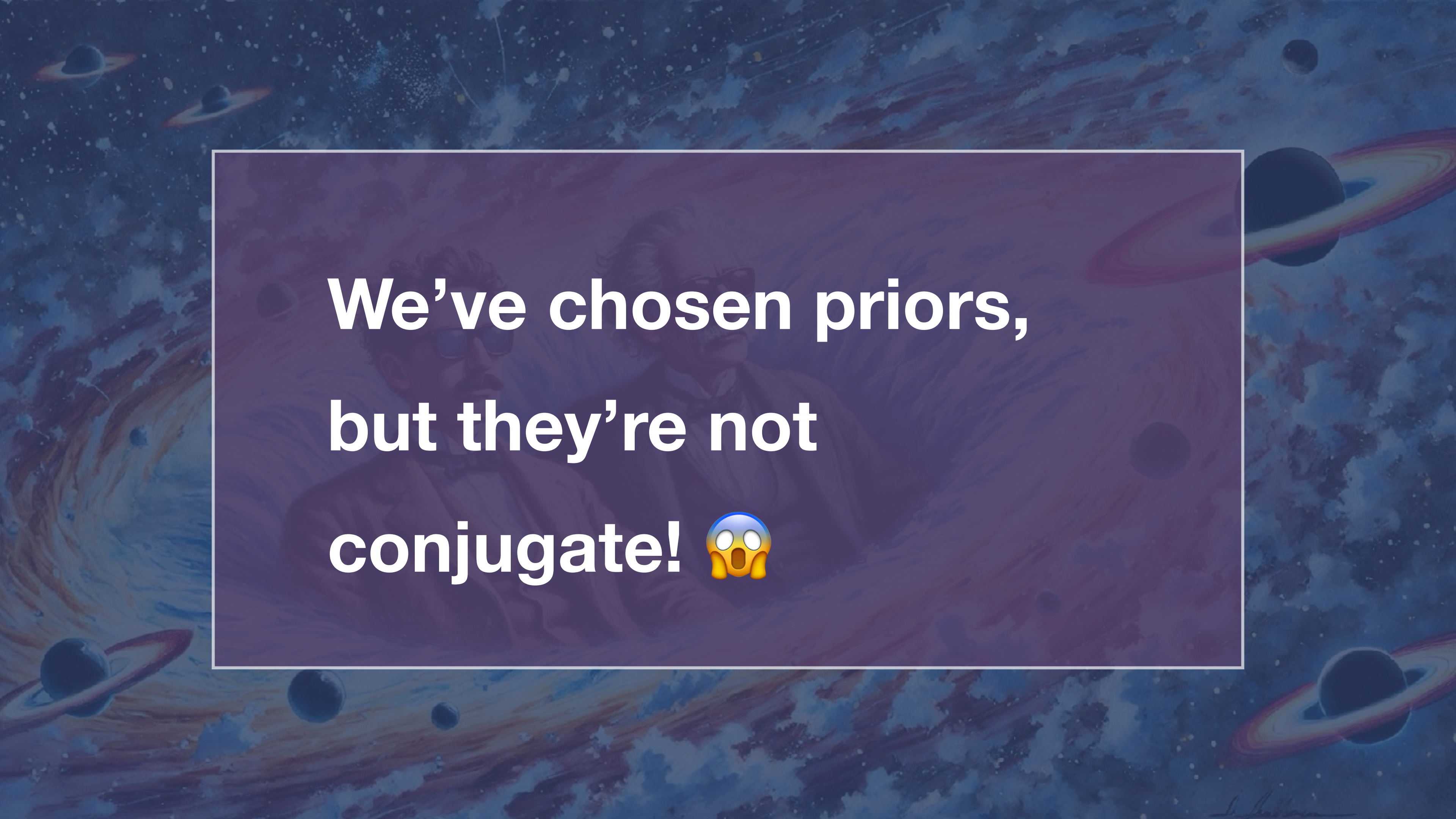
Empirical Distributions

“My prior is actually someone else’s
posterior”

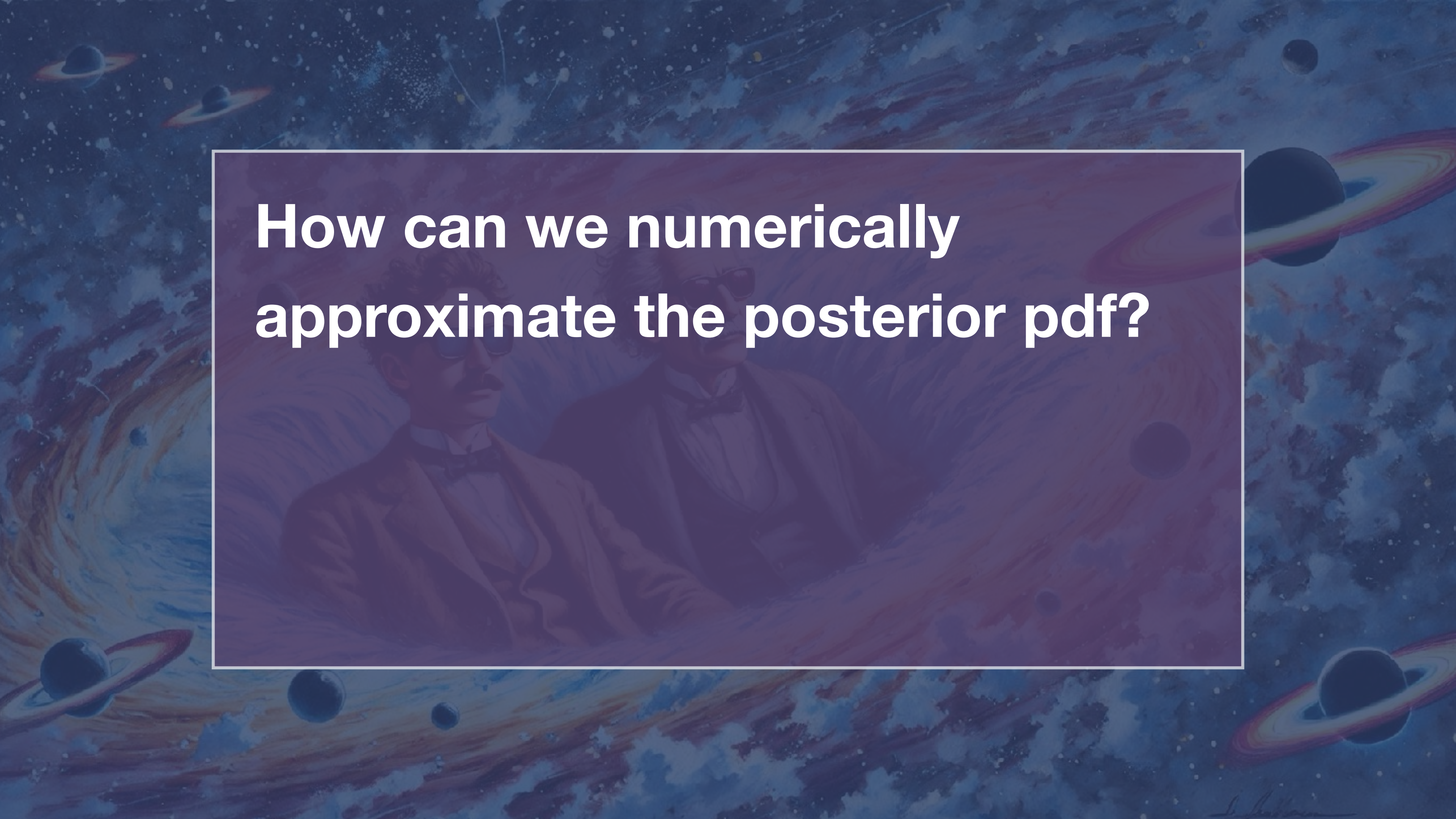


Kernel Density Estimation

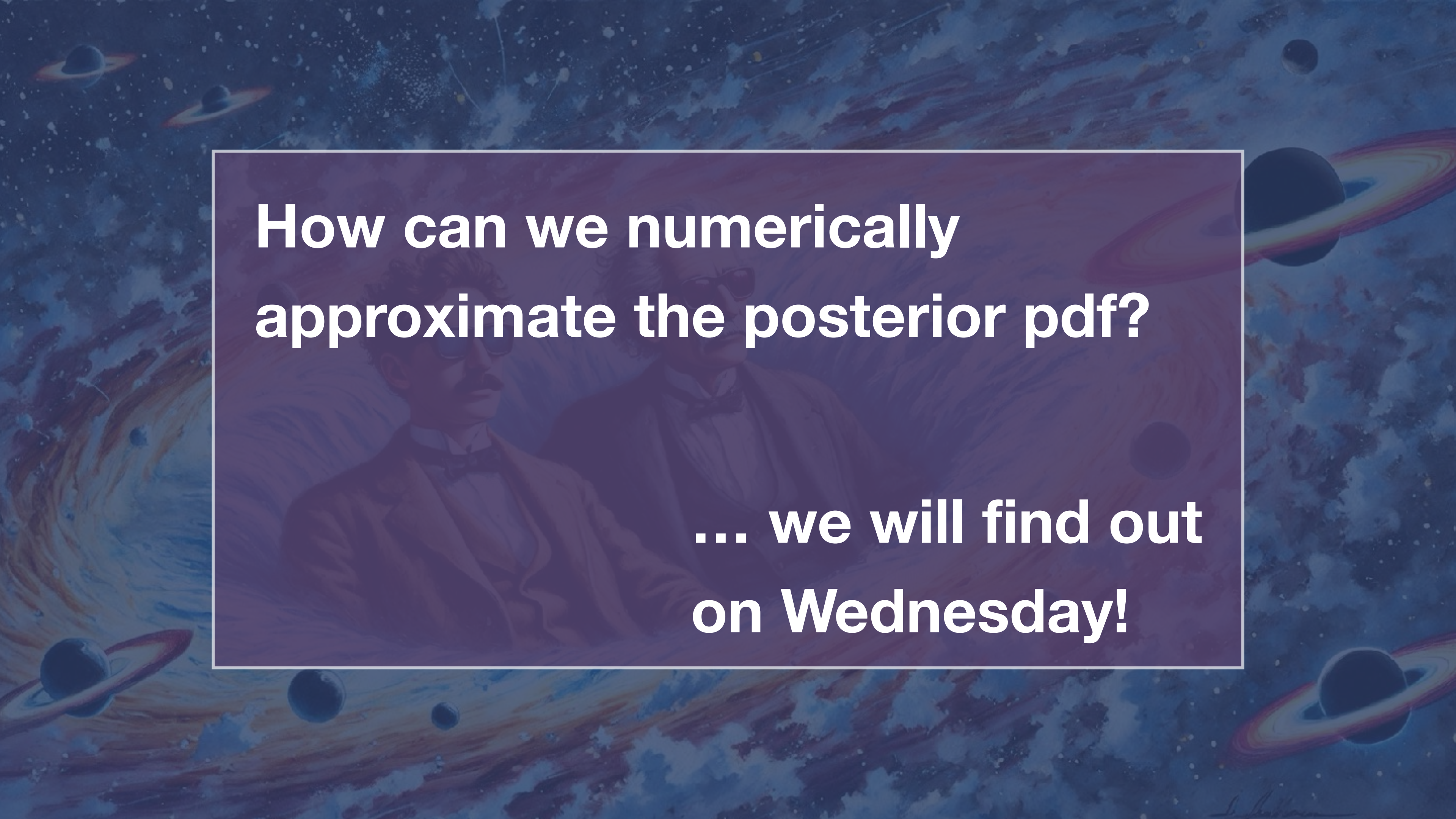




**We've chosen priors,
but they're not
conjugate! 🤯**



**How can we numerically
approximate the posterior pdf?**



**How can we numerically
approximate the posterior pdf?**

**... we will find out
on Wednesday!**